

# Ross Casebook 2025-26



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# Welcome to the Ross 2025-26 Casebook!

Dear CC@R Member,

Welcome to the Ross 2025 Casebook. This book focuses on the case interview portion of the consulting interview and is to be used in conjunction with other case-oriented club training materials.

The elements tested in a case interview are core to firms' hiring decisions. These cases, or mini-business problems, are a glimpse into a consultant's life as they are frequently taken from real client experiences. Cases become a natural way of thinking about how to structure approaches and solutions to nearly any type of problem. Along the way, we hope you enjoy solving problems in this manner and would enjoy performing this type of work for a living.

To facilitate your preparation, your Consulting Club at Ross has created the following cases with customized frameworks and solution elements. These cases act as a strong reference point for what to expect during a consulting interview but are in no way exhaustive. Since each case is a conversation between the interviewer and the candidate, it is very plausible that one candidate could receive the same case from two different interviewers and have two very different conversations about the business problem. In fact, we encourage this.

Finally, you may have noticed that you are reading this compilation in landscape format. This is intentional. Consultants think in terms of PowerPoint slides much more often than essay-style documents. They also constantly work to devise the most succinct way to illustrate and frame-out a problem, necessary action steps, and a solution.

Good luck, and remember, your fellow club members are here to help!

**2025-2026 Board**  
**Consulting Club at Ross (CC@R)**

# Special thanks to our 2025 case writers!

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**Any issues on cases (incorrect math, feedback to improve)?  
Provide feedback using the QR code to the right:**



# How to Use the Ross 2025 Casebook

# Case Structure

How to case as an interviewee

## Understand the Question (~1-2 minutes)

- **LISTEN**
- Summarize the problem statement and ensure you understand the problem and goal(s)
- Ask ~2-4 clarifying questions around the topic and/or metrics to be used for the analysis
- The questions posed should necessitate a short response

## Develop Framework (~1-2 minutes)

- Ask for a moment to plan your structure/roadmap to solve the problem at hand
- Develop ~3-4 areas to analyze with sub-topics tailored to the problem
- Structure the framework logically, with most important topic first
- Provide the interviewer with a roadmap of where you plan to take the case

## Exhibits, Analysis, Brainstorming (~25 minutes)

- Refer back to your framework as you move through each main area
- Tie back each piece of analysis to the main objective or problem statement (i.e., “so what?”)
- Walk through the calculations and analysis, driving insights
- Answer brainstorming questions using structure
- Keep notes organized throughout the case!

## Synthesize Recommendations (~1-2 minutes)

- State your recommendation as a direct response to the problem/objective – it should not come as a surprise to the interviewer
- Incorporate key metrics/findings as part of recommendation
- Include risks and next steps
  - Risks and next steps should tie 1:1 (i.e., each risk must have a corresponding next step)

# Administering Cases

The interviewer's interaction with the candidate and ability to convey information can change the style and experience of a case. Given the interviewer's position of power in the discussion, there are several things to keep in mind prior to, during, and after a case interview.

## Preparing for Interview

- Read the case over 2-3 times
- Familiarize yourself with the relevant numbers and details
- Prepare for how you will address irrelevant questions or requests for data you do not have
- Prepare for how to pressure test interviewees, and to redirect interviewees as needed

## During Interview

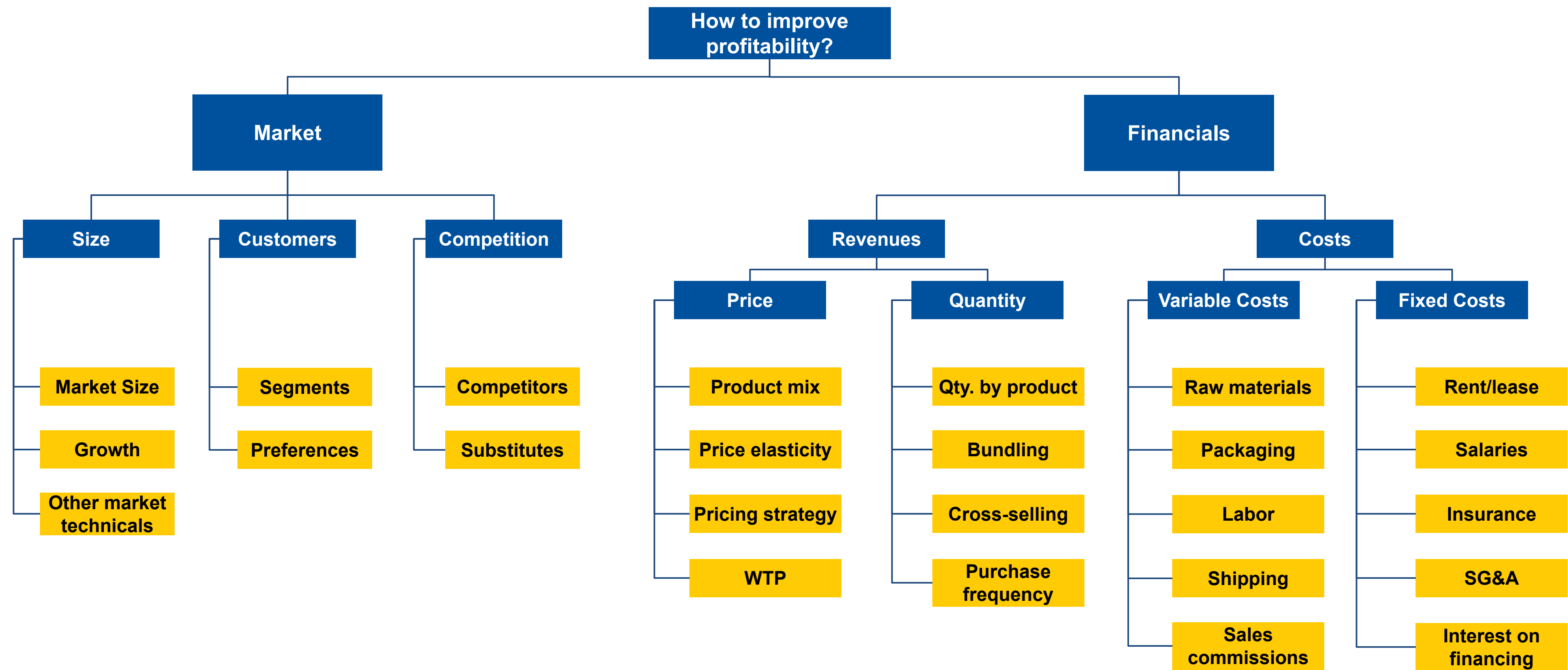
- Track time (~25 minutes is average) to balance finishing case and letting candidate struggle
- Candidates often think of very different approaches to cases. Before discounting questions as wrong, ask the candidate for their thinking... if it makes sense, go with it
- Consider what a consultant would be looking for in each candidate
  - Presentation: can I put this person in front of a client?
  - Aptitude: can this person do the work?
  - Interest: does this person like what they are doing?

## After Interview

- Provide feedback
  - This is possibly the most critical step of the case interview process
- Honestly let the candidate know strengths, but more importantly areas for improvement
- Without honest feedback and constructive criticism, it is very difficult to improve

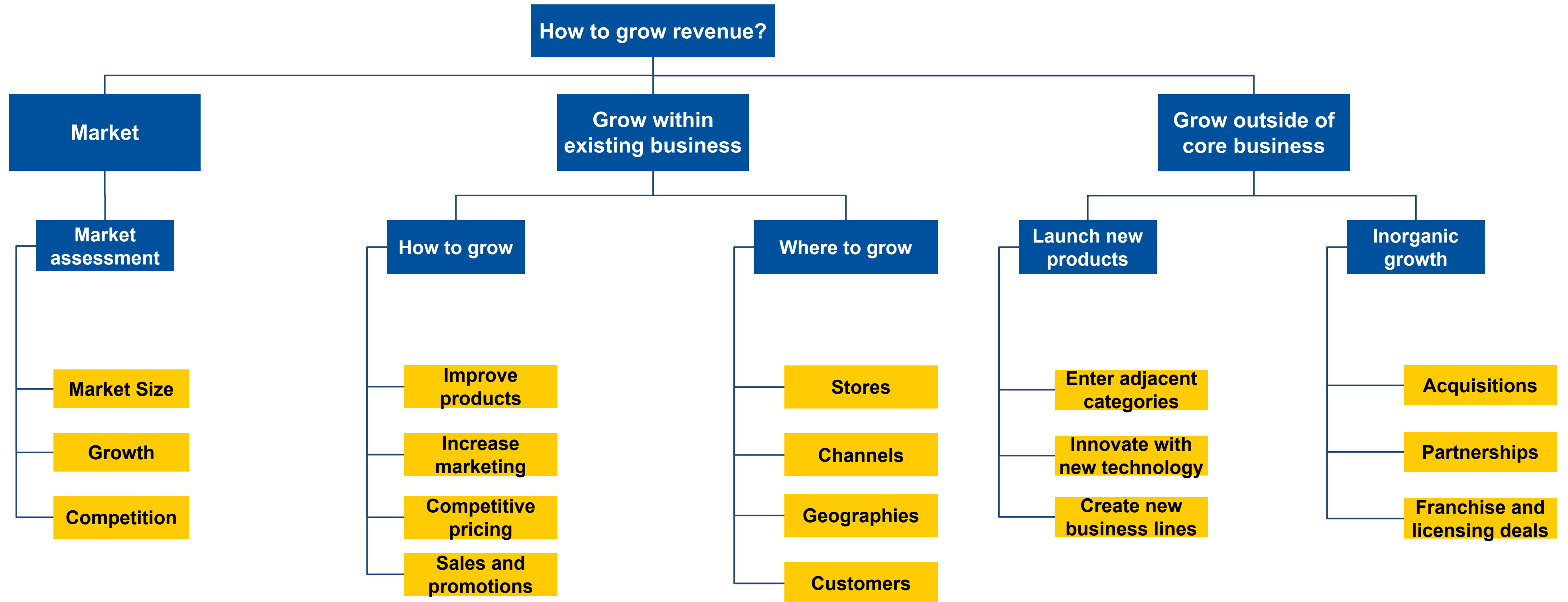
# Sample Frameworks

# Profitability Sample Framework



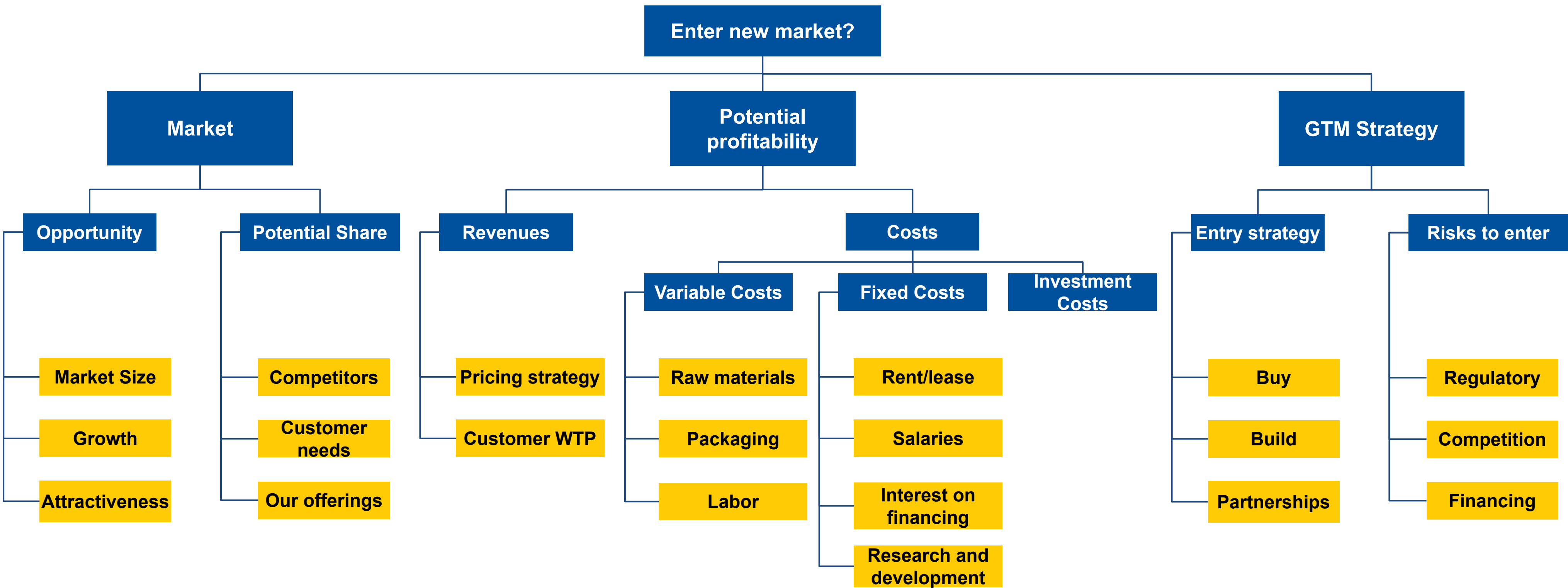
Note: Only core sample framework displayed. Other areas (e.g., internal capabilities, implementation, go-to-market, risks and considerations) may be needed depending on case specifics. These should be used as guides, not crutches.

# Revenue Growth Sample Framework



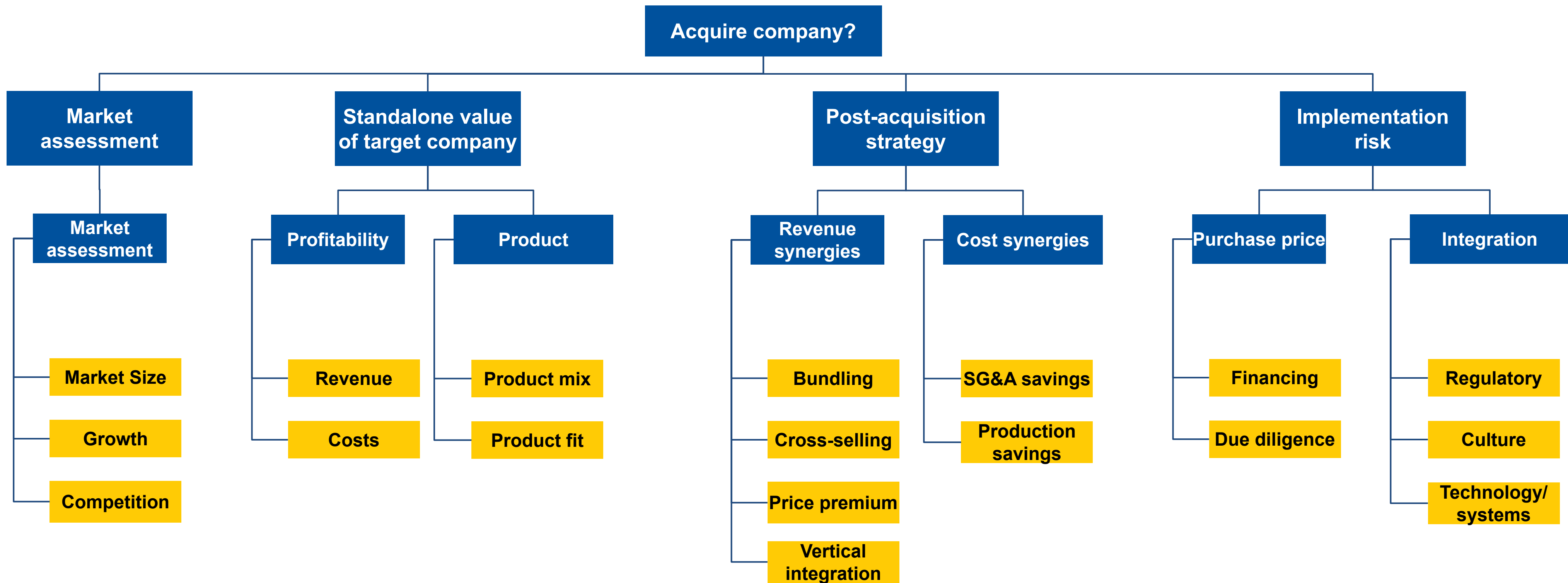
Note: Only core sample framework displayed. Other areas (e.g., internal capabilities, implementation, go-to-market, risks and considerations) may be needed depending on case specifics. These should be used as guides, not crutches.

# Market Entry Sample Framework



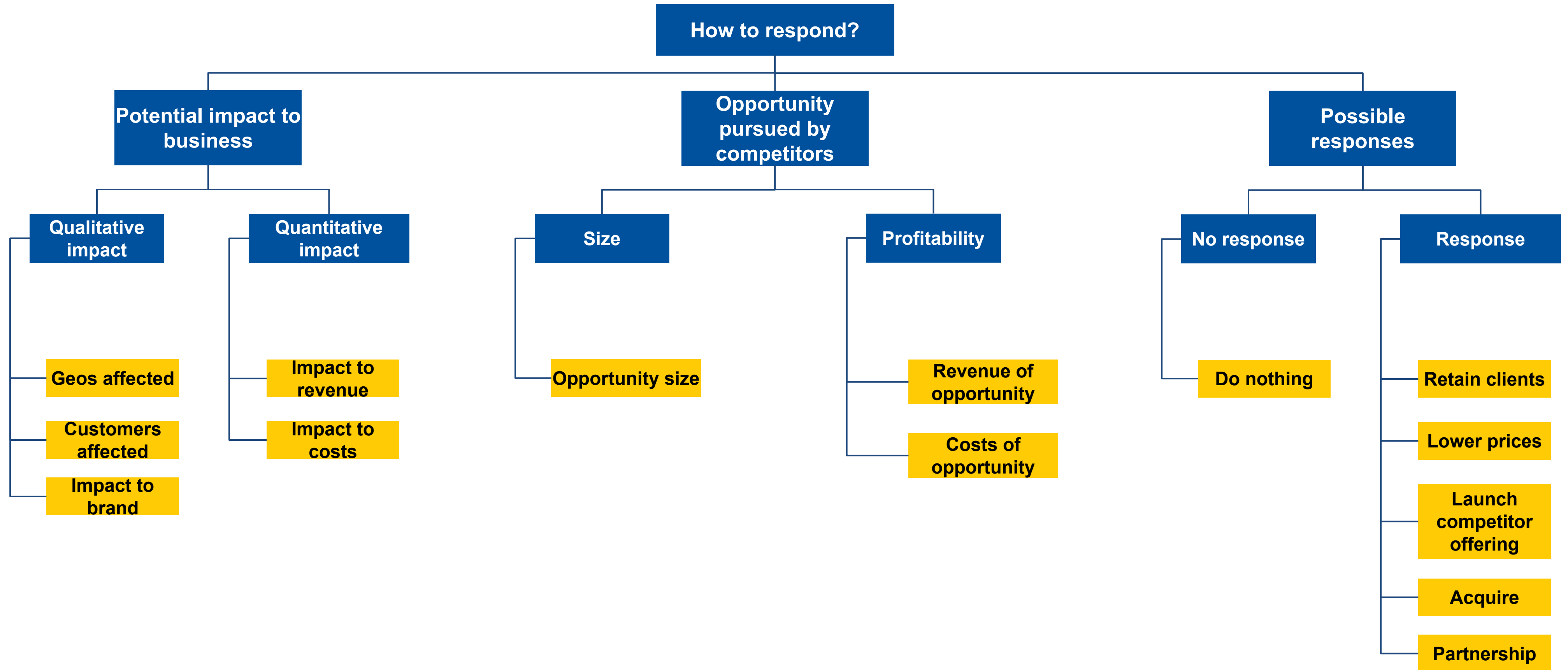
Note: Only core sample framework displayed. Other areas (e.g., internal capabilities, implementation, go-to-market, risks and considerations) may be needed depending on case specifics. These should be used as guides, not crutches.

# M&A Sample Framework



Note: Only core sample framework displayed. Other areas (e.g., internal capabilities, implementation, go-to-market, risks and considerations) may be needed depending on case specifics. These should be used as guides, not crutches.

# Competitive Response Sample Framework



Note: Only core sample framework displayed. Other areas (e.g., internal capabilities, implementation, go-to-market, risks and considerations) may be needed depending on case specifics. These should be used as guides, not crutches.

# Case Math

# Key Market Sizing Figures

## Population

- Global Population: 8B
- US Population: 330M (~320M for estimation)
- China Population: 1.4B (~1.5B)
- India Population: 1.4B (~1.5B)
- Japan Population: 125M

## Economics

- US GDP: \$30T
- US GDP per capita: \$85K
- US median individual income: \$47K (~50K)
- US median household income: \$85K
- Top 1% of income: \$400K
- US Work hrs/yr: 2,000 hours

## Demographics

- Every 20 years, equal breakdown of population:
  - 0-19 years: 64M
  - 20-39: 64M
  - 30-59: 64M
  - 60-79: 64M
  - 80+: 64M
- US life expectancy: ~80 yrs
- No. of households: 120M
- People per household: 2.5

| Topic                          | Formula                                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| NPV or Valuing Money Over Time | $\text{Value to Perpetuity} = \frac{\text{Value of Asset}}{\text{Discount Rate}}$ $\text{NPV} = \sum_{t=0}^n \frac{\text{Annual Cash Flow}}{(1+r)^t}$ |
| Rule of 72                     | $\text{Time for Invested Principle} = \frac{72}{r}$ <p><math>r</math> = Rate of Return (%)</p>                                                        |
| Little's Law                   | $\text{Inventory} = \text{Throughput} \times \text{Flow Time}$                                                                                        |
| Inventory                      | $\text{Inventory Turns} = \frac{\text{COGS}}{\text{Average Inventory}}$ $\text{Days of Inventory} = \text{Inventory Turns} \times 365$                |
| Profitability                  | $\pi = Q (P - VC) - FC$                                                                                                                               |
| Breakeven                      | $\text{Breakeven} = \text{Investment Price} - \text{Cost}$                                                                                            |
| Margin                         | $\text{Gross Margin} = \frac{\text{Revenue} - \text{Cost}}{\text{Revenue}}$ $\text{Net Margin} = \frac{\text{Net Income}}{\text{Sales Margin}}$       |
| Markup                         | $\text{Markup} = \frac{\text{Price} - \text{Cost}}{\text{Cost}}$                                                                                      |

| Topic                  | Formula                                                                                                                                                                                                                                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Return on Assets (ROA) | $ROA = \frac{\text{Net Income}}{\text{Total Assets}}$                                                                                                                                                                                                                                                                      |
| Return on Equity (ROE) | $ROE = \frac{\text{Net Income}}{\text{Total Shareholders' Equity}}$                                                                                                                                                                                                                                                        |
| DuPont Analysis        | $ROE = \frac{\text{Net Profit}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}$                                                                                                                                                                                        |
|                        | $ROE = \text{Operating Efficiency} * \text{Asset Utilization} * \text{Leverage}$                                                                                                                                                                                                                                           |
| Working Capital        | $ROI = \frac{\pi}{K}$ <p>K = Capital Invested (Assets, Working Capital, etc.)<br/>Working Capital = Assets - Liability</p>                                                                                                                                                                                                 |
| Income Statement       | $\begin{aligned} &\text{Sales} - \text{COGS} \\ &= \text{Gross Profit} \\ &\quad - \text{SG\&A} \\ &= \text{EBITDA} \\ &\quad - \text{Depreciation/Amortization} \\ &= \text{Operating Profit} \\ &\quad - \text{Interest Expense} \\ &= \text{EBIT} \\ &\quad - \text{Tax Expense} \\ &= \text{Net Income} \end{aligned}$ |

# Industry Primers

# Airlines

## Key Ideas

- Consolidation in industry
- Capital intensive, regulated industry
- Low cost carriers and fare competition on competitive routes
- Expansion of domestic & international routes
- Capacity optimization (Load Factor)
- Narrow profit margins
- Passenger volume back to pre-covid levels

## Revenue Streams

- Ticket sales to economy and business passengers with dynamic pricing models
- Charges for baggage and on-board services (up-selling)
- Cargo transportation
- In-flight purchases
- Credit cards and loyalty programs

## Cost Drivers

- Aircraft purchases/leases
- Fuel (price is often volatile)
- Labor (often unionized)
- Aircraft maintenance
- Takeoff/landing (airport) fees
- Insurance/legal fees
- Depreciation (non-cash, but still impacts)

## Customer Segments

- Leisure travelers – (generally price sensitive)
- Business travelers – (very important to airlines due to higher margins and services purchased)
- Freight/Cargo transportation

## Channels

- Internet - online travel sites, airline websites
- Airline sales team: call centers, online, or kiosk
- Channel partners: Travel management companies, travel agents, credit card travel portals

## Risk

- Changes in fuel prices have a major impact on profitability (especially if unhedged)
- Macro-economic conditions greatly impact amount of leisure travelers
- Intensely competitive market with narrow margins; many airlines partly government subsidized

## Key Economic Drivers

- World price of crude oil (very volatile, typically follows economic cycles but can be disrupted due to other macroeconomic factors)
- Trips by US residents (especially abroad for tourism) - disposable income is important here
- Optimization of capacity (to ensure planes are as filled as possible)

## Key Companies

- Delta Airlines
- American Airlines
- United Airlines
- Southwest Airlines
- Emirates
- Air France
- KLM
- British Airways
- RyanAir
- AeroMexico
- Singapore Airlines

# Automotive & Manufacturing

## Key Ideas

- Automakers, Original Equipment Manufacturers (OEMs), replacement parts production, rubber fabrication
- Highly capital and labor intensive
- High competition from foreign automakers
- Unions (strikes, higher wages, etc.)
- Tariffs impact supply chain and assembly
- Autonomous driving is an emerging trend

## Revenue Streams

- New car sales
- Auto part sales (typically highest margin)
- Services offered with vehicle purchase
- Financing
- Extended warranties
- Leasing

## Cost Drivers

- Labor
- Materials
- Transportation costs (parts & finished goods)
- Advertising
- Financing costs
- Recall costs
- Research & Dev. (especially for EVs)
- Customs/Taxes (from tariff impact)

## Customer Segments

- Cars, vans, pickup trucks and SUVs
- Personal car buyers
- Rental car companies
- Commercial and government (e.g., law enforcement) purchasers

## Channels

- Automobile dealerships
- Secondary automobile market (i.e., auctions)
- Automotive parts/services outlets

## Risk

- Globalization of the industry enables more ease of foreign competition
- Extensive competition impact on already low margins
- Changes in consumer trends and tastes
- Impact of autonomous and ride share may impact future car sales / reduce car ownership

## Key Economic Drivers

- GDP growth, income growth, and disposable income
- Price of commodities (crude oil, steel)
- Consumer Confidence Index
- Regulatory changes (e.g., EV mandates)

## Key Companies

- Ford
- General Motors
- Toyota
- Honda
- Tesla
- General Electric
- Boeing
- Airbus
- Embraer
- Lockheed Martin
- Siemens
- Honeywell

# Financial Services/Banking

## Key Ideas

- High consolidation and acquisitions
- Increased mobile banking
- Channel innovation in digital & physical
- Offshoring of call centers, back office functions
- Digitization of processes
- Cross-selling (e.g., loans and credit cards)
- Rise of FinTechs disrupting industry

## Revenue Streams

- Loan interest
- Loan types (real estate, auto, education, personal)
- Service fees (maintenance, reporting)
- Spread between interest rate charged and Fed rates
- Foreign exchange fees and trade finance
- Investment income (from investing deposits)

## Cost Drivers

- Wages
- Bad debt expense
- Technology investments e.g., compliance systems, core banking systems
- Interest rates on deposits
- Branch and compliance costs
- Branding, advertising and marketing
- Regulatory/compliance costs (reporting)

## Customer Segments

- Retail customers (individuals) with varying preferences based on income/wealth
- Corporate clients needing complex financing, risk management, financial advisory
- Government/nonprofits
- For mobile banking / FinTechs: typically younger consumers

## Channels

- Online (through mobile app/website)
- Branches
- Call centers
- Relationship managers (for corporate clients / high net-worth individuals)

## Risk

- Macroeconomic risks: impact of economic conditions, inflation, geopolitical stability
- Market risks: interest rates, FX rates, and securities valuations
- Operational risks: cybersecurity attacks, fraud, system downtime
- Regulatory risks: fines, changes to regulation/compliance

## Key Economic Drivers

- Consumer confidence
- Household debt
- Employment statistics
- Urbanization
- Disposable income
- Interest rates
- Government regulations
- Technology advancements

## Key Companies

- Wells Fargo
- Citicorp
- US Bank
- HSBC
- Bank of America
- JP Morgan Chase
- Barclays
- Deutsche Bank
- Credit Suisse
- Morgan Stanley

# Healthcare

## Key Ideas

- High M&A activity/consolidation
- Highly fragmented: top 50 organizations account for only 15% of revenues
- Aging population driving increased revenues
- Rise of digital and telehealth services
- Continuous impact of global health events (COVID, rollback of vaccine mandates)

## Revenue Streams

- Hospital care
- Insurance reimbursements
- Out-of-pocket payments
- Physician and clinical services
- Prescription drugs
- Monetization of proprietary technology
- Government subsidies/funding

## Cost Drivers

- Labor costs
- Medications, supplies, equipment
- Research & development
- Technology implementation costs
- Regulatory/compliance costs
- Malpractice insurance
- Administrative overhead (billing, claims)

## Customer Segments

- Patients/consumers
- All generations and segments of the population require different products/services
- Employers requiring healthcare packaging for staff
- Pharma and device companies (for partnerships, distribution)

## Channels

- Hospitals / urgent care centers
- Doctors offices
- Nursing homes
- Outpatient surgery centers
- Pharmacies
- Medical equipment
- Telemedicine platforms
- Public / community health centers

## Risk

- New legislation (impact of Affordable Care Act still uncertain, changes to Medicare/Medicaid)
- Funding availability
- Supply chain shortages for large global health events (e.g., COVID-19 impact on PPE availability)
- Technology disruption (including expensive technology becoming obsolete)

## Key Economic Drivers

- Demographic shifts (aging population)
- Rates of insurance coverage impact consumer use of healthcare
- Federal funding for Medicare and Medicaid
- Advances in medical care and technology

## Key Companies

- HCA Healthcare
- Ascension Health
- Trinity Health
- CommonSpirit Health
- Advent Health
- Dignity Health

# IT & Infrastructure

## Key Ideas

- Cloud based platforms vs. on-premise infrastructure
- User centric IT solutions – IT depts want to enhance usage and productivity
- Open platforms / integrating and partnering with other providers

## Revenue Streams

- Hardware sales (servers, network)
- Cloud services (storage, SaaS, IaaS)
- Consulting and system integration services
- Maintenance contracts
- Connectivity services (VPN, enterprise Wi-Fi)
- Managed IT services (outsourced)

## Cost Drivers

- Labor
- Data center operations and maintenance
- Software upgrades
- Licensing fees
- Vendor management
- Hardware manufacturing
- Sales and marketing

## Customer Segments

- Commercial customers (multinational companies needing global IT solutions/outsourced models)
- Government (emphasis on security, local hosting, and compliance)
- Regulated industries (needs critical infrastructure)
- Individual customers (price-sensitive, value-seeking for software and cloud services)

## Channels

- Direct sales through sales teams for enterprise accounts
- Channel partners and value-added resellers, managed service providers and system integrators
- Online self-service ordering, onboarding, and provisioning platforms.

## Risk

- Competition risk: startups and new entrants able to disrupt
- Environmental risk: data center energy usage and carbon emissions
- Political risk: tariffs, geopolitical tensions affecting global supply chains

## Key Economic Drivers

- Cybersecurity threats increase need for investment in infrastructure
- Demand for enterprises to go digital creates need for IT solutions
- Remote and hybrid work environments require robust IT infrastructure

## Key Companies

- Microsoft
- HP
- IBM
- Dell-EMC
- SAP
- Oracle

# Non-profits

## Key Ideas

### Intended Impact

- Define success criteria
- Think big picture (e.g., society, people you are working for/with)
- Consider trade-offs
- Depth vs. breadth of reach
  - Quality vs. quantity of program initiative
  - Intended impact should align with strategic goals

### Theory of Change

- Define specific actions steps to achieve the intended impact
- Define timelines, initiative priorities and ownership responsibilities

### Implementation Feasibility

- Revenue impact (self sustaining model, grants)
- HR costs: creating new roles, hiring new staff, train existing and new staff, modify existing organization structure
- New infrastructure cost: IT systems, office space
- Indirect costs
  - Impact on culture of organization
  - Impact on scale on quality of outcomes

### Performance Measures & Reporting Impact

- Measure performance vs. peers
- Set milestones for financial and operational goals
- Monitor and modify plan accordingly
- Consider performance during and after implementation of initiatives

### Case Topics

- Growth through existing platforms
- Growth through new partnerships
- Growth driven by policy changes
- Thought sharing to strengthen the industry
- Growth using technology

# Oil & Gas

## Key Ideas

- Capital intensive industry
- Highly regulated industry
- Ongoing consolidation/M&A to reduce costs and achieve economies of scale
- Pressure to pursue renewable energy
- Global geopolitical tensions affect supply and demand for oil/gas

## Revenue Streams

- Upstream (Sale of crude oil, natural gas)
- Midstream (Fees from transportation storage, gathering, and processing)
- Downstream (Sale of refined gasoline)
- Gas stations: gas and convenience products

## Cost Drivers

- Exploration: seismic studies, drilling rigs
- Production: refining
- Pipeline construction and maintenance
- Transportation and logistics
- Regulatory compliance (testing, emissions controls)
- Insurance
- Labor

## Customer Segments

- Utilities, manufacturers, and chemical producers
- Refineries buying crude as input for processing
- Wholesalers and retailers of gas products
- Businesses and individuals purchasing refined products (via retail or service stations)

## Channels

- Retail gas stations and stores
- Fuel wholesalers and marketers expanding end-market reach.

## Risk

- Price risk: Volatile oil prices driven by global forces
- Geopolitical risk: Supply disruptions, sanctions, nationalization, political instability
- Regulatory Risk: Spills, emissions, and non-compliance
- ESG Risk: Transition to clean energy may decrease demand for oil/gas

## Key Economic Drivers

- Government regulation (or lack thereof recently) for renewable energy vs. oil/gas production
- Advances in technology, incl. fracking
- US dollar strength (oil is priced in USD)

## Key Companies

- Exxon Mobil
- Baker Hughes
- Saudi Aramco
- Gazprom
- Pemex
- Philips 66

# Pharmaceutical

## Key Ideas

- Strong innovation and investment in R&D
- Regulated environment (FDA)
- Frequent M&A activity / consolidation
- AI has improved discovery, clinical trial optimization, and patient engagement
- Demand for ethically-sourced materials/manufacturing
- Multi-sourcing supply chains after COVID-19

## Revenue Streams

- Prescription drug sales
- Licensing/royalties from IP
- Federal government grants for R&D
- Vaccines
- Over the counter (OTC) products

## Cost Drivers

- Research & Development
- Manufacturing cost (the largest share of the industry's costs)
- Sales and marketing costs
- Wages (scientists, experts)
- Liability insurance and legal fees
- Intellectual property protection costs (legal)

## Customer Segments

- Healthcare providers (physicians, clinics)
- Payers and insurers (public and private insurance companies)
- Pharmacies/distributors
- Patients

## Channels

- Over-the-counter at pharmacies and retailers
- Online and mail-order pharmacies
- Partnerships (for exploring new geographies)

## Risk

- Competitive risk: generic manufacturers pose major threat following patent expiration
- Trade risk: imposed tariffs could impact sales
- Regulatory risk: New drugs/products subject to clinical and marketing approval of federal agencies
- Supply chain risk: Shortages, recalls, and quality issues

## Key Economic Drivers

- Aging populations driving demand for more pharmaceutical drugs and products
- M&A and partnership activity can affect growth and efficacy of drug development
- Patent protection laws
- Insurance and regulatory landscape

## Key Companies

- Roche
- Pfizer
- Johnson & Johnson
- Merck
- Novartis
- Abbvie
- Moderna

# Private Equity

## Key Ideas

- Record levels of uninvested capital at funds (“dry powder”)
- Increased focus on value creation (operational improvements, transformation)
- Ongoing regulatory challenges incl. transparency, fees, and ESG reporting
- Difficulty turning around without strong management to run portfolio companies

## Revenue Streams

- Invested capital
- Transaction and management fees (1.5-2% of committed capital)
- Carried interest (typically 20%)
- Consulting fees to portfolio companies
- Proceeds from exits

## Cost Drivers

- Personnel compensation (high salaries and bonuses)
- Due diligence costs (legal, advisory, and consulting services)
- Fund operations (admin, compliance)
- Regulatory costs (reporting)

## Customer Segments

- Institutional investors: pension funds (largest share), endowments, sovereign wealth funds, banks
- Private investors: high net-worth individuals, family offices
- Portfolio companies: post-acquisition firms invested in and managed by PE funds

## Channels

- Large firms focus on deals ~ \$1.0B; middle market firms cover deals between \$15.0M- \$1.0B
- Average holding period before sale has increased from 3 years to 6 years in the past 15 years
- Borrowing can typically range from 65.0% to 85.0% of the purchase price of the firm

## Risk

- Market risk: economic downturns affect portfolio value and exits
- Execution risk: PE firm may not be able to turn around the company / exit profitably
- Regulatory risk: higher costs from increased reporting needed
- Reputational risk: public scrutiny for layoffs, short-term mindset

## Key Economic Drivers

- Interest rates and borrowing conditions affect access to leverage (debt)
- Exit opportunities (or lack thereof) can impact investment decisions
- Higher regulations can impact investment decisions

## Key Companies

- Blackstone
- KKR
- CVC Capital Partners
- The Carlyle Group
- EQT
- TPG Capital

# Retail

## Key Ideas

- E-commerce impacting brick and mortar
- Direct to consumer through online platforms vs. third-party retailers
- Personalization of marketing using AI
- Rising demand for ethical sourcing, sustainable products
- Growth in subscription models
- Tariffs impact on supply chains

## Revenue Streams

- Product sales (apparel, cosmetics, furniture and household appliances, toys, footwear)
- Subscription model (Amazon Prime, Costco)
- Services (warranties, delivery)
- Loyalty programs
- Financial services (credit card partnerships, buy now pay later)
- Advertising (in store and online) sold to brands

## Cost Drivers

- Cost of Goods Sold (very large cost)
- Supply chain/logistics (transportation, warehousing, fulfillment, returns)
- Labor (wages, benefits)
- Rent and utilities
- Technology investments (POS systems, e-commerce)
- Shrinkage (losses from theft and damage)

## Customer Segments

- Mass market consumers (price sensitive, seek value)
- Affluent shoppers (less price sensitive, seek premium goods)
- Corporate clients (for office supply, uniforms, furniture, etc.)

## Channels

- Brick and mortar (Department stores/big box retailers, discount retailers)
- E-commerce marketplace (Amazon, eBay, retailer online website)
- Catalogs and direct mail
- Mobile apps

## Risk

- Demand risk: changes in economic conditions, consumer taste and preferences
- Supply chain risk: inventory shortages, geopolitical issues
- Financial risk: eroding margins from price wars, competition, and increase in costs
- Labor risk: labor shortages, strikes, store closures

## Key Economic Drivers

- Tariffs impacting global supply chains
- Changes in demographics and population growth driving consumption
- Strong revenue concentration in holiday months
- Increased digital adoption enables e-commerce growth

## Key Companies

- Walmart
- Target
- Costco
- CVS
- Walgreens
- Kroger

# Telecommunications

## Key Ideas

- Deregulation led to spur of new companies
- Bottlenecks: high capital, scarce operating skills and management experience
- Bundling of services (convergence of telco, cable, internet and content)
- 5G / Fiber rapid expansion
- AI for network security and predictive maintenance

## Revenue Streams

- Subscription (mobile, internet, etc.)
- Usage-based fees (international roaming)
- Leasing of towers to smaller operators
- Media / streaming
- Accessories
- Cloud services for enterprises
- Government contracts

## Cost Drivers

- Infrastructure (acquisition, construction, maintenance of towers, fiber, etc.)
- Technology upgrades (5G, fiber)
- Energy costs
- License and regulatory fees
- Wages (for operations and maintenance)
- Marketing and advertising

## Customer Segments

- Residential and small business (price sensitive and seeking reliable service)
- Large multinationals (price insensitive, demand for customized secure offerings)
- Governments (critical infrastructure and public safety)
- Other industries needing telecom connectivity in its assets/services (e.g., technology industry)

## Channels

- Retail stores - carriers and mass retailers
- Online (billing, customer service)
- Third-party retailers (authorized dealers, electronics outlets)
- Direct sales (especially for B2B)

## Risk

- Technology risk: rapid advances requiring constant upgrades and old tech becomes obsolete
- Regulatory risk: changes in regulations, telecom policy, privacy rules, net neutrality
- Competitive risk: saturated market with high competition leading to price wars
- Climate risk: natural disasters creating more costly maintenance and operational expenses

## Key Economic Drivers

- Population growth and increasing usage per user fueling network demand and innovation
- Enterprises merging to cloud requiring stronger telecom solutions
- Economic expansion fuels more demand for network connectivity

## Key Companies

- Verizon
- AT&T
- T-Mobile
- Vodafone
- NTT Corp
- Telefonica SA
- Spectrum

# Utilities

## Key Ideas

- Increase in energy consumption (climate change, population growth)
- High investment costs and regulation
- Industry disintegrating into smaller segments
- Seasonality (climate affects usage)
- Clean energy transition
- Infrastructure renewal / upgrades
- Global rollback of sustainability initiatives

## Revenue Streams

- Electricity sales (kWh-based billing)
- Water and gas sales
- Connection and service fees (new hookups, metering)
- Government subsidies/grants
- Ancillary services (energy efficiency upgrades)

## Cost Drivers

- Purchased power accounts (costs of buying from renewables or spot markets)
- Capital investment (infrastructure upgrades)
- Fuel (natural gas, coal-based generation)
- Debt servicing (highly leveraged)
- Insurance (liability, environmental risk)
- Wages
- Vegetation/disaster management

## Customer Segments

- Residential: households, apartments (smallest per-customer usage)
- Commercial: businesses, retail, institutions (mid-tier consumption)
- Industrial: factors, mines (heaviest users, often have negotiated fees)
- Government: Municipal services, city infrastructure, military bases

## Channels

- Transmission lines/pipelines
- Upstream electricity generators
- Wholesale markets (for purchase/sale of surplus)

## Risk

- Regulatory risks: clean energy mandates, carbon pricing, rollback of clean energy initiatives
- Price risk: Fluctuations in fuel costs, utility price due to seasonality
- Climate risk: Natural disasters becoming more common lead to higher maintenance costs
- Market risk: rising interest rates can impact debt servicing, access to financing

## Key Economic Drivers

- Population growth increasing demand for electricity, water and gas
- Climate/seasonality
- Emerging industries (e.g., AI) requiring substantial electricity
- Falling prices of renewables could shift usage to alternate sources

## Key Companies

- PG&E
- AES
- Duke Energy
- Exelon
- National Grid
- Nextera Energy
- DTE Energy
- Southern California Edison
- Florida Power and Light (FPL)
- Dominion Energy

# Cases

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# GPG

Non-Profit | Market Entry & New Product  
*Interviewer Led*

Ratings: Quant - 1

Qual - 1

Overall - 1

# Prompt

Case 1 – GPG

Our client is Bass Industries, a global commercial real estate developer headquartered in New York City. Bass Industries needs to expand its CSR portfolio in its home city and has decided to focus on public restroom availability.

Bass Industries believes that this initiative relies on existing core competencies such as site selection, zoning management, and dispersed maintenance. Additionally, the issue of restroom availability has been a topic of debate at the New York City Council this year. Bass Industries has reached out to us for help assessing this proposed project.

## Clarifying Information

**What is Bass Industries' specific objective in pursuing this project?**

- Bass Industries is seeking to generate a positive social return and contribute to New York City while adhering to their core corporate principle of robust return on capital.

**Why is Bass Industries pursuing CSR investments now?**

- Our client recently elected a new CEO whose priorities are not identical to his predecessor's.

**What did the NY City Council Debate entail?**

- The Council identified that NYC has many fewer public restrooms per capita than peer cities around the world. Currently, NYC has approximately one public toilet per 7,820 residents. The council set a target ratio of one public toilet per 2,000 residents.

**Is the new CEO Chuck Bass?**

- Hard to say, really.

# Question 1: (Potential Framework)

Case 1 – GPG

What are the factors that Bass Industries should consider as it creates a plan for their CSR initiative?

## Feasibility

- Access to real estate
- Zoning and public policy
- Maintenance and cleaning access
- Costs
  - Construction
  - Leasing
  - Maintenance
  - Cleaning

## Impact

- Identify best locations for placement
- Establish KPIs to evaluate projects
- Launch pilot and assess impact level and relevance to local community
- Continuous evaluation and adaptability of on-going project
- Connect with relevant communities
- Develop robust stakeholder relationships

## Public Relations

- Methods of ensuring positive reception
- Controlled narrative
- Influencer relationship
- Minimization of downside risk associated with project failure

# Question 2

**Bass Industries has identified two potential routes to move forward: permanent installation construction with sewer access and “ThroneLab” mobile restrooms. From a financial standpoint, how would you assess these options?**

Read to candidate only if probed by interviewee:

## **Permanent Construction:**

25 Units  
10 Toilets per Unit  
\$500,000 per Unit Construction Cost  
\$75,000 Annual Rent per Unit  
\$0 Annual Leasing Fee

## **ThroneLab:**

250 Units  
1 Toilet per Establishment  
\$0 Construction Cost  
\$5,000 Annual Rent Per Unit  
\$10,000 Annual Leasing Fee

## **Solution:**

### **Permanent Construction:**

25 Units \* 10 Toilets = **250 Toilets**  
 $\$500,000 * 25 = \$12.5\text{M}$  / 50 Year Lifespan  
= **\$250,000**/yr amortized construction cost

$\$75,000 / \text{Year} * 25 = \$1,875,000 / \text{Year}$   
(Rent)

$\$250,000 + \$1,875,000 = \$2,125,000$  total  
annual cost / 250 toilets  
= **\$8,500** / Toilet / Year

### **Solution:**

#### **ThroneLab:**

$\$10,000 / \text{Toilet} + \$5,000 / \text{Toilet}$   
= **\$15,000** / Toilet

## **Intended Takeaways:**

(1) Candidates should ask about the expected lifespan of the constructed toilets in order to compare like units.

**Each establishment will last 50 years with maintenance.**

(2) Candidates should perform the calculations listed under solution and identify that permanent construction yields much lower cost / toilet.

## **Excellent Candidates:**

(1) Quickly identify that permanent construction has much higher initial costs but much lower unit costs

(2) Rapidly identify that the most relevant metric will be annual cost / toilet (and not cost / establishment)

(3) Will note that although the annual cost / toilet is lower for permanent construction, it does require **vastly** more upfront investment than ThroneLab.

# Exhibit 1

## Public Survey Data

| Question                                                                   | Manhattan | Brooklyn | Queens | Bronx | Staten Island |
|----------------------------------------------------------------------------|-----------|----------|--------|-------|---------------|
| I am satisfied with access to public restrooms in my neighborhood.         | 1.6       | 2.0      | 1.5    | 2.1   | 1.9           |
| My neighborhood would be improved by greater presence of public restrooms. | 2.1       | 4.2      | 4.0    | 3.1   | 4.3           |
| Access to public restrooms is an important issue to me.                    | 3.9       | 3.4      | 4.1    | 2.9   | 1.4           |

# Question 3: Exhibit 1 Guidance

Case 1 – GPG

To assess the impact this investment might have, the client commissioned an opinion poll. Respondents were asked to rate their agreement with three statements on a 1-5 scale, with 1 being “strong disagree” and 5 being “strongly agree.” What are your takeaways from this data?

## Interviewer Notes:

**Intended Takeaways:** Manhattan residents are opposed even though they aren’t satisfied. They feel reasonably strongly. Brooklyn residents are in favor and unsatisfied, and they feel somewhat strongly. Queens residents are in favor and are not satisfied, this is the borough where people feel the most strongly about the issue. Bronx residents are relatively neutral, and this is not a big priority. Staten Island residents are dissatisfied and believe that access would improve their neighborhoods but are indifferent.

**Excellent Candidates:** Will identify that Brooklyn and Queens are the best locations, balancing desire and fervor of residents for more public restroom access.

# Conclusion

Case 1 – GPG

## Recommendation

- Bass Industries should invest in **permanently constructed toilets** in **Brooklyn**.
- These choices maximize return on investment at \$8,500 / toilet / year while placing the toilets in the borough where they are most wanted.

## Risks

- Polling errors
- Errors in projecting the lifespan of the permanently constructed installations

## Next Steps

- Conduct more robust public opinion surveys
- Buy insurance

# Maize and Blue Cement

Engineering/Construction | Profitability / Cost Optimization

*Interviewee Led*

Ratings: Quant - 1

Qual - 2

Overall - 1

# Prompt

A cement manufacturer, Maize and Blue Cement (MBC), operates in Country X. Over the last three years, their profitability has stagnated, and MBC is currently under pressure to increase it.

MBC has three main customer segments: private-owned general contractors, government/state-owned general contractors, and retail. The reason for differentiating between private-owned and state-owned general contractors in Country X is that it is generally believed to be necessary to provide special treatment when dealing with government/state-owned companies officially.

The CEO believes that an opportunity lies in their distribution strategy, particularly in transitioning from using third-party service providers to an in-house approach. Should they pursue this change?

## Clarifying Information

- Third party service providers handle end-to-end logistics once the product leaves the MBC's distribution center to hardware stores and the sales and relationship with hardware stores is done by the third party
- Revenue per channel: Retail: 10%, Private-owned contractors: 40%, State-owned contractors: 50%
- There is no single industry standard for distribution. One big competitor does the distribution by themselves, whereas another big competitor also uses third party services.
- There is no specific profitability target. Any increase of profitability with reasonable risk is accepted.

# Potential Framework

## Financial Analysis

### Expected change in cost

- Increase in salaries and transportation
- Reduced commission/fees to third party

### Projected revenue change

- Potentially higher revenue through better distribution and relationship with customers

### Planned capex

- Potential capex for vehicles and sales equipment, etc.

## Strategic Value

### New opportunities

- Geographic expansion not limited to third party coverage

### Advantages in operations

- More flexibility, potential shorter lead times

### Improvement in customer experience

- Potentially stronger relationship with customers

Better understanding of customers

## Risks

### Financial risks

- Lack of scale
- Overestimate revenue / underestimate cost projection

### Operational risks

- Long time to develop capabilities
- Unable to get favorable funding for investment

# Profitability Calculation

**Prompt:** How much extra profit from private-owned general contractors segment will MBC get by converting the distribution to inhouse?

## Information Provided by Interviewer

MBC wants to focus on the private-owned general contractors at the moment as they believe they have a competitive advantage in this segment in the good relationship of the CEO with the contractors. Their internal analysis indicates that by doing end-to-end distribution process in this segment will impact the financials as follows:

| Financial Impact                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Increase the segment revenue by 35% as current third party does not focus on this customer segment.</li><li>• Current commission paid to third party: 10% of segment revenue</li><li>• Extra cost for internal operations: \$7M.</li><li>• Current total revenue: \$100M</li><li>• Current revenue Proportion per Segment: Retail: 10%, Private-owned contractors: 40%, State-owned contractors: 50%</li></ul> | <p><b>Important note if prompted by candidate:</b> Currently third-party service provider handle all distribution process for all segments</p> |

## Solution From Interviewee

Additional profit = Revenue Increase – Additional Cost =  $(35\% * 40\% * \$100M) - [\$7M - (10\% * 40\% * \$100M)] = \$11M$

# Brainstorming Question

**Prompt:** What would be major risks in continuing with the plan?

## Note for Interviewer

Good interviewee should notice that the dependency to the current third party service provider is very high, therefore the current third party service provider has high bargaining position that taking away some of their business may risk in their services in other segments.

## Solution From Interviewee

As third party service provider may not be happy with the new distribution strategy, it will risk MBC's business in the other 2 segments, especially the State-owned contractors segment which is bigger and require very good customer relationship that currently being managed by the third party

Revenue increase may be lower than expected and/or cost increase may be higher than expected

MBC may not be able to execute the operational distribution channel smoothly which may bring disruption to the private-owned contractor segment

# Strategic Analysis

**Prompt:** Assess the feasibility of the new plan.

## Information Provided by Interviewer

MBC CEO is afraid taking some of the current third-party service provider revenue will make them turn away to work together with other competitor that will result in lower revenue of the state-owned contractors segment by 50%. After consulting with internal experts, it is believed that the most optimal distribution model for this segment is to do sales and customer relationship inhouse, while keeping the physical logistics operation handled by third party. Initial analysis show the new model will impact the financials as follows:

### Financial Impact

- Increase of the segment revenue remains at 35%
- New commission paid to third party: 9% of segment revenue, decreased from currently at 10% of the segment revenue
- Extra cost for internal operations: \$5M.
- Current total revenue: \$100M
- Current revenue Proportion per Segment: Retail: 10%, Private-owned contractors: 40%, State-owned contractors: 50%

### Solution from Interviewee

- Additional MBC profit = Revenue Increase – Additional Cost =  
 $(35\% * 40\% * \$100M) - [\$5M + (9\% * 40\% * \$100M * 1.35) - (10\% * 40\% * \$100M)] = \text{\$8.14M}$
- Change in third-party service provider revenue =  
 $(9\% * 40\% * \$100M * 1.35) - (10\% * 40\% * \$100M) = \text{\$0.86M}$

The new plan is feasible as MBC still get additional profit of \$8.14M and third-party service provider get additional revenue of \$0.86M and therefore there is no risk in undermining other segments by implementing this plan.

# Conclusion

## Recommendation

- We would like to recommend for MBC to go ahead and make transition in private-owned contractors segment to do sales and customer relationship inhouse, while keeping the physical logistics operation handled by third party.
- Reasoning
  - Despite the additional cost, the revenue will increase more significantly and therefore will result in higher profit of \$8.14M
  - It will also reduce MBC dependency with current third party service provider

## Risks

- Current lack of distribution expertise
- Competitive response from third party distributors

## Next Steps

- Confirm analysis assumption
- Put together acquisition target attribute list
- Conduct due diligence

# New Terrain

Auto & Mobility | New Product / Market Entry

*Interviewee Led*

Ratings: Quant - 1

Qual - 2

Overall - 1

# Prompt

ABC Motors, a market leader in motorcycle manufacturing, is headquartered in Nigeria with a strong presence across India and Southeast Asia. Recently, ABC expanded operations into Italy as part of its broader European growth strategy. ABC Motors is known for being an exceptional, high-quality manufacturer.

Following the success of its three-wheeled vehicles (used as tuktuks) in India, ABC Motors is considering entering the U.S. market — but not with motorcycles, as they are certain they can't compete in this space in the U.S. Instead, they are evaluating whether to manufacture and sell snowmobiles, capitalizing on growing demand for winter outdoor recreation across the northern United States.

ABC plans to launch only one model initially, a four-seat recreational snowmobile, and to sell through U.S. dealers, who will take a standard margin. Fleet sales to rental operators or resorts are not being considered.

ABC believes it can competitively position itself between premium U.S. and Canadian brands (Polaris, Ski-Doo, Yamaha) and the lower-end Chinese brands, which flooded the U.S. market with aggressively priced snowmobiles pre-COVID, but have faced severe supply and reliability issues since then.

You have been hired to help ABC Motors evaluate whether to pursue this U.S. snowmobile market entry.

# Clarifying Information

## Objective & Timing

- ABC Motors plans to make a go/no-go decision within 3 months.
- If approved, the launch would occur in the next winter season (~12–15 months from today).

## Market Focus

- ABC's proposed model is designed for recreational use by individual buyers, not rental fleets or commercial operators.
- The target market includes northern U.S. states with active winter recreation (e.g., Minnesota, Michigan, Maine, Vermont).

## Go-to-Market Strategy

- Sales will occur exclusively through U.S. dealers. ABC will not pursue direct-to-consumer distribution.
- ABC has no current U.S. dealer relationships.
- Dealers typically only carry one or two established brands, and are cautious about onboarding unknown international players.

## Brand & Operations

- Brand recognition is extremely low in the U.S. ABC has no current customer awareness in this category or geography.
- Manufacturing will initially remain overseas, leveraging ABC's existing production capabilities.

## Competitive Dynamics

- The U.S. snowmobile market is highly consolidated, with Polaris, Ski-Doo, and Yamaha controlling the vast majority of share.
- Chinese brands previously disrupted the market with aggressive low-price entries, though many suffered major reliability and delivery issues during COVID.
- Several Chinese brands have relocated production to the U.S., improving speed, reliability, and brand perception, regaining share.

# Potential Framework

## Market Attractiveness

- Size and growth rate of the U.S. recreational snowmobile market
- Seasonal demand patterns and regional concentration
- Long-term demand trends (e.g., weather shifts, outdoor recreation habits)
- Market share and positioning of Polaris, Ski-Doo, Yamaha
- Risk of incumbent retaliation (e.g., pricing, dealer lockouts, lobbying)

## Customer Preferences

- Purchase drivers for individual buyers (brand trust, safety, price)
- Dealer willingness to carry new brands; exclusivity norms
- Importance of service, parts availability, and warranty expectations

## Financial Viability

- Net revenue after dealer cut and full cost stack
- Sensitivity to tariffs, seasonality, and lower-than-expected sales
- Margin headroom to support marketing, service, and incentives

# Question 1

**ABC Motors projects selling 2,000 units in Year 1 at a retail price of \$12,000 per unit. What is ABC's projected Year 1 gross profit?**

Additional information provided to interviewee only if asked:

- **Dealer Margin:** 20% of retail price
- **Manufacturing Cost:** \$5,000 per unit
- **Transportation Cost:** \$800 per unit
- **U.S. Assembly Cost:** \$400 per unit
- **Tariff:** 25% tariff applies to the manufacturing cost upon import

(To make this case more challenging you can prompt the caser for what information they'd need in order to calculate the profit.)

Solution From Interviewee

- Dealer margin =  $20\% \times \$12,000 = \$9,600$  = Net revenue to ABC
- Tariff =  $25\% \times \$5,000 = \$1,250$
- Total cost per unit =  $\$5,000 + \$800 + \$400 + \$1,250$  (Tariff) =  $\$7,450$
- Gross profit per unit =  $\$9,600 - \$7,450 = \$2,150$
- Total gross profit =  $2,000 \times \$2,150 = \mathbf{\$4.3M}$

# Question 2

**ABC Motors is looking to differentiate itself against both premium brands and low-cost Chinese brands. How could they do this?**

Interviewee should focus on structured response. Possible response include:

## Against Premium Brands: Polaris, Ski-Doo, Yamaha

- Offer longer warranties (e.g., 5 years full coverage)
- Include smart features standard (app connectivity, navigation assist)
- Introduce creative ownership models (subscription, winter-only leasing)
- Sponsor regional winter events to build brand recognition

## Against Low-Cost Chinese Brands

- Emphasize consistent delivery timelines and post-sale support
- Highlight third-party safety testing and certifications
- Offer dealer-friendly programs (higher margins, co-op marketing)
- Launch eco-friendly variants (quieter engines, lower emissions)

# Exhibit 1

## U.S. Snowmobile Market Position



# Exhibit 1 Guidance

**Take a look at this chart. Based on the positioning of the current competitors, what challenges would ABC Motors face if they were to enter the U.S. snowboarding market?**

## **What a Strong Casing Candidate Should Notice**

- Strong candidates should recall that ABC is planning to enter at a price point (\$12k) that sits between the low-cost Chinese brands and the premium players like Polaris and Ski-Doo. Prompt them with this reminder should they need help.
- Candidates should recognize that there's no meaningful brand presence in that middle space: despite high growth in the value tier and dominance in the premium tier.
- The market appears to be bifurcated: customers either want trusted, high-end brands or low-cost utility, not something in between.
- Thus, ABC risks being stuck in the middle: not cheap enough to win price-sensitive buyers, not trusted enough to steal from Polaris or Ski-Doo.
- If candidates ask about ABC's projected market share/growth you can inform those projections don't exist yet.

# Conclusion

## Recommendation

- Despite healthy projected gross profit in Year 1, ABC Motors should not pursue a full-scale U.S. launch at this time with a price point of \$12k.
- The U.S. snowmobile market is bifurcated: customers overwhelmingly choose between premium, trusted brands and low-cost imports.
- ABC is aiming for a middle ground that no brand currently occupies, and the market structure suggests that's for good reason.
- Instead, ABC should consider a limited regional pilot to test brand receptivity, dealer interest, and serviceability before making broader investments – or reconsider their pricing strategy.

## Risks

- Brand Trust Gap: ABC has zero brand equity in U.S., and new buyers unlikely to switch from high-trust incumbents / low-cost known options
- Dealer Resistance: Without strong U.S. support, dealers may hesitate to onboard a brand that is neither premium nor cost-leading.
- Margin Sensitivity: A few points of cost inflation, discounting, or operational delays could wipe out profitability quickly.
- Category Expertise Gap: ABC has no experience in cold-weather, powersport vehicles. Entering the U.S. snowmobile market from Nigeria stretches far beyond its core product expertise or environmental familiarity.

## Next Steps

- Conduct Research to Quantify Mid-Price Range Demand: Survey potential customers to evaluate willingness to pay and switching triggers. Size the potential market for buyers seeking “value-tier quality” with “premium-tier reliability.”
- Conduct a Small-Scale Pilot: Target high-density snowmobile regions and launch with ~10 dealers and offer strong short-term incentives to test buyer response and dealer engagement
- Invest in Local Service and Parts Network: Ensure post-sale support is reliable to build trust with early buyers and dealers.

# Revenue Streaming

Media & Technology | M&A  
*Interviewee Led*

Ratings: Quant - 1

Qual - 2

Overall - 2

Our client, Vidnet, started as an online DVD rental company. Thanks to an early pivot into online movie and television streaming, the company found massive success as internet infrastructure improved and enabled widespread video streaming. Over the last ten years, Vidnet has amassed customers around the globe, growing to \$20B in annual revenue. Today, however, the company is faced with increasing competition as new video streaming platforms have emerged.

The CEO is considering the purchase of a popular music streaming platform, DotMuse, to diversify revenues and fend off competition. She has asked our firm what factors she should be considering, and whether Vidnet should make the purchase?

## Clarifying Information

- **Who are the new competitors?** These are a mix of video streaming services with features ranging from low price to original content and will be explored later in the case. None of them are streaming music at the current moment.
- **Does Vidnet have past M&A experience?** The company has not completed an acquisition in the past, which is why they are engaging our firm.
- **How do users access the services?** Vidnet and DotMuse are both accessible via computers and mobile devices. Vidnet also built partnerships with several major electronics companies to have its platform built into SMART TVs.
- **Where does DotMuse operate?** DotMuse currently makes \$9B in annual revenue, also from customers around the world. We'll provide a more specific breakdown later.
- **Does Vidnet still sell DVDs?** The company has limited DVD sales in North America and Europe, where sales total to about \$.5B. For the purposes of this engagement, we will focus on streaming.

# Potential Framework

## Music Streaming Market

- Magnitude and growth
- Competition
  - Number and fragmentation
- Bargaining power of music providers
- Number and fragmentation of major labels, artists
- Broader market conditions
- Is this a favorable time to buy a company?

## Revenue Synergies

### Cross selling

- Advertising across platforms or markets (DotMuse may be more popular in places with poor internet infrastructure)

### Pricing

- Bundling may lead to more revenue from existing customers

### IP & Unique Assets

- Shared data improves rec. algorithms
- Differing content relationships may enable acquisition of new content

## Cost Synergies

### Redundancies

- HR, IT, and other back office functions can be consolidated

### Economies of scale

- Bargaining power with content companies
- Fixed assets like servers and office space can be shared DVD and CD materials for hard copy sales

## Risks

### Integration costs

- Cultural differences
- Employee training for policies, IT, etc.

### Brand dilution

### Better alternatives

- Other music platforms
- Other diversification strategies

### Due diligence

- Overestimating synergies, especially revenue synergies

### Valuation

- Similar acquisitions, DCF, asset value methods

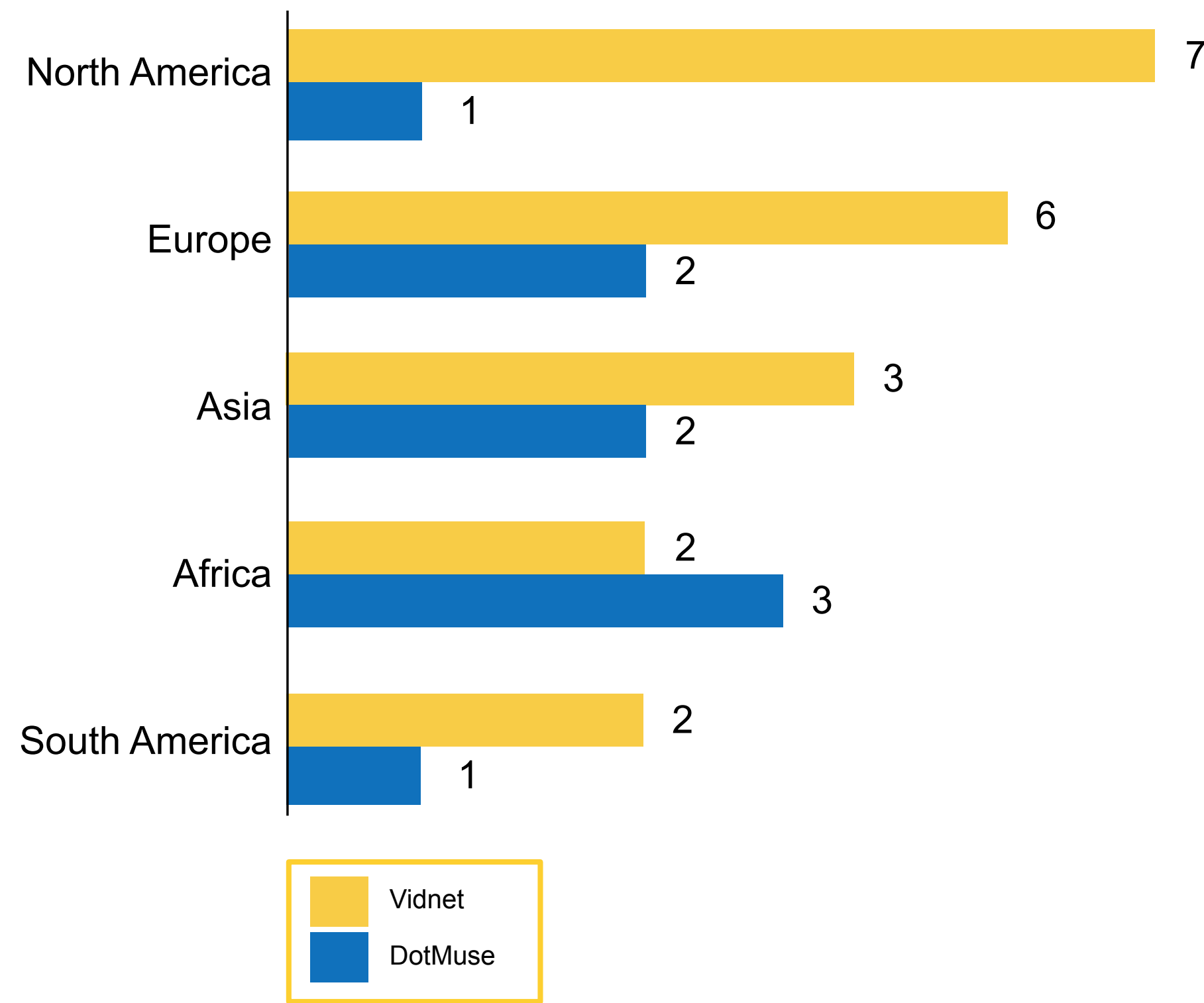
## Notes for interviewer:

- This case takes place early in the target selection phase. Candidate should not fixate on price or financials.
- The general case flow is exploring synergies (Exhibit 1, Calculation), Strategy (Exhibit 2), Alternatives (Brainstorm).

# Exhibit 1

## Revenue and Cost Information for Vidnet/DotMuse

Revenues by Geography (\$B)



Cost by Category as Percent of Revenue

|                        | Vidnet | DotMuse |
|------------------------|--------|---------|
| <b>Cost of Revenue</b> |        |         |
| Content Acquisition    | 23     | 46      |
| Data Streaming         | 15     | 8       |
| <b>Operating Costs</b> |        |         |
| User Acquisition       | 28     | 25      |
| Admin & Staff Salaries | 23     | 24      |
| HR, IT, & Back Office  | 9      | 8       |
| Rent & Overhead        | 12     | 19      |
| <b>Other Costs</b>     | 6      | 7       |

# Exhibit 1 Guidance

## Revenue and Cost Information for Vidnet/DotMuse

### Revenues

- Vidnet and DotMuse are dominant in different geographic markets. Both companies could cross-sell to the counterpart's customers to increase market share. E.g. Vidnet to DotMuse's customers in Africa and DotMuse to Vidnet's customers in North America.

### Costs

- Identify percentage imbalances, particularly in Content Acquisition, Data Streaming, and Rent & Overhead.
- Recognize cost redundancies, particularly in back-office functions. The NewCo will not require the same number of HR, accounting, etc. staff and tools, which could potentially result in a reduction below 8%.

## Best candidates will:

### Revenues

- Talk through examples on how to cross-sell the services. E.g. Advertisements for a free trial of DotMuse when you open up the Vidnet app, and vice versa. Or a service bundle with both subscriptions at a discounted price.

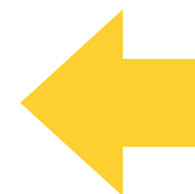
### Costs

- Talk through examples on how the companies can help each other lower costs. E.g. Vidnet can leverage connections with content producers and provide economies of scale to help DotMuse lower Content Acquisition costs. DotMuse can share best practices to help Vidnet lower streaming costs. DotMuse offices can be moved if they are in a more expensive city or consolidated if there is space in Vidnet offices.
- Highlight acquisition-specific nuances. E.g. Staff from both companies need to be retained when examining redundancies to maintain company-specific knowledge and create a sense of fairness in the NewCo. Synergies will be highly impacted by the level of integration (i.e. whether this will be a tuck-in or a standalone acquisition.)

# Calculation

Vidnet has strong relationships with content producers and is confident that they can bring DotMuse’s cost of Content Acquisition to a similar percentage as their own. What would be the cost savings in dollars for this scenario?

|                                                   | Current        | Projected |
|---------------------------------------------------|----------------|-----------|
| DotMuse Revenue                                   | \$9B           | \$9B      |
| % of Revenue                                      | 46             | 23        |
| Cost of Content Acq.                              | \$4.14B        | \$2.07B   |
| <b>Difference between Current &amp; Projected</b> | <b>\$2.07B</b> |           |



Candidate should recognize that this is just half of the current percentage

## Exhibit Guidance

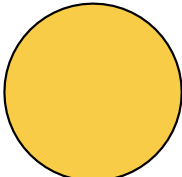
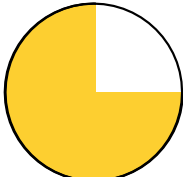
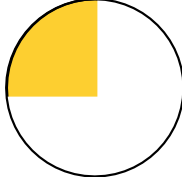
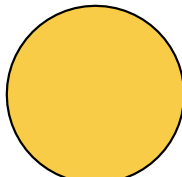
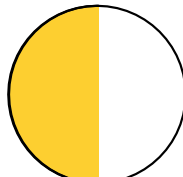
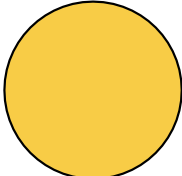
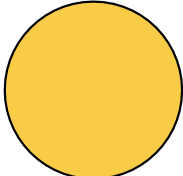
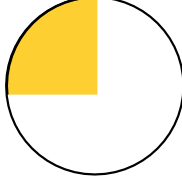
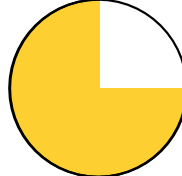
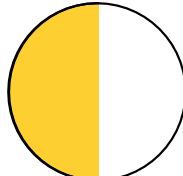
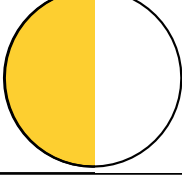
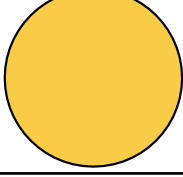
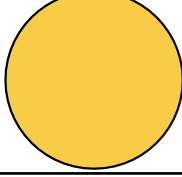
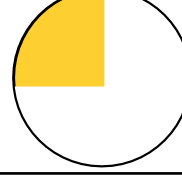
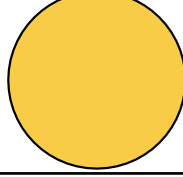
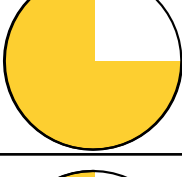
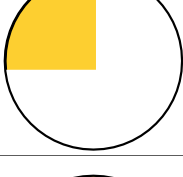
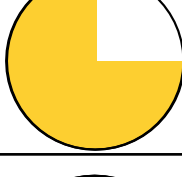
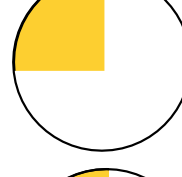
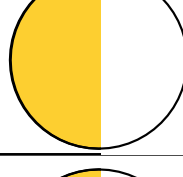
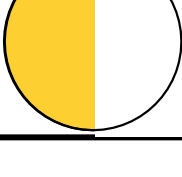
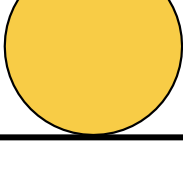
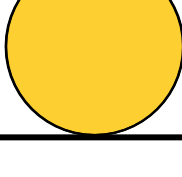
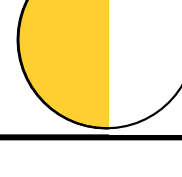
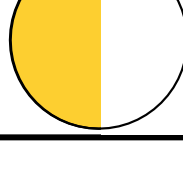
- Candidate should recognize that they have all necessary data and complete the calculation easily.
- Tip: \$9B can be written as \$9,000M. 50% of this is \$4,500M. Minus \$360M is \$4,140M.
- Candidate should acknowledge that \$2.43B is a sizable chunk of the current revenue stream, making DotMuse an attractive target.

## Best candidates will:

- Recognize cost synergy calculation is highly dependent on assumption that Vidnet’s relationships can reduce DotMuse’s costs to a similar percentage of revenues. They should question this assumption and suggest further due diligence.
- Connect high savings number to valuation of the target. How much of this synergy each party captures is dependent on whether target company has produced similar estimates through due diligence, number of other potential buyers, etc.

# Exhibit 2

## Features of Vidnet, Vidnet Competitors, and DotMuse

| Company    | HD Streaming                                                                         | Content Quality                                                                       | Original Content                                                                      | Suggestion Algorithm                                                                  | Low Price                                                                             |
|------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Vidnet     |    |    |    |    |    |
| Rainforest |    |    |    |    |    |
| MeTube     |   |   |   |   |   |
| Pidgeon    |  |  |  |  |  |
| DotMuse    |  |  |  |  |  |

# Exhibit 2 Guidance

## Features of Vidnet, Vidnet Competitors, and DotMuse

### Exhibit Guidance

- Candidate should examine the companies and observe that Vidnet and Rainforest are very similar in features. The acquisition of DotMuse, may be an opportunity to differentiate the platform and diversify revenues as the CEO requested.
- They should notice that HD Streaming, Content Quantity, and Suggestion Algorithm are all features that are relevant to music.
- The acquisition of DotMuse would be helpful for retaining or drawing additional revenue from current Vidnet customers if DotMuse's HD Streaming and Suggestion Algorithm features can be improved via knowledge or resource transfer.
- Vidnet can use the acquisition of DotMuse to develop capabilities around producing original content. In addition to diversifying revenue by adding a new type of content, knowledge or resource transfer can help them diversify within their current medium.

### Best candidates will:

- Consider how strategy affects level of integration. Knowledge transfer is generally easier through a tuck-in acquisition, but a standalone acquisition would be sufficient and less complicated if the goal is simply to diversify the types of content offered.
- Point out that Vidnet and DotMuse are not the low-priced options in the market. This means options designed to draw more revenue out of current customers (e.g. bundling) are more viable. However using the same customer base may be bad from a diversification perspective, in which case Vidnet may consider targeting other, low-priced streaming services.
- Notice that the differences in data streaming costs in Exhibit 1 are likely rooted in differences in HD Streaming features.

# Brainstorming

Aside from acquisition of DotMuse, what are other ways that Vidnet can diversify revenue streams?

## Brainstorming Guidance

- **Streaming Platforms**
  - Other video platforms, Other music streaming platforms, other media (e.g. video games).
  - Tiered pricing (e.g. regular and HD streaming, family plans)
- **Services**
  - Original content, Merchandising, DVD/Blu-ray Sales.
- **M&A alternatives**
  - Organic growth (i.e. start their own music streaming platform), Joint venture, Partnership, purchase assets (e.g. streaming exclusivity rights)

## Best candidates will:

- Consider methods of diversification both in terms of other streaming platforms and other services for Vidnet.
- Think of alternatives to M&A, acknowledging the complicated process and permanence of integration.

# Conclusion

## Recommendation

- Vidnet should acquire DotMuse because projected cost synergies are above \$2B, which is a significant proportion of DotMuse's revenues.
- This provides a profitable way for Vidnet to diversify revenues.

## Risks

- Vidnet might not be able to bring down Content Acquisition costs to the projected value.
- Synergy capture may be lost due to integration challenges (loss of key staff, culture clashes, service interruptions).
- Vidnet may experience negative brand impacts as it strays from video streaming.

## Next Steps

- Conduct proper due diligence to confirm synergies.
- Consider alternative streaming platforms or services in place of DotMuse.
- Develop integration strategy, in particular, whether this will be a tuck-in or a standalone acquisition.

# BuildSemi

Engineering & Construction | New Product  
*Interviewer Led*

Ratings: Quant - 1

Qual - 2

Overall - 2

# Prompt

Your client, BuildSemi, is a semiconductor equipment manufacturer who produces equipment to create Memory and ASIC (Application-Specific Integrated Circuit) semiconductor chips. BuildSemi has recently seen an increase in worldwide demand for GPU (Graphics Processing Unit) chips, so the CEO has approached you to determine if manufacturing GPU production equipment is the best for the company.

The CEO believes that by offering this new product, they can capture a share of a growing market. What factors should BuildSemi consider when expanding their portfolio into this new market?

## Clarifying Information

- Where is the headquarters located? Where are the Memory / ASI factories located?
  - Headquartered in northern California, but production of their equipment is completed in Taiwan
- What are GPU chips used for?
  - GPU chips are traditionally used for accelerating computer graphics and rendering. They are increasingly used in fields for artificial intelligence (AI) and machine learning (ML) applications.
- Can Memory/ASI equipment be used to make GPU chips? What type of equipment does BuildSemi produce?
  - No, the products BuildSemi would create are different. If asked, say eqp used in the chemical vapor deposition steps.
- Does BuildSemi make semiconductor chips?
  - No, only the equipment in the semiconductor fabrication process.

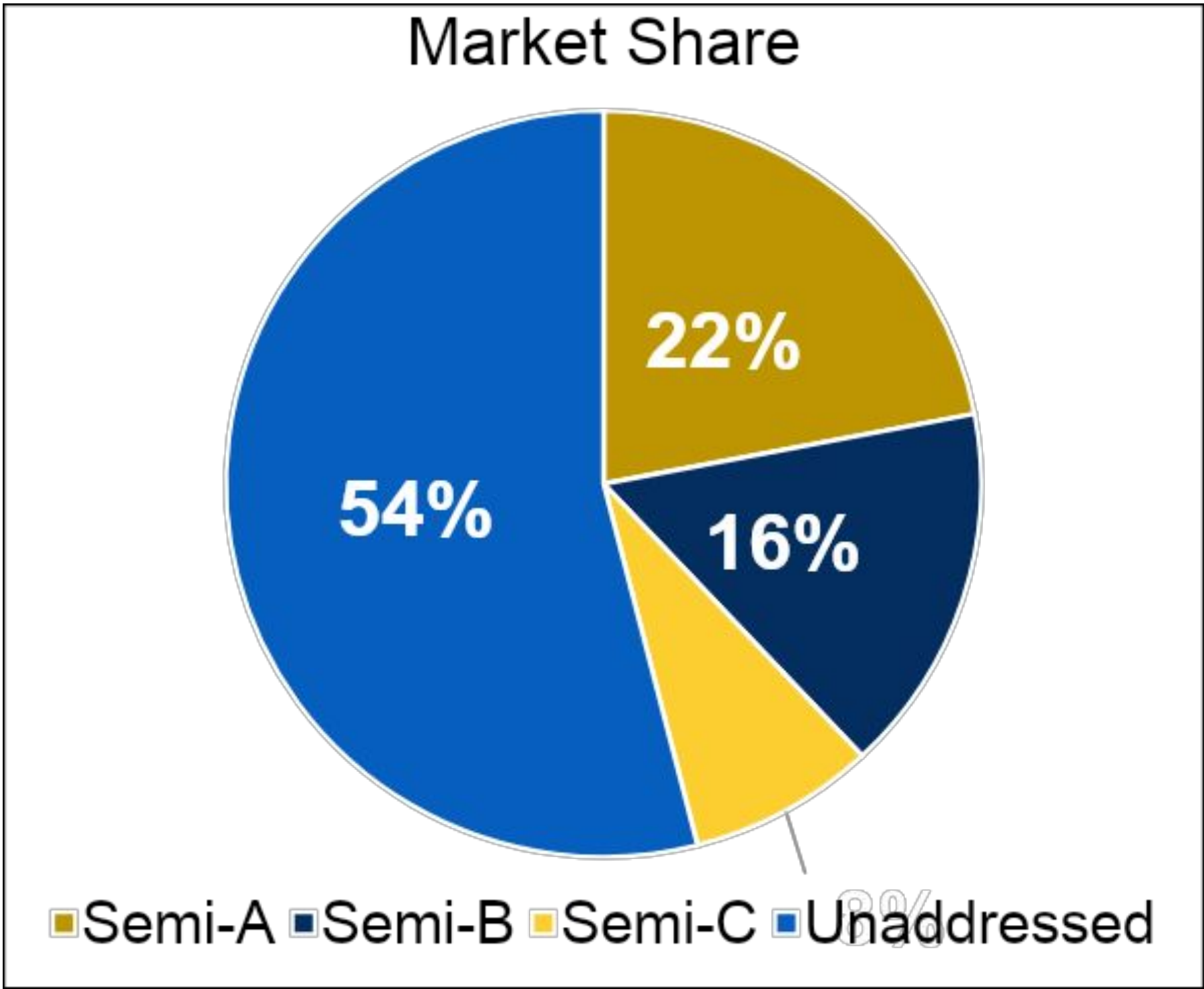
# Potential Framework

Case 5 – BuildSemi

| Profitability                                                                                                                                                                                                                                                                                                                                                                                               | Market Potential                                                                                                                                                                                                                                                                                                                                                                                    | Entry Strategy                                                                                                                                                                                                          | Client Factors                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Revenue<ul style="list-style-type: none"><li>• Market size/growth</li><li>• Equipment price / qt</li></ul></li><li>• Costs<ul style="list-style-type: none"><li>• Fixed Costs: IP and R&amp;D, licenses,</li><li>• VC: material, labor, customer acq.,</li><li>• Upfront costs: new manufacturing site, new line in existing factory, retooling</li></ul></li></ul> | <ul style="list-style-type: none"><li>• Existing competition</li><li>• Customers:<ul style="list-style-type: none"><li>• Targeted segmentation</li><li>• Drivers – AI</li></ul></li><li>• Trends:<ul style="list-style-type: none"><li>• Semiconductor boom or bust?</li><li>• Geopolitical tensions? trade? Tariffs?</li><li>• CHIP and Science act, supplement semiconductors</li></ul></li></ul> | <ul style="list-style-type: none"><li>• Green field (build a new line or new factory)</li><li>• Acquisition (M&amp;A)</li><li>• Partnership (JV)</li><li>• Geographical location for supply chain + customers</li></ul> | <ul style="list-style-type: none"><li>• Brand perception amongst B2B clients</li><li>• Existing IP</li><li>• Existing partnerships with chip customers</li><li>• Capacity constraints</li><li>• Loss of focus on core competencies</li><li>• </li></ul> |

# Exhibit 1

Semiconductor industry landscape



| Year | Market Size (USD) |
|------|-------------------|
| 2025 | \$79.3B           |
| 2026 | \$80.9B           |
| 2027 | \$82.5B           |
| 2028 | \$84.2B           |
| 2029 | \$85.8B           |

# Exhibit 1 Guidance

**Prompt:** The client has some public information about the semiconductor equipment market. What do you notice about this figure and table?

## Notes for the Interviewer:

A good interviewee will give an overview of what they see in the exhibit, then provide major insights – both positives and negatives, then relate back to the case prompt. A good interviewee will also jot some of these numbers to provide a market capture potential. If they discuss a number, ask them to explain why it's important.

## Potential ideas:

Much of the GPU market's demand is unmet, and is expected to grow 2% YoY, therefore entering this market seems slightly attractive. Understanding what the challenges are to enter this market is key, challenges could be because...

**Prompt:** What else do you see? . . . . . What else?

## Notes for the Interviewer:

A good interviewee will collect their thoughts when pressured and be able to provide additional organized thoughts: pushes the interviewee to think beyond what data is presented.

## Potential semiconductor specific insights:

Entering this market may take years due to the long time to build a factory. It is possibly a good idea to look into acquiring a startup, or plans to build a new factory, or add on additional lines to the existing plants.

# Exhibit 2

- ★

= BuildSemi Headquarters
- 🏭

= Optional Plant Location
- 🏠

= Existing BuildSemi Plant
- 👥

= Customer Cluster

| Location Option | Cost     | Workforce Skill | Other Notes                |
|-----------------|----------|-----------------|----------------------------|
| Texas, US       | \$\$\$\$ | 3/5             | Gov subsidize, close to HQ |
| Bogota, CO      | \$       | 1/5             | Good trading partner       |
| Lagos, NG       | \$       | 1/5             | Developing economy         |
| Shangha, CN     | \$\$     | 4/5             | Unstable gov relations     |
| Chengdu, CN     | \$\$     | 4/5             | Unstable gov relations     |
| Taipei, TW      | \$\$\$   | 5/5             | Good US relations          |
| Seoul, KR       | \$\$\$   | 5/5             | Good US relations          |
| Hanoi, VN       | \$       | 2/5             | Good US relations          |



# Exhibit 2 Guidance

**Prompt:** BuildSemi has decided to investigate building their own GPU plant (ground up), and they need to determine the best location. Please review the exhibit and let me know what you think.

## Notes for the Interviewer:

A good interviewee notices all influential factors, and weighs both positives and negatives to fully determine the best location choice. Possible influential factors are, noticing the location of the headquarters, the existing BuildSemi fabrication plant location, the new plant's country, the cost of the new factory, the training level needed (highly technical skill needed), the relationship with the US (as HQ is there), and the closeness of the plant options to the major customer clusters.

- If asked, “customer cluster” is a group of major customers who would purchase the GPUs once fabricated.

## Potential ideas:

- Taiwan is the best choice for building the new fabrication facility because of its proximity to the existing plant, and its ability to cross train to build the competency of its new employees. Taiwan is prone to geographical weather risk (typhoons and earthquakes), but the cost for ensuring a consistently safe plant is worth it the return because of the very skilled workforce accessible.
- Texas is the best choice for building the new fabrication facility because of its ability to upskill its workforce (due to local community colleges and trade centers), the new plant would be located close to the headquarters and the risk of international conflict is low. It is the most expensive but could be mitigated through partnerships and aggressive customer acquisition.

# Question 3

**Prompt:** There are many different types of equipment sold to chip fabrication facilities, and BuildSemi could likely produce GPU equipment focused on the three steps: lithography, chemical vapor deposition, and etching steps. If lithography equipment was priced at \$25M per unit, deposition at \$7M, and etching at \$5M, how many units would need to be sold to pay off a \$1.2B facility?

Information provided if asked:

Do not consider any other costs besides the facility.

Equipment sales projections for BuildSemi would be:

- 1/14 of all units sold would be Lithography EQP,
- 5/14 of all units sold would be Deposition EQPs, and
- 4/7 of all units sold would be Etching EQPs.

Solution from Interviewee:

- 1 Lithography EQP + 5 Deposition EQP + 8 Etching EQP = \$25M + (5 x \$7M) + (8 x \$5M) = \$100M & (\$1.2B / \$100M = 12)
- (12 x 1 Lithography EQP) + (12 x 5 Deposition EQP) + (12 x 8 Etching EQP) = **12 Lithography EQP, 60 Deposition EQP, and 96 Etching EQP**

A good interviewee would:

- Recognize how the percentage of units sold breaks down into a ratio used to equate to 14 units costing \$100M
- Provide the “so-what” after the number of units would be sold: how long it might take to sell this quantity of units, ask for further understanding if these purchase quantities are typical, ask how long it takes to produce one equipment, who are the customers (TSMC)

# Question 4

**Prompt:** The CEO is leaning towards building a facility from scratch. What factors would you consider when opening a new plant?

Notes for the Interviewer:

A good interviewee structures their thoughts, and creates more major buckets wherever possible.

Potential ideas:

- Physical considerations: layout (space utilization), semiconductor clean room environment
- Processes and methodologies: lean or six sigma, creating SOPs, hiring specialized talent for this, quality inspections, existing start-up knowledge, existing local tier 2/ 3 suppliers
- Automation and technology: robots to palletize + logistical support, managing demand and forecasting, warehousing

**Prompt:** What else? . . . . . Anything else?

# Conclusion

## Recommendation

- Interviewee should be able to justify whether CEO should create new GPU equipment facility in which location

Example:

- Build in Taiwan because of the advanced labor pool available in Taiwan, the proximity to its potential customer clusters, and its proximity to its existing ASI/memory equipment plant.

## Risks

- Risks highlighted in brainstorming questions
  - Supply chain concentration
  - Global tariffs
  - Client knowledge gaps

## Next Steps

- Next steps should address potential risks outlined in recommendation
- Begin discussions with manufacturing contractors
- Get land quotes
- Start engaging with new GPU customer clusters to understand product demands

# Desert City

Water & Utilities | Investment Decision

*Interviewer Led*

Ratings: Quant - 2

Qual - 1

Overall - 2

# Prompt

Our client is an investment firm that is looking to invest in a project in Middle East. A interested target is a Middle Eastern government's plan to construct a new city in the country that attracts new industries, business, and residents. The nearest existing city to this site is 25 kilometers. The government wants our client to jointly invest in a for-profit water utility company that is going to serve the new city. You are hired to determine whether it should make the investment.

## Clarifying Information

What are characteristics of this country?

- This country is extremely wealthy but has a lack of water given its arid climate. It recently underwent severe financial distress during financial crisis, which led to recent political unrest.

Does this investment firm have any experience in similar investments?

- The investment firm has other investments in water and utilities, so we can assume they have expertise / knowledge in this space.

Is there a particular goal or target that the investment firm is looking to achieve?

- Nothing in particular. We can assume they just want a positive return on their investment.

Are there alternatives to be aware of, or should we consider alternative investments in this case?

- No.

### Interviewer Notes

1. It is a typical McKinsey-led case. Interviewer is supposed to ask questions and prompt interviewees when needed
2. Ask questions in sequence, provide numbers only when asked

# Question 1

Please walk me through your framework.

## Project Valuation

- Cost
  - Investment Costs
  - Ongoing costs
- Benefit
  - Projected demand
  - Revenue
- Feasibility
  - Supply source
  - Pipeline construction
  - Distribution
  - Timeline

## Risks and Considerations

- Supply risk
  - Long term water supply
- Construction Risk
  - Risk linked to construction of the project, affiliated infrastructure and city itself
- Economic Risk
  - Can the proposed city attract enough business and residents (demand of water)
- Political Risk
  - Political stability; public opinion on running water utility for profit
- Risk of not doing it

# Exhibit 1

Case 6 – Desert City

## Estimated Water Demand

|                    |                      | 3 <sup>rd</sup> year | 10 <sup>th</sup> year | Remarks                     |
|--------------------|----------------------|----------------------|-----------------------|-----------------------------|
| <b>Industrial</b>  | <b>Construction</b>  | 23,550               | 18,000                | Wastewater treatment needed |
|                    | <b>Manufacturing</b> | 10,550               | 19,000                | Various qualities needed    |
|                    | <b>Agriculture</b>   | 8,000                | 12,000                | No additives                |
|                    | <b>Airport</b>       | 1,340                | 2,450                 |                             |
| <b>Commercial</b>  | <b>Hospital</b>      | 40                   | 100                   | High quality                |
|                    | <b>Office</b>        | 30                   | 130                   | Wastewater treatment needed |
|                    | <b>School</b>        | 20                   | 160                   | High quality                |
|                    | <b>Tourism</b>       | 1,200                | 16,000                |                             |
| <b>Residential</b> | <b>Residential</b>   | Unknown              | Unknown               | High quality                |

# Question 2

What are your thoughts on Exhibit 1? How do the demand figures and remarks column impact cost and feasibility?

- Demand
  - Majority of the demand is from industrial
  - Demand growth from commercial (hence possibly residential) sector is much faster than industrial sector
  - Demand from construction will peak within next 10 years; while demand from non-construction activities will continue to grow
- Specification
  - Different quality requirements
  - Waste water treatment capability is needed for certain sectors
  - Commercial and residential tend to require higher quality water

# Question 3

Estimate residential water demand in 3rd year and 10th year

Please provide below information **only when interviewees ask specifically**

1. The city is expected to have 70,000 workers by the 10<sup>th</sup> year
2. Average family size is 3
3. On average 1.2 members per family will work
4. Each person consumes about 5 L water per day

## **Step 1**

*Calculate water demand in 10<sup>th</sup> year*

Calculate number of families =  $70,000 / 1.2 = 58333$

- (i) Total population =  $58333 * 3 = 175,000$
- (ii) Total water demand per day =  $175,000 * 5 = 875,000$
- (iii) Per year demand =  $875,000 * 365 = \mathbf{319 \text{ million L}}$

## **Step 2**

*Estimate water demand in 3<sup>rd</sup> year*

Use Exhibit 1, it is observed that water demand from school grows 8 times; it is reasonable to assume that water demand from school correlates with school size (number of teachers and students); also, it is reasonable to assume that number of teachers is proportional to number of students.

Therefore, we estimate number of students grows 8 times.

We assume demographic profile remains relatively constant over years, i.e.

number of students correlates with number of families in this city.

Therefore, we estimate population grows 8 times; hence water demand.

Result: we estimate residential water demand in 3<sup>rd</sup> year =  $319 \text{ ML} / 8 = \mathbf{40 \text{ ML}}$

# Conclusion

## Recommendation

- Should consider the following: (1) the financial return of the project, and (2) the feasibility of the project.
- The projected water demand indicated that there is a healthy growth of water demand in the future, a positive sign.

## Risks

- However, serious due diligence on various risks needs to be conducted. These include but not limit to (1) risk of security of water supply in the middle of nowhere; (2) risk associated with the construction of the city itself (editor: use Masdar City as a case in point); (3) risk associated with macro economy, especially the inflow of talent and businesses; (4) risk associated with political situation, and (5) risk associated with not doing this project

## Next Steps

- Next steps should focus on addressing risks outlined earlier in recommendation

# Rubicon Co.

Airline | Post-Acquisition / Profitability  
*Interviewee Led*

Ratings: Quant - 2

Qual - 2

Overall - 2

# Prompt

Your client is Rubicon Co, a low-cost airline operating in the US. A couple of years ago, Rubicon Co acquired Scarlet Air, an airline based primarily on the West Coast. The post-acquisition profits do not meet the executive committee's expectations, and the CEO of Rubicon Co has brought you in to understand the causes and improve profitability.

## Clarifying Information

- The CEO wants to increase profit by \$100M
- Assume Rubicon Co operates a single aircraft type across their fleet with similar seat layouts
- Rubicon Co is an all economy airline, not looking to change the business model
- At this point Rubicon Co is not looking to expand operations internationally
- The industry has remained relatively stable, with single digit profit % growth YoY

# Potential Framework

Case 7 – Rubicon Co.

## Revenue (Synergies)

- Ticketing (all metrics should  $\geq$  the combined metrics of the individual airlines)
- # of tickets sold
- Price of tickets
- Dependent on routes
  - Consolidation should decrease competition, resulting in increased prices
- Seasonality
- Fees (all metrics should  $\geq$  the combined metrics of the individual airlines)
- Wifi
- Food and Bev
- Luggage
- Change Fees

## Cost (Synergies)

- Variable Costs (all metrics should  $\leq$  the combined metrics of the individual airlines)
- Hourly Labor
  - Aircraft operations
    - Fuel
    - Maintenance & Servicing
    - Parking/Storage
    - Landing fees
- Fixed Costs
- Central functions
    - Marketing
    - IT
    - HR
    - Etc
  - Plant, Property, Equipment
    - Airport gate rental agreements
    - Corporate building rentals
    - Hangar space

## Competitive Response

- Other airlines consolidating in the space could lead to greater pricing pressure as cost structures are reduced
- Alternatively, less players in the space could lead to more favorable pricing as consumers have less choice

## Notes / Advice to Interview / How to Move Forward:

- Candidate should look to understand the financials of the airline(s) pre and post merger – if asked, provide Exhibit #1
- Alternatively, the candidate may ask to look at competitors financials as a comparable – if asked, let candidate know that pre-merger Rubicon Co. and Scarlet both met industry averages on financials
- Ensure the candidate mentions that airline routes differ in terms of profitability – if not, ask candidate to think about how the metrics for a flight from DTW to JFK are different than JFK to LAX

# Exhibit 1

Case 7 – Rubicon Co.

Pre and post acquisition financials

| <i>In Millions of Dollars</i> | Rubicon Co (Pre) | Scarlet Air (Pre) | Rubicon Co (Post) |
|-------------------------------|------------------|-------------------|-------------------|
| Revenue (Tickets)             | 749              | 690               | 1439              |
| Revenue (Non-Tickets)         | 251              | 210               | 461               |
| Operations, Variable          | 650              | 605               | 1255              |
| Marketing                     | 50               | 40                | 90                |
| IT                            | 75               | 70                | 145               |
| HR                            | 20               | 14                | 25.5              |
| PPE (Airport Slots, offices)  | 115              | 90                | 153.75            |

# Exhibit 1 Guidance

Case 7 – Rubicon Co.

Pre and post acquisition financials

## Questions for Candidates about Exhibit that the Interview should ask:

- Are there areas of opportunity to positively impact profitability?
- If candidate is struggling, ask: although VC may need to remain at that level, does FC of the combined entity need to be the combined individual FCs?

## Question/Exhibit Guidance:

- Candidate should recognize that while synergies were found in HR and Property costs, Marketing and IT costs are equivalent to the sum of the original airline line items – these are two opportunities for cost cutting
- Strong candidates will recognize that the combined company was able to find 25% net cost savings in property and HR – this can be used as a rough baseline for potential cost savings in the IT and marketing buckets. If they do not recognize this, let them know that the company can have a net cost savings of 25% in marketing and IT, like we see in HR & PPE.

# Exhibit 1 Guidance

Case 7 – Rubicon Co.

Pre and post acquisition financials

| <i>In Millions of Dollars</i> | Rubicon Co (Pre) | Scarlet Air (Pre) | Rubicon Co (Post)       | Rubicon Co (Post) w/ Potential Synergy Gains |
|-------------------------------|------------------|-------------------|-------------------------|----------------------------------------------|
| Revenue (Tickets)             | \$749            | \$690             | \$1439                  | Same                                         |
| Revenue (Non-Tickets)         | \$251            | \$210             | \$461                   | Same                                         |
| Operations                    | \$650            | \$605             | \$1255                  | Same                                         |
| Marketing                     | \$50             | \$40              | \$90                    | \$67.50 (.75*\$90)                           |
| IT                            | \$75             | \$70              | \$145                   | \$108.75 (.75*\$145)                         |
| HR                            | \$20             | \$14              | \$25.50 (.75*\$34)      | Reductions Already Made                      |
| Property (Airport, offices)   | \$115            | \$90              | \$153.75 (.75*205)      | Reductions Already Made                      |
| Total Revenue                 | \$1,000          | \$900             | \$1,900                 | \$1,900                                      |
| Total Cost                    | \$910            | \$819             | \$1,669.25              | \$1,610                                      |
| Profit                        | \$90             | \$81              | \$230.75                | \$289.50                                     |
| Margin                        | 9.00%            | 9.00%             | 12.14%                  | 15.24%                                       |
|                               |                  |                   | Increased Profitability | <b>\$58.75</b>                               |

## Note:

- Case is looking for profitability in terms of dollars. Percentages are given as a nice to have

# Brainstorming

Now that we have identified a potential for cost reductions in the Marketing and IT functions, what are some ways this can be accomplished?

## Brainstorming Guidance:

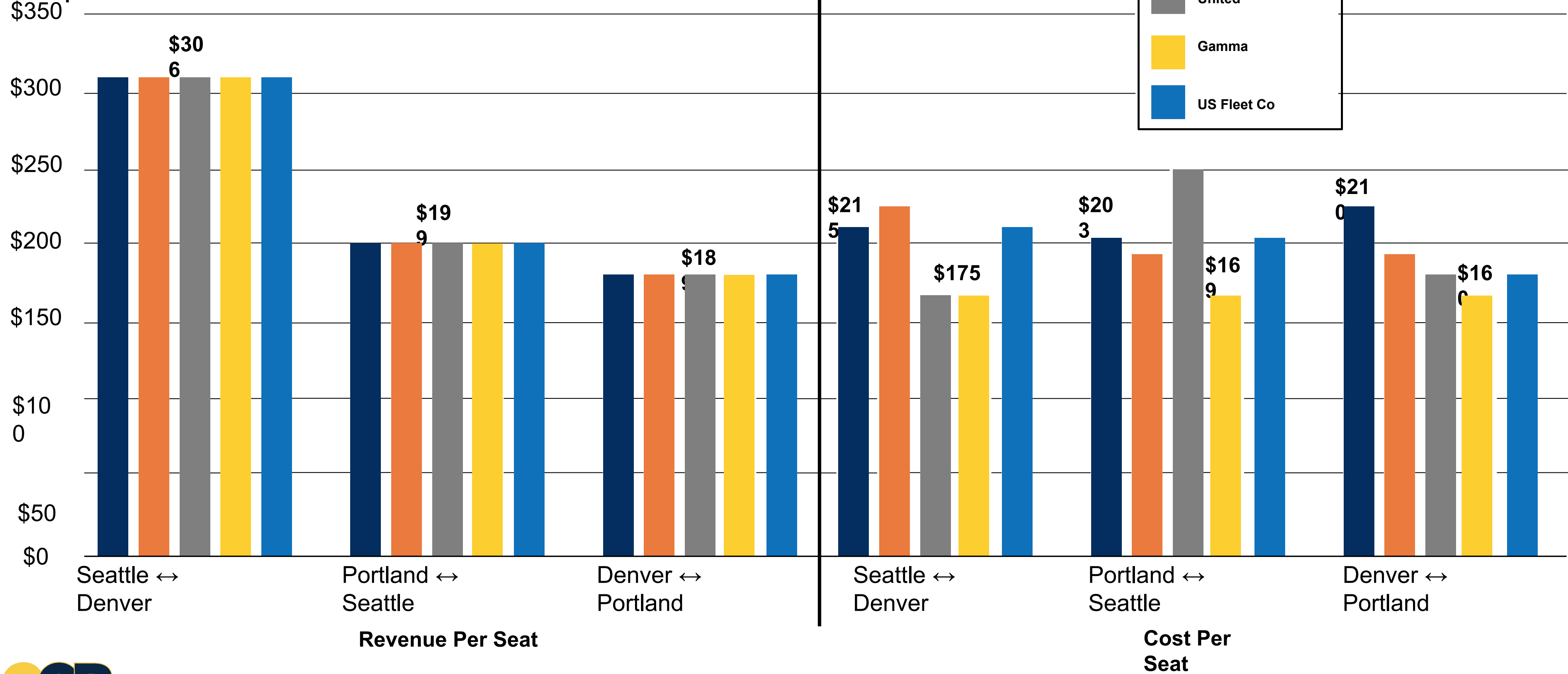
- Candidates should list a few (2 – 3) ways costs could be reduced in each area
- Marketing: Reduce headcount, leverage digital solutions to streamline campaigns, outsource specific functions
- IT: Reduce headcount, outsource specific functions, merge disparate systems

## Best candidates will:

- Identify risks associated with their recommendations and plan mitigation steps
- Notice that roughly half of the \$100MM improvement target has been achieved, will organically think
- about other areas to find profit

# Exhibit 2

## Northwest Region Route Comparison



\*\*Rubicon Co flies ~500k passengers per route per year

# Exhibit 2 Guidance

Case 7 – Rubicon Co.

Northwest region route comparison

## Questions for Candidates:

- One of our analysts flagged the Pacific NW as an area of concern. What are your initial thoughts on the routes?

## Question/Exhibit Guidance:

- Candidates should recognize that Portland ↔ Seattle and Denver ↔ Portland are not profitable.
- Strong candidates will calculate the annual profit/loss the airline experiences from the routes and suggest trying to cut costs and match competitors cost structure across all three routes

# Exhibit 2 Guidance

Case 7 – Rubicon Co.

Northwest region route comparison

|                    | Profit (Loss) Per seat | Total Passenger Volume | Net Profit (Loss)                                                       | Profit Potential (if matching most profitable competitors) | Net Profit                                |                                   |
|--------------------|------------------------|------------------------|-------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------|-----------------------------------|
| Seattle ↔ Denver   | $\$306 - 215 = \$91$   | 500,000                | $\$91 * 500,000 =$<br><br>$\$45,500,000$                                | $\$306 - \$175 =$<br><br>$\$131$                           | $\$131 * 500,000 =$<br><br>$\$65,500,000$ |                                   |
| Portland ↔ Seattle | $\$199 - 203 = (\$4)$  | 500,000                | $\$4 * 500,000 =$<br><br>$(\$2,000,000)$                                | $\$199 - \$169 =$<br><br>$\$30$                            | $\$30 * 500,000 =$<br><br>$\$15,000,000$  |                                   |
| Denver ↔ Portland  | $\$189 - 210 = (\$21)$ | 500,000                | $\$21 * 500,000 =$<br><br>$(\$10,500,000)$                              | $\$189 - \$160 =$<br><br>$\$29$                            | $\$29 * 500,000 =$<br><br>$\$14,500,000$  | Net Improvement                   |
|                    |                        | <b>Total</b>           | $\$45,500,000 -$<br>$12,500,000 =$<br><b><math>\\$33,000,000</math></b> |                                                            | <b><math>\\$95,000,000</math></b>         | <b><math>\\$62,000,000</math></b> |

**Prompt:** While Rubicon Co. is considering options to lower costs on these routes, some within the company are thinking of cutting the loss-making routes completely. What concerns do you have with this?

## Brainstorming Guidance:

Candidates can think about the issue in terms of qualitative and quantitative buckets

- **Quantitative:**
  - Loss of revenue from fees, inflight purchases, freight/shipping
- **Qualitative:**
  - Knockback effect -> passengers flying these routes may be connecting to other profit positive flights,
  - and cutting routes may make other routes less attractive
  - Political issues with city leadership
  - Airport contractual obligations

## Best candidates will:

- Think about potential changes in demand for the routes, and ask the airline to consider that before trimming routes

# Conclusion

## Recommendation

- To achieve the target profitability improvement, Rubicon Co should reduce costs in Marketing & IT to match synergies achieved in other departments.
- In addition, the airline should look to reduce costs on their Pacific NW routes to match competitors' cost structure

## Risks

- Detriment in morale with headcount reductions/outourcing
- Changes in demand on routes, impacting pricing and profitability

## Next Steps

- Research competitors' cost basis for PNW routes and begin matching methodology
- Consider cross training employees to reduce volume of layoffs
- Identify strategy to reduce costs in marketing and IT by 25%

# Alternative Milk

Consumer Packaged Goods | M&A

*Interviewer Led*

Ratings: Quant - 2

Qual - 2

Overall - 2

# Prompt

Your client is FoodCo a multi billion dollar company looking to enter the growing alternative milk market. FoodCo wants to understand the best way to enter the market through acquisition of a growing brand, Smilk. Smilk's profits have doubled in the last 3 years in both the US and Europe. FoodCo has expressed interest in purchasing the US business of Smilk. FoodCo's main objective is to increase Smilk market share in the US to 8% in the next 5 years and increase profitability by 6x in the next 5 years. How would you help FoodCo think about a potential acquisition of Smilk?

## Clarifying Information<sup>9</sup><sub>6</sub>

- **What products does Smilk sell:** drinkable alternative milk products (like oat, almond and soy milk)
- **Value Chain:** Smilk handles everything from manufacturing to distribution in supermarkets
- **Specific financial target for FoodCo:** 6x profit and increase market share to 8%

# Potential Framework

## Smilk Profit

### Revenues in US

- # of Sales
- Current distribution channels (Supermarkets, ecommerce)
- Pricing

### Costs in US

- Variable Costs
  - COGS
  - Distribution costs
- Fixed Costs
- Manufacturing costs
- Facilities cost

## Market of Alternative Milk

- Growth + size of market in US
- Competition in the space
- Customers preferences / trends in market

## FoodCo Company

- Revenue Synergies
- Cost Synergies
  - Savings in manufacturing
  - Economies of scale
  - Supplier cost savings
- Expertise

## Notes / Advice to Interview / How to Move Forward:

### Expectations of a good candidate:

- Candidate mentions the above three buckets or other reasonable buckets.

### Expectations of an outstanding candidate:

- Candidate mentions the three buckets, sufficiently expands on each bucket, and remembers that the goal is to increase profit.

# Question 1

What is the total market size today and in 5 years of the alternative milk market?

## Note for Interviewer

Provide candidate with 5 year growth rate of 40%. Give all other information verbally. Candidate can take a minute to develop approach.

| Crop                      | Smilk Market Size                  | Smilk Market Share | Alternative Milk Market Size |
|---------------------------|------------------------------------|--------------------|------------------------------|
| Today                     | \$5M                               | 4%                 | $=5M/4\% = \$125M$           |
| 5 year growth rate is 40% |                                    |                    | $=125M*1.4=\$175M$           |
| **8% of Alt Milk Market   | <b>Solve For</b><br>(Answer \$14M) | 8%                 | $=\$175*.08=\$14M$           |

\*\*Candidate should remember that FoodCo wants to increase Smilk market share in the US Alt Milk market to 8% in the next 5 years and therefore note that to be successful Smilk must be able to obtain a market size of \$14M in 5 years to reach FoodCo target.

# Question 2

Our client is curious in understanding the profitability of the US SMILK business. What is Smilk's profit for the US? What would you expect the growth rate need to be to obtain 6x profit?

| Question 1      | US Smilk Business  | Calculation                    | Notes                                                    |
|-----------------|--------------------|--------------------------------|----------------------------------------------------------|
| Sales Worldwide | 2.5M products sold |                                |                                                          |
| % of business   | 40% (provided)     | $2.5M \times .4 = 1M$ products | Candidate should solve for number of products sold in US |
| Sales Price     | Average price \$5  |                                |                                                          |
| Revenue         | SOLVE FOR          | $= 1M \times \$5 = 5M$         |                                                          |
| Profit Margin   | 15%                |                                |                                                          |
| Profit          | SOLVE FOR          | $5M \times .15 = 750K$         |                                                          |

| Question 2     | US Smilk Business | Calculation                   | Notes |
|----------------|-------------------|-------------------------------|-------|
| Profit         | 750K              |                               |       |
| Profit 5 Years | SOLVE FOR         | $\$750k \times 6 = \$4.5M$    |       |
| Growth Rate    | SOLVE FOR         | $(\$4.5 - 750K) / 750k = 5\%$ | **    |

\*\*A good candidate will contextualize the 5% growth rate indicate if they believe this is reasonable or not.

## Note for Interviewer

Provide candidate with information verbally. Candidate can take a minute to develop approach.

# Brainstorming

What are some synergies we can expect to see if FoodCo acquires SMILK?

## Solution

### Revenue Synergies

- Diversify product mix with alternative milk products from SMILK
- Cross-sell SMILK with complementary FoodCo products

### Cost Synergies

- Supply Chain & Operations – Greater buyer power and economies of scale by producing products in the same location
- SG&A – reduce headcount to share marketing, finance, sales, and IT teams

# Conclusion

## Recommendation

- Yes, FoodCo should acquire Smilk because they will be able to obtain their goal of increasing market share by 8% and profitability by 6x.

## Risks

- Culture clashes may occur between FoodCo and Smilk
- Government approval of acquisition
- Brand Cannibalization

## Next Steps

- Send Smilk a term sheet with details for the acquisition
- Start preparing internal teams to onboard and transition the Smilk team
- Conduct a survey to further understand the alternative milk market and remain a leader in the category

# German Luxury Car Maker

Auto & Mobility | Market Entry

*Interviewee Led*

Ratings: Quant - 2

Qual - 2

Overall - 2

# Prompt

Our client is a German luxury car maker. They have come to us as they want to grow their business and are looking into selling their cars in Bangladesh.

The GDP growth in Bangladesh is 5% per year. Currently, the only luxury car sold in Bangladesh is Mercedes-Benz, and they have been in the market for the past 10 years.

The CEO wants to find out if the company enters the market, can they break even in three years?

# Potential Framework

| Market                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Financial Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                              | Risks                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Industry<ul style="list-style-type: none"><li>○ Size</li><li>○ Growth</li><li>○ Trends</li></ul></li><li>• Competition<ul style="list-style-type: none"><li>○ Strengths &amp; weaknesses</li><li>○ Barriers to entry</li></ul></li></ul> <p><b>Product</b></p> <ul style="list-style-type: none"><li>• How are cars similar or different?</li><li>• Do we have same branding?</li></ul> <p><b>Customers</b></p> <ul style="list-style-type: none"><li>• Brand loyalty</li><li>• Interest in new car?</li><li>• Features that drive WTP</li></ul> | <p><b>Revenue</b></p> <ul style="list-style-type: none"><li>• Price/Vehicle<ul style="list-style-type: none"><li>○ Pricing strategy</li></ul></li><li>• # of Vehicles sold</li></ul> <p><b>Cost</b></p> <ul style="list-style-type: none"><li>• VC<ul style="list-style-type: none"><li>• Materials</li><li>• Hourly labor</li></ul></li><li>• FC<ul style="list-style-type: none"><li>• Overhead</li><li>• SG&amp;A</li></ul></li><li>• Set-up costs</li></ul> | <p>Geopolitical Risks</p> <ul style="list-style-type: none"><li>• Tariffs or customs</li><li>• Political instability</li></ul> <p>Competition risk</p> <ul style="list-style-type: none"><li>• Price war</li><li>• New Entrants</li></ul> |

# Clarifying Information and Case Guidance (*Market*)

## Market

- Mercedes-Benz imported and sold 10,000 cars in this market over the past 10 years, and has their own dealership in Bangladesh.
- There are 1,000 new luxury vehicle buyers each year.
- The average selling price of a luxury vehicle in Bangladesh is \$100,000.
- Existing owners replace their car every 10 years
- We will have a 30% market share (*don't give this info out right away, ask candidate to brainstorm how to estimate market share*) each year.\*\*

## Interviewer Guidance

- Ask candidate how to estimate market share  
Possible answers:
  - Understand customer needs through survey and estimate how well we could meet those needs and therefore how much market share we could gain.
  - Find benchmark
    - It turned out that our client already entered Vietnam and other markets similar to Bangladesh and on average gained 30% market share in each of those markets.

## Solution Guide

- Market Size: (the interviewee should calculate that 1,000 cars are sold to existing owners per year, and therefore the total market size per year is 2000 new cars)
- **# of cars sold per annum x price/ vehicle**
- 2000 vehicles \* \$100,000/veh =
- **\$200,000,000**

# Clarifying Information and Case Guidance (*Financials*)

## Financials

### Cost Structure

- Initial Investment \$7M

### Cost / Car

- |                  |       |                                |
|------------------|-------|--------------------------------|
| • Manufacturing  | \$20K |                                |
| • Transportation | 120%  | Manufacturing cost             |
| • Customs/Taxes  | 95%   | Manufacturing + transportation |
| • SG&A           | 12%   | of all above costs             |

## Solution Guide

- Manufacturing = \$20k
- Transportation = \$24k
- Customs / Taxes = \$41.8k
- SG&A = \$10.3k

Variable per car rounds up to \$96k/car Therefore profit on each car is \$4k

**With a 30% market share, the client's annual profit will be \$2.4M (2K cars/yr \* 30% \* 4K) and will break even in 3 years (\$7M/\$2.4M = ~3 yrs).**

# Brainstorming

Evaluate potential risks of entering this market

## Solution

- Assumptions might be inaccurate
- Bangladesh is not stable politically and economically, therefore our client will bear more risk
- There could be other new entrants
- Mercedes-Benz could react to our entry by
  - Reducing price (the best answer might also mention that with the low income level in the country, customers who can afford luxury cars might not be price sensitive)
  - Ask candidate what price Mercedes-Benz will reduce their car to.
    - Answer: \$96k/car or less
  - Improving their services if that's the differentiating factor of our product

# Conclusion

## Recommendation

- The profit on each car is \$4k
- With 30% market share, the client's annual profit will be \$2.4M and will break even in 3 years

## Risks

- Those gained from brainstorming

## Next Steps

- Market analysis on competitive response and target customer
- Determine the best way to enter the market (JV, greenfield, etc.)
- Look into ownership structure (franchise or owned showrooms)

## Excellent Case

- Excellent candidates will inquire on their own about ways to estimate market share
- Should recommend a price sensitivity analysis to determine the correct price

# Rack+

Entertainment & Media | Profitability / Cost Optimization

*Interviewee Led*

**Ratings: Quant - 2**

**Qual - 2**

**Overall - 2**

# Prompt

Case 10 – Rack+

Rack+ is a newly launched venture that recently secured exclusive streaming rights to professional pickleball tournaments from the USA Pickleball Association. Pickleball has experienced explosive growth in the US over the past five years, and the company is eager to capitalize on this momentum by launching a new streaming platform focused on showcasing pickleball content. With distribution being a critical success factor, Rack+ is evaluating two potential monetization models:

- **AVOD (Advertising Video-On-Demand):** A free, ad-supported model similar to YouTube
- **SVOD (Subscription Video-On-Demand):** A paid subscription model, akin to Netflix

In addition, the company is also considering expanding into other sports as a way to scale its platform and accelerate growth. The CEO of Rack+ has hired your consulting firm to help answer two key questions: Which monetization model would best position the company for long-term profitability and growth? Should Rack+ pursue expansion into other sports today? If so, which sport?

## Clarifying Information

**What is the primary goal of Rack+?**

- Rack+ is focused on driving long-term profitability, with success being the ability to generate the highest net profit within the next two years.

**Is there a target region Rack+ is targeting in its new streaming service?**

- Rack+ is only focused on the U.S. region for now

**Can you clarify what is AVOD and SVOD?**

- AVOD stands for Advertising Video On-Demand, where users can watch content for free with ads. SVOD stands for Subscription Video On-Demand, where users must pay a subscription to access content

**What other sports has Rack+ look into for its streaming platform?**

- Rack+ is interested in exploring Tennis, Cricket, and Rugby

**How does Rack+ acquire sports content?**

- Rack+ strategy is to secure exclusive streaming rights through direct licensing agreements with event organizers or governing bodies. These agreements are structured through upfront fees for exclusive access to broadcast pickleball tournaments or Revenue Sharing Deals

# Potential Framework

Case 10 – Rack+

A strong candidate should understand the objectives, then dive into the key questions at hand, which is *AVOD vs SVOD profitability* and *Sports Expansion Strategy*

## Current Streaming Market

### Competitive Landscape

- AVOD vs. SVOD players in niche sports streaming

### Consumer Behavior & Trends

- Demographics of streaming audiences
- Willingness to Pay for Streaming Services

### Content Consumption Channels

- Viewership Habits

## AVOD vs SVOD Profitability

### Revenue Potential

- Market Size for Pickleball
- AVOD Revenue (Impressions, CPM, Fill Rates)
- SVOD Revenue (Subscription Price, Churn, LTV)

### Expected Costs

- User Acquisition Cost
- Tech Costs
- Content Acquisition Costs
- SG&A

## Sports Expansion Strategy

### Market Attractiveness

- Size and growth of potential target sports
- Rights Availability, Costs, and Competition

### Strategic Fit & Synergies

- Revenue Synergies
- Cost Synergies

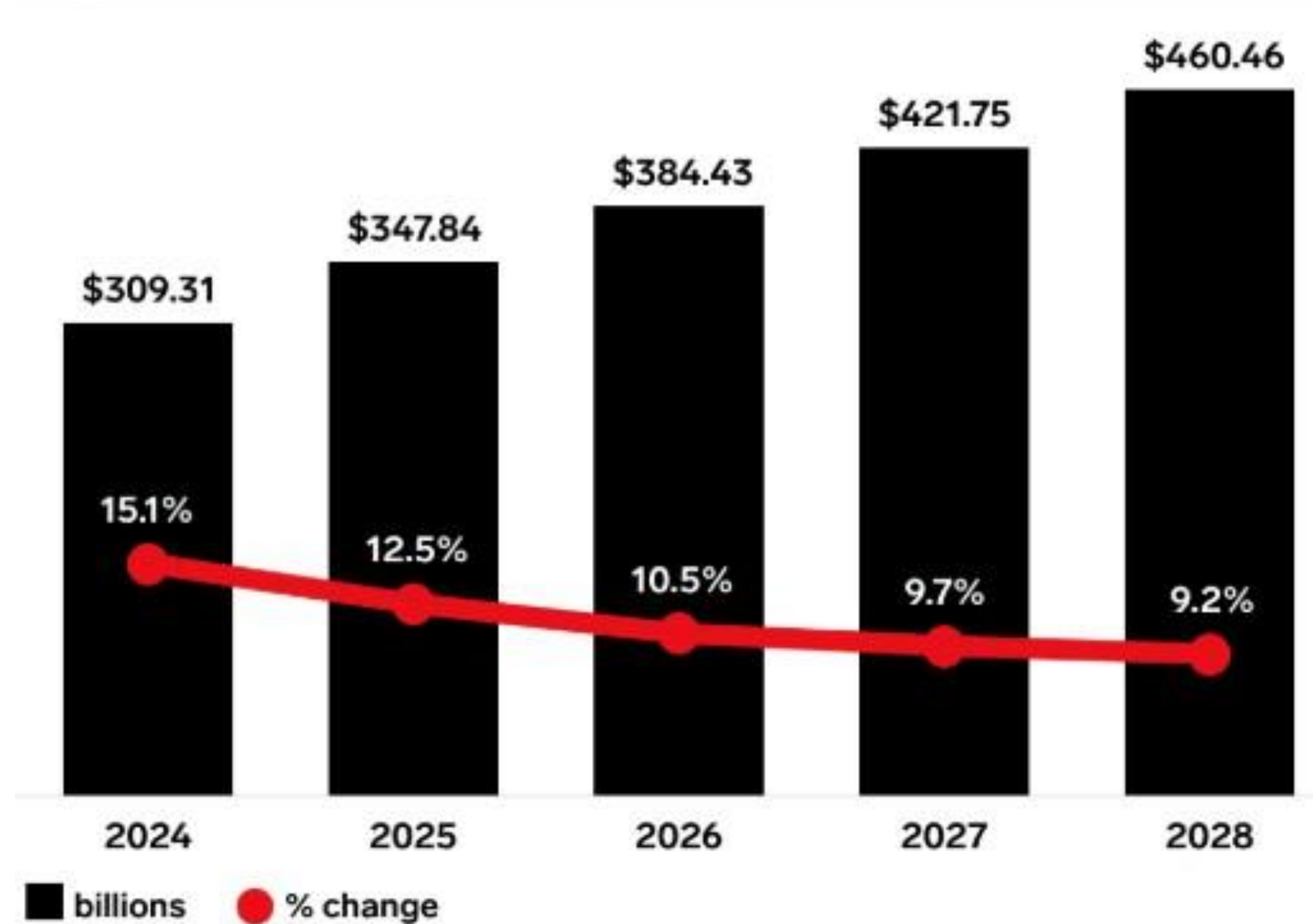
### Risk Assessment

- Dilution/Cannibalization

# Exhibit 1

## Digital Ad Spending in the US

Case 10 – Rack+



Note: includes advertising that appears on desktop and laptop computers as well as mobile phones, tablets, and other internet-connected devices, and includes all the various formats of advertising on those platforms

# Exhibit 1 Guidance

Case 10 – Rack+

Ask interviewer their thoughts on Exhibit 1.

Candidates should recognize the growth of Digital Ad Spending as favorable. While the growth rate is slowing slightly Y/Y, it is still growing at/near double-digits rate for the next four years. This is favorable for an AVOD monetization model.

Additional information to be provided only if probed by interviewer

## Streaming Market

- Currently there are no pickleball streaming competitors in the US market (*Remember, Rack+ has exclusive rights to pickleball streaming*)
- There are 40M US population in the Pickleball Ecosystem today. This number is expected to grow, but exact growth rates are not provided

## AVOD vs SVOD Market

- **AVOD Revenue:**  $\text{Revenue} = \text{Impressions} \times (\text{CPM} \div 1,000) \times \text{Fill Rate}$
- **SVOD Revenue:**  $\text{Revenue} = \text{Subscribers} \times \text{Monthly Price}$

## Metrics

- **Fill Rate** is a key metric that measures how effectively an ad network or ad server is delivering ads to available ad spaces. It represents the percentage of ad requests that result in a successful ad serving, providing a gauge of how much ad inventory is being utilized
- **CPM** is Cost Per Mille. This is how much the platform earns from ads per 1000 impressions

# Brainstorming

**What initial factors would you consider when evaluating AVOD vs. SVOD. Can you think of some benefits and disadvantages for each?**

A strong candidate will list out some following considerations in a structured manner

## AVOD

### Benefits:

- **Larger Potential Audience Reach:** Free access appeals to a wider demographic, including cost-sensitive users
- **Lower User Acquisition Cost:** Easier to drive trials and adoption without paywall friction
- **Faster User Growth Potential:** Scalable model, especially for early-stage or niche content
- **Leverages Digital Ad Spend Trends:** Can benefit from growing digital ad budgets

### Drawbacks:

- **Higher Churn Risk:** Viewers may drop off due to ad annoyance or poor targeting
- **CPM Volatility:** Revenue is susceptible to fluctuations in ad rates and inventory
- **Dependency on Ad Demand:** Performance is tied closely to external advertiser interest

## SVOD

### Benefits:

- **Higher ARPU:** Consistent monthly revenue per user increases profitability
- **Avoids Ad-Related Churn:** Offers a cleaner, uninterrupted viewing experience
- **Increased Stability**

### Drawbacks:

- **Higher User Acquisition Cost:** Consistent monthly revenue per user increases profitability
- **Price Sensitivity:** Risk of Cancellation if value proposition weakens
- **High Content Expectations:** Subscribers expect frequent, high-quality content to justify the monthly fee
- **Lower Reach & Scale:** Paywalls can limit user acquisition, especially in early-stage growth

# Question 2

What is the profitability of AVOD vs. SVOD after 2 years?

*Only provide formula if asked by interviewee*

- **AVOD Revenue:** Revenue = Impressions × (CPM ÷ 1,000) × Fill Rate
- **SVOD Revenue:** Revenue = Subscribers × Monthly Price

## AVOD Revenue

|                                                  |      |
|--------------------------------------------------|------|
| Avg Year 1 MAU (Monthly Active Users)            | 1M   |
| Avg Year 1 Impressions per MAU                   | 20   |
| CPM (Cost Per Mille) (Cost Per 1000 Impressions) | \$20 |
| Year 1 Fill Rate                                 | 80%  |
| Expected Avg MAU Increase (Y/Y)                  | 50%  |
| Expected Impression per MAU Increase (Y/Y)       | 0%   |
| Expected CPM Increase (Y/Y)                      | 25%  |
| Year 2 Fill Rate                                 | 80%  |

|                                         |     |
|-----------------------------------------|-----|
| Avg Year 1 Monthly Subscribers          | 50K |
| Avg Year 1 Subscription Price Per Month | \$8 |
| Y/Y Avg Monthly Subscriber Increase     | 30K |
| Y/Y Churn % (From Avg Yr 1 Subscribers) | 20% |
| Y/Y Subscription Price Increase         | 0%  |

## SVOD Revenue

## Solution:

| Year 1 Revenue                                                                 | Year 2 Revenue                     |
|--------------------------------------------------------------------------------|------------------------------------|
| \$3.84M                                                                        | \$7.2M                             |
| 1M x 20 x (\$20/1000) x 80% x 12                                               | 1.5M x 20 x (\$25/1000) x 80% x 12 |
| Avg MAU x Avg Impressions per MAU x (CPM/1000) x Fill Rate x # of Months in Yr |                                    |

| Year 1 Revenue                             | Year 2 Revenue |
|--------------------------------------------|----------------|
| \$4.8M                                     | \$6.72M        |
| 50K x \$8 x 12                             | 70K x \$8 x 12 |
| Monthly Subscribers x Price Per Month x 12 |                |

# Question 2

Case 10 – Rack+

What is the profitability of AVOD vs. SVOD after 2 years?

Let's assume startup costs for both AVOD and SVOD are same. The only costs considerations today are operational costs. Below are costs for each monetization model (provide verbally)

## AVOD Costs

| Cost Item                 | \$        |
|---------------------------|-----------|
| User Acquisition Costs    | \$800K/Yr |
| SG&A                      | \$1.1M/Yr |
| Tech Stack                | \$600K/Yr |
| Content Acquisition Costs | \$1M/Yr   |

*\*\* Assume User Acquisition Cost to increase by 12.5% and SG&A to increase by \$300K in Year 2*

## SVOD Costs

| Cost Item                 | \$        |
|---------------------------|-----------|
| User Acquisition Costs    | \$1.5M/Yr |
| SG&A                      | \$1.2M/Yr |
| Tech Stack                | \$700K/Yr |
| Content Acquisition Costs | \$1.6M/Yr |

*\*\* Assume UA Costs to increase by 33%, SG&A Costs to increase by 25%, and Content Acquisition Cost to increase by 25% in Year 2*

## Solution:

| Year 1 Cost                     |         | Year 2 Cost                 |  |
|---------------------------------|---------|-----------------------------|--|
| \$3.4M                          |         | \$3.9M                      |  |
| \$800K + \$1.1M + \$500K + \$1M |         | \$900K + \$1.4M+\$600k+\$1M |  |
| Just Add All Cost Items Up      |         |                             |  |
|                                 | Year 1  | Year 2                      |  |
| Total Revenue                   | \$3.84M | \$7.2M                      |  |
| Total Costs                     | \$3.4M  | \$3.9M                      |  |
| Total Profit                    | \$0.44M | \$3.2M                      |  |

| Year 1 Cost                       |          | Year 2 Cost                   |  |
|-----------------------------------|----------|-------------------------------|--|
| \$5.0M                            |          | \$6.2M                        |  |
| \$1.5M + \$1.2M + \$700K + \$1.6M |          | \$2M + \$1.5M + \$700K + \$2M |  |
| Just Add All Cost Items Up        |          |                               |  |
|                                   | Year 1   | Year 2                        |  |
| Total Revenue                     | \$4.8M   | \$6.72M                       |  |
| Total Costs                       | \$5.0M   | \$6.2M                        |  |
| Total Profit                      | (\$0.2M) | \$0.52M                       |  |

# Exhibit 2

Case 10 – Rack+

## Alternative sports opportunities

| Sport   | Market Size | Competition | U.S. Interest                                                                         | Similarity to Pickleball                                                              |
|---------|-------------|-------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Tennis  | \$16B       | Medium      |    |    |
| Cricket | \$3.5B      | Low         |    |    |
| Rugby   | \$3B        | Low         |  |  |

## Solution (Qualitative Question):

- There is no single correct answer. The goal is for candidates to carefully read the chart, form a clear hypothesis, and support their recommendation based on the tradeoffs shown
- i.e. Tennis stands out with the largest global market size (~\$16B), high U.S. interest (●●●○), and strong Pickleball affinity (●●●●), making it the most natural extension to drive short-term growth

**Also, strong candidates will have good additional considerations to further explore:**

- **Content Acquisition Costs:** Tennis rights are likely more expensive than cricket or rugby
- **Seasonality:** Tennis offers year-round events, while rugby is more seasonal
- **International Growth:** Cricket and rugby could help diversify into new geographies if international expansion is a goal, but remember the team is only focused on US region for now
- **Platform Brand Strategy:** Tennis would reinforce the platform's focus on racquet sports
- **Competitive Landscape:** Any other streaming platforms offering these sports?

# Conclusion

Case 10 – Rack+

## Recommendation

- Pursue AVOD-first monetization model and expand into [insert sport] as the first adjacent sport
- AVOD is the better starting point financially, delivering positive profitability in Year 1 (\$0.44M vs. a (\$0.2M) loss for SVOD) and stronger scaling by Year 2 (\$3.2M profit vs. \$0.52M under SVOD). It also supports faster user acquisition by reducing friction to join.
- On expansion, the decision to add a new sport should be conditional on achieving early platform adoption goals. If initial traction is strong, Tennis is the logical next move: it offers the largest global market (~\$16B), strong U.S. interest, and high affinity with Pickleball users

## Risks

- Exposure to CPM volatility and ad fill rates; to mitigate this, the platform should closely track CPM trends and prioritize building a broad, high-engagement user base.

## Next Steps

- The platform should prioritize closely monitoring AVOD performance metrics — particularly CPM rates, fill rates, and MAU growth — to validate early profitability assumptions
- The team should begin piloting Tennis-related content through low-risk initiatives such as highlight reels, instructional clinics, and exhibition matches. This phased approach would allow the platform to test user demand without incurring full content acquisition costs upfront

# GreenTech Laptops

Technology | Market Entry  
*Interviewee Led*

Ratings: Quant - 2

Qual - 3

Overall - 2

GreenTech Laptops is a leading mid-sized laptop manufacturer that has invested heavily in sustainability initiatives over the past three years. The company has committed over \$50 million to using recycled materials, reducing carbon footprint in manufacturing, implementing take-back programs for end-of-life products, and redesigning packaging to minimize waste. Despite these significant investments, GreenTech has not seen the anticipated market perception changes or sales increases. CEO Jamie Chen has brought you in to evaluate GreenTech's sustainability strategy and determine the optimal approach to maximize both business impact and environmental benefits.

## Clarifying Information

**What is GreenTech's current position in the market, including market share?**

GreenTech is a leading mid-sized laptop manufacturer with a stagnant market share of 11%. Competitors who have invested less in sustainability but marketed it more aggressively have seen growth.

**What are GreenTech's sustainability initiatives to date, and how much has been invested in each category (recycled materials, carbon reduction, etc.)?**

GreenTech has invested \$50 million total: approximately \$15 million in recycled materials, \$20 million in carbon footprint reduction initiatives, \$10 million in take-back programs for end-of-life products, and \$5 million in redesigning packaging to minimize waste.

**What are the specific actions that are being considered for GreenTech, and what technologies are they investigating?**

GreenTech is considering acquiring EcoInnovate for \$20 million, which would provide patented biodegradable laptop component technology that could reduce environmental impact by 40% and provide sustainable competitive advantage for 3-5 years.

**How does GreenTech's pricing structure compare to competitors across different market segments (budget, mainstream, premium), and what premium is currently charged for sustainable features?**

GreenTech currently charges a uniform 10% premium for sustainable features across all segments, making their laptops cost \$550 (budget), \$990 (mainstream), and \$1,650 (premium), which is approximately 4-6% higher than competitors' comparable sustainable offerings.

# Potential Framework

## Market & Customer Analysis

- Current market share and competitive positioning
- Customer segmentation and sustainability preferences
- Willingness to pay premium for sustainable features by segment
- Gap analysis between customer values and purchasing behavior

## Financial & ROI Analysis

- Current investment in sustainability initiatives by category
- Financial returns on existing sustainability investments
- NPV analysis of potential investment options (acquisition vs. R&D)

## Positioning & Marketing

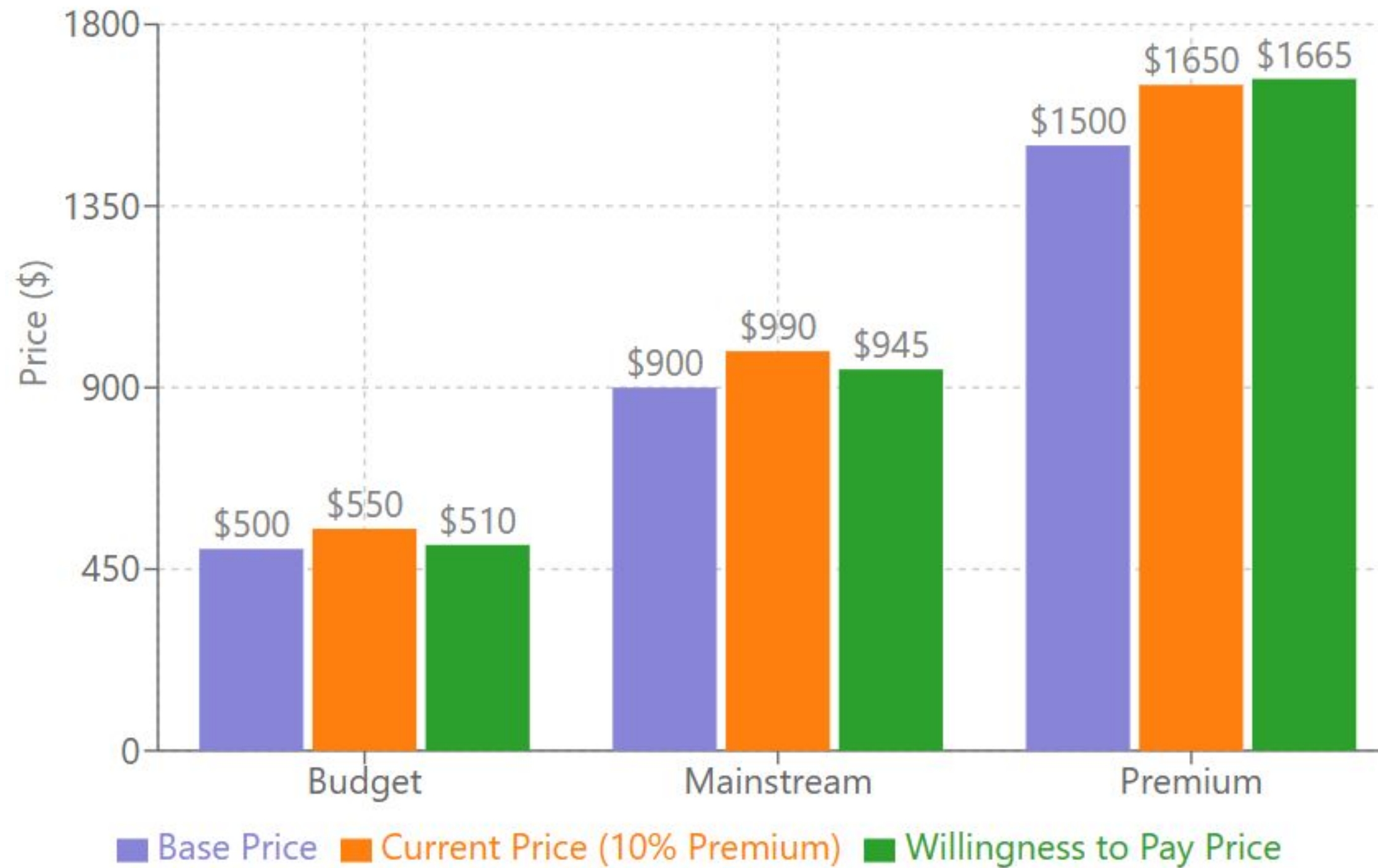
- Strategic Positioning & Marketing
- Competitive differentiation through sustainability
- Marketing effectiveness of sustainability messaging
- Alignment of sustainability initiatives with brand values

## Operational Considerations

- Integration capabilities for potential acquisition
- R&D timelines and probability of success
- Supply chain and manufacturing implications
- Regulatory considerations and compliance
- Talent and organizational capabilities

# Exhibit 1

**GreenTech Laptop Pricing Analysis by Segment**



Note: Price in USD. Assume willingness to pay is derived from customer surveys.

# Exhibit 1 Guidance

**Based on the information in Exhibit 1, should GreenTech maintain its current pricing strategy for sustainability features?**

**For your analysis, consider (only provide this if asked for by interviewee):**

- Current monthly sales volume: approximately 10,000 units per segment
- 10% decrease in units for every \$25 increase in price over willingness to pay
- Profit per segment: **\$20 for budget; \$50 for mainstream; \$150 for premium**

| Segment           | Price Difference                                                                           | Impact on units sold                                                                                                                                                                                            | Change in profitability                                                                                                                                           |
|-------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Budget</b>     | Current price: \$550<br>Willingness to pay (WTP): \$510<br>Price reduction needed: \$40    | Price reduction: \$40<br>Impact: 10% increase in units for every \$25 decrease<br>Units increase: $40 \div 25 \times 10\% = 16\%$ increase<br>Current units: 10,000<br>New units: $10,000 \times 1.16 = 11,600$ | Current profit: $\$20 \times 10,000 = \$200,000$<br>New profit: $\$20 \times 11,600 = \$232,000$<br>Profit increase: \$32,000                                     |
| <b>Mainstream</b> | Current price: \$990<br>Willingness to pay (WTP): \$945<br>Price reduction needed: \$45    | Price reduction: \$45<br>Impact: 10% increase in units for every \$25 decrease<br>Units increase: $45 \div 25 \times 10\% = 18\%$ increase<br>Current units: 10,000<br>New units: $10,000 \times 1.18 = 11,800$ | Current profit: $\$50 \times 10,000 = \$500,000$<br>New profit: $\$50 \times 11,800 = \$590,000$<br>Profit increase: \$90,000                                     |
| <b>Premium</b>    | Current price: \$1650<br>Willingness to pay (WTP): \$1665<br>Price increase possible: \$15 | Per instructions: No volume change (remains at 10,000 units)                                                                                                                                                    | Current profit: $\$150 \times 10,000 = \$1,500,000$<br>New profit with price increase: $(\$150 + \$15) \times 10,000 = \$1,650,000$<br>Profit increase: \$150,000 |

**Therefore, by aligning all segment prices with their respective willingness to pay values, GreenTech can capture an additional \$272,000 in profit monthly OR \$3.26M annual increase in profit**

# Question 2

As GreenTech looks at their sustainability approach, they are considering two different approaches to maintaining their product leadership in the space.

**Option A:** Acquire EcoInnovate, a leading startups in the space, for \$20 million

- Provides biodegradable laptop components immediately
- Annual benefits: \$6 million per year

**Option B: Internal R&D Investment** of \$15 million

- Will take 2 years to develop the technology
- Annual benefits: \$7 million per year

Calculate the return on investment for each option over a 5-year period and recommend which path GreenTech should pursue.

# Question 2 Guidance

## Option A: Acquisition Analysis

### Costs:

- Initial investment: \$20 million

### Benefits:

- Annual benefit: \$6 million
- Benefit period: 5 years
- Total benefits:  $\$6 \text{ million} \times 5 \text{ years} = \$30 \text{ million}$

### Net Return:

- Total benefits - Total costs =  $\$30 \text{ million} - \$20 \text{ million} = \$10 \text{ million}$

## Option B: Internal R&D Analysis

- Initial investment: \$15 million

### Benefits:

- Annual benefit: \$7 million
- Benefit period: 3 years (years 3-5 only due to time required for R&D)
- Total benefits:  $\$7 \text{ million} \times 3 \text{ years} = \$21 \text{ million}$

### Net Return:

- Total benefits - Total costs =  $\$21 \text{ million} - \$15 \text{ million} = \$6 \text{ million}$

# Question 3

GreenTech has identified a gap between consumers' stated preference for sustainable products and their actual purchasing behavior. Market research indicates that while 78% of consumers say sustainability is "important" or "very important" in their purchasing decisions, only 14% report having paid a premium for sustainable features in their most recent laptop purchase. Develop a creative, multi-channel marketing campaign for GreenTech that addresses these challenges and effectively communicates the value of their sustainability investments.

## Interviewer Guidance

Lots of potential options here – focus on structured and creative responses that align with the recommendation from above.

**Campaign Concept: "Feel the Difference, Make the Difference"** Bridges the value-action gap by making sustainability tangible, emotional, and personally relevant to consumers

### Three Key Tactics:

1. **Interactive "Sustainability Impact Visualizer"**: Translates technical specs into compelling real-world equivalents. Example: "This laptop prevents 35 plastic bottles from entering oceans and saves enough energy to power your home for 3 days"
2. **"Experience Sustainability" Retail Strategy**: Creates multi-sensory product displays contrasting conventional vs. sustainable materials. Features transparent product sections revealing eco-components with digital kiosks sharing sourcing stories
3. **"Sustainability as Standard" Pricing Approach**: Integrates sustainable features across all product tiers rather than as premium add-ons. Offsets partial costs through supply chain efficiencies and long-term brand value growth
4. **Highlight planned acquisition – if recommending to acquire Ecolnnovate based on the Q2 ROI**

### Measurement Framework:

- Track conversion lift between product views and sustainable purchases
- Target: Reduce value-action gap by 50% within 18 months
- Monitor competitor price differentials for sustainable products
- Measure earned media value and sentiment from campaign storytelling
- Conduct quarterly consumer perception studies to refine messaging

# Conclusion

## Recommendation

- GreenTech should pursue a three-pronged strategy to maximize the ROI of its sustainability investments
- **Pursue Strategic Acquisition:** Acquire EcoInnovate to gain immediate access to revolutionary tech. The NPV shows option is superior to internal R&D over 5-year horizon
- **Optimize Pricing Strategy:** Implement segment specific pricing that aligns with customers' willingness to pay for sustainability features. This will increase total revenue by approximately \$3.2M annually while expanding GreenTech's market share, particularly in the budget and mainstream segments where current pricing is misaligned with customer preferences.
- **Transform Marketing Approach:** Launch a marketing campaign to address the value-action gap in consumer behavior. By making sustainability tangible through the campaign, GreenTech can differentiate its brand and convert stated sustainability preferences into purchasing decisions.
- The revised strategy addresses the fundamental challenge GreenTech faces. By optimizing pricing, acquiring innovative technology, and transforming marketing, GreenTech can achieve both business growth and environmental impact.

## Risks

- **Acquisition Integration:** Manage cultural alignment, retain key talent, and seamlessly incorporate new technology into existing manufacturing processes.
- **Pricing Strategy Implementation:** Introduce segment-specific pricing gradually with clear value communication while preparing for competitive responses.
- **Marketing Effectiveness:** Ensure substantiated sustainability claims with consistent messaging and continuous measurement against defined KPIs.

## Next Steps

- Think about managing acquisition integration
- Market study for pricing strategy
- Any other steps that address risks outlined by interviewee

# Midwest Hospital

Healthcare | Profitability

*Interviewer Led*

Ratings: Quant - 2

Qual - 3

Overall - 2

## General Information:

- The case of a turnaround of a loss-making business to a profitable business. It would require the candidate to understand customer mix of the client and understand competitive benchmarking.
- Strong candidates should identify the competitive advantage of client's competition. Strong candidate should also understand the difference between gross profit and net profit to decide the way forward.

## Exhibit Summary

- Exhibit 1: Patient Mix
- Exhibit 2: Joint Replacement Department P&L
- Exhibit 3: Competitive Benchmark

# Prompt

Midwest Hospital is a research-based hospital and takes pride in its joint replacement surgery department. Recently Midwest Hospital did a P&L analysis for all departments and found that the joint replacement surgery department is providing losses. The CEO has asked us to help out.

## Clarifying Information

- There are no financial targets.
- Focus of this case is only on joint replacement surgery.
- Give the exhibits in the subsequent slides only when the candidate asks for the relevant data.

# Potential Framework

| Revenue                                                                                                                                                                                                                      | Cost                                                                                                                                                                                                                                                                                                                                | Competition                                                                                                         | Customers                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• No. of Surgeries</li><li>• Complexity of surgery</li><li>• Patient Mix</li><li>• Price</li><li>• Other Charges (bed, medicines)</li><li>• Post Surgery (visits, medicines)</li></ul> | <ul style="list-style-type: none"><li>• Fixed Costs<ul style="list-style-type: none"><li>○ Hospital</li><li>○ Doctors</li><li>○ Equipment</li><li>○ Insurance</li><li>○ License</li></ul></li><li>• Variable Costs<ul style="list-style-type: none"><li>○ Visiting Doctors /Surgeons</li><li>○ Govt. regulation</li></ul></li></ul> | <ul style="list-style-type: none"><li>• Price</li><li>• Patient Mix</li><li>• Better facilities/equipment</li></ul> | <ul style="list-style-type: none"><li>• Age group</li><li>• Paying with insurance</li></ul> |

## Notes / Advice to Interview / How to Move Forward:

- At some point near the start of the case, interviewer should take the lead
- Make Exhibit 1 and 2 available for the interviewee ahead of question 1 and 2

# Exhibit 1

Patient Mix

| Payer Type                  | # Surgeries | List Price | Invoiced Price |
|-----------------------------|-------------|------------|----------------|
| Commercial<br>(Enterprises) | 100         | \$40,000   | \$40,000       |
| Insurance                   | 300         | \$40,000   | \$20,000       |
| Medicare<br>(Government)    | 600         | \$40,000   | \$15,000       |

# Exhibit 2

## Joint Replacement Department P&L

| P&L           |            | \$Mn   |
|---------------|------------|--------|
| Revenue       |            | \$19M  |
| Variable Cost | Physician  | \$5M   |
|               | Materials  | \$5M   |
|               | Others     | \$4M   |
| Fixed Cost    | Facilities | \$3.5M |
|               | Others     | \$3.5M |
| Total Costs   |            | \$21M  |
| Profit        |            | (\$2M) |

# Question 1

Would it be advisable to not cater to Medicare patients (assuming no backlash)? Show Exhibit 1 and 2.

## Solution:

- On a fully cost allocated basis, Medicare patients are unprofitable but they are still paying \$1k above the variable cost (marginal cost). This helps cover the fixed costs of the department. So, it is not recommended to stop conducting surgeries for Medicare patients.
- Total Cost / # of Surgeries =  $\$21\text{M} / 1000 = \textbf{\$21,000 / surgery}$
- Marginal Cost / # of Surgeries =  $\$14\text{M} / 1000 = \textbf{\$14,000 / surgery}$
- Revenue per Medicare patient = **\$15,000 / surgery**

# Question 2

What is the number of surgeries that Midwest needs to conduct in a year to breakeven?

## Solution:

- Average revenue per patient is 19k. Average variable cost is 14k. Gross margin per patient is 5k. Fixed cost are 7M, so 1400 surgeries are required to breakeven. Assuming same proportion as in Exhibit 1 the hospital requires 140 commercial, 420 insurance, and 840 Medicare Patients

Avg. Revenue Per Surgery = Total Revenue / # of surgeries = \$19M / 1000 = **\$19,000 per patient**

Avg. Variable Cost = Total Variable Cost / # of Surgeries = **\$14,000 per patient**

Fixed Costs = **\$7M**

**Breakeven** = FC / (Rev - VC) = \$7M / (\$19000 - \$14000) = **1400 Surgeries Patient Mix:**

- Commercial = 10% \* 1400 = **140 surgeries per year** Insurance = 30% \* 1400 = **420 surgeries per year** Medicare = 60% \* 1400 = **840 surgeries per year**

**\*\*Percentages based on share of surgeries from Exhibit 1\*\***

# Exhibit 2

Competitive benchmark

|                  | Surgeries | Commercial | HMO | Medicare | Profitable |
|------------------|-----------|------------|-----|----------|------------|
| Midwest Hospital | 1000      | 10%        | 30% | 60%      | No         |
| Comp A           | 1200      | 20%        | 20% | 40%      | Yes        |
| Comp B           | 800       | 30%        | 20% | 50%      | Yes        |
| Comp C           | 900       | 10%        | 20% | 70%      | Yes        |
| Comp D           | 1000      | 5%         | 25% | 75%      | Yes        |

# Question 3

What is Competitor D able to stay profitable despite having fewer patients and unfavorable patient mix? (Show Exhibit 3).

## Solution:

- Competitor D might have a lower cost structure or was able to negotiate better pricing from payers.

# Question 3

Evaluate potential risks to increasing number of surgeries.

## Brainstorming Guidance:

- There might not be enough market demand and increasing surgeries would mean stealing marketing share from competitors
- The competitors might reduce the price and enter a price war
  - Quality of surgery may be reduced, impacting reputation.
  - Surgeons may resent higher workload

# Conclusion

## Recommendation

- Increase total number of patients to cover
- Change mix of patients to have a higher proportion of commercial and insured customers
- See if you can negotiate with insurers and Medicare for higher reimbursement.

## Risks

- Those gained from brainstorming

## Next Steps

- Analyze scope for cost reduction, starting with competitive benchmarking
- Analyze scope for increase in price, starting with competitive benchmarking
- Analyze profitability of post care services
- Start conversations with reimbursement providers

## Excellent Case Answers

- Star candidates quickly identify that competitor D has a similar patient mix but is still profitable.
- Candidate should provide a reason as to why the client should not eliminate medicare patients
- Candidate should bring up the idea that joint replacement department may be a loss leader and provides synergies with other department offerings

# EuroRail

Hospitality | Growth

*Interviewee Led*

**Ratings: Quant - 3**

**Qual - 2**

**Overall - 3**

Our client, EuroRail, is a railway company that operates luxury trains across Europe. It is a very different kind of railway company where tourists sign up for a 24-hour trip on the train. The trains are like hotels on wheels with various amenities on board - ranging from a ballroom, jacuzzi, spa, and a bar. These trains are very popular in the Europe and run at 100% capacity throughout the year. Its end of 2022 and the company is looking to quickly increase its yearly revenue for year 2023. What should they do?

## Clarifying Information:

- Goal: Increase revenues by at least €44M for the year 2023
- Current revenue can be calculated later (but for reference current revenue is €365M, so €44M is about 12% growth)
- The company charges €100 per person per trip (note, this is average price and candidate should ideally ask about different tiers or segments in the train. If they ask about segments tell them ticket prices by segments from exhibit 1)
- The company operates 20 trains across Europe
- The trains run at 100% capacity and 365 days a year.
- Demand for Luxury currently perfectly matches with capacity but the demand for economy is so high that company has to deny many registrations everyday.
- No competitor in the market

# Potential Framework

## Increase Average Spend

- Increase prices
- Change mix of cabins tiers
- Introduce new tiers in train
- Improve quality of onboard services to drive more onboard sales

## Find New Revenue Stream

- Introduce casinos on-board similar to casinos on cruise ships
- Place advertisements inside and outside the trains to get ad- revenues
- Partner with luxury companies to drive commission-based sales onboard
- Add more coach to the train
- Any practical idea is acceptable...

## Grow Inorganically

- Add new trains to the feet
- Expand to other geographies and introduce new routes
- Acquire competitors
- Explore partnering with other railway operators

### What is acceptable performance?

- A good candidate will create a framework that explores ways to grow revenue and will not include other out of scope levers such as costs or profits

### What is expected from an exceptional performance?

- An exceptional candidate will acknowledge the fact that the trains are running at 100% capacity and will not suggest revenue growth strategies that aim at increasing volume (Example: decrease price to drive more sales, target new customers, improve marketing and sales, etc.)
- Additionally, an excellent candidate will not only discuss inorganic growth strategies but will also acknowledge that EuroRail wants to increase revenues within the next year so acquisition, partnerships, buying new trains, etc. might not be feasible options due to time constraints

### Notes for Interviewer

- Shoot down ideas such as new routes / new trains (not possible in one year) or add coach (no engine capacity)
- Push candidate to towards ideas such as increase price or change mix of cabin tiers (the ideal idea at this point is to explore price increase of one or both segments)
- Ask candidate to hold the thought on new revenue stream ideas
- If candidate asks for information regarding willingness to pay, current pricing and tiers, etc. show them Exhibit 1
- In any case, do not show Exhibit 2 without going through (and solving) exhibit 1

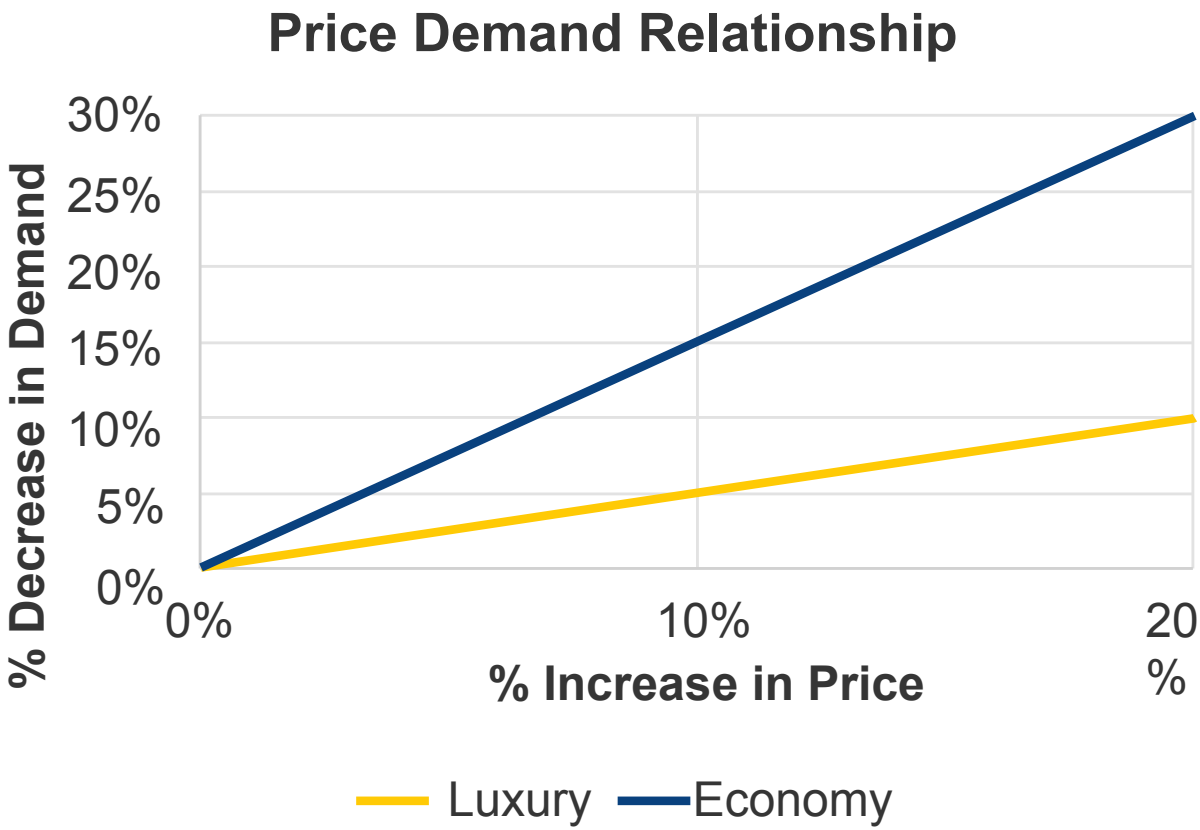
# Exhibit 1

Data from consumer insights team

EuroRail Train



| EuroRail Revenue Streams | Economy | Luxury |
|--------------------------|---------|--------|
| Ticket per PAX           | €75     | €125   |
| Onboard spend per PAX    | €25     | €25    |



**Economy Coach** 10 cabins per coach 2 PAX per cabin

**Luxury Coach** 10 cabins per coach 2 PAX per cabin

*PAX = passenger*

## Notes

- This is only a graphical representation of train to show split of economy and luxury cabins.
- Actual trains also have more coaches dedicated to onboard amenities such as spa, jacuzzi, bar, etc.
- EuroRail does not want to increase prices by more than 20% due to fear of public outrage

# Exhibit 1

## Pre and post-change financials

### Notes for Interviewer

- This case is difficult to solve at yearly level. Candidate should bring everything down to per train per day level (ok to use 360 days / year)
- A good candidate will indicate that economy price cannot be increased because of high demand elasticity (higher quantity decrease per percent increase in price). Therefore, increase price of luxury segment.
- An excellent candidate will realize that increase prices will free up some cabins due to reduced demand. These empty cabins can then be used for other purposes.
- **Mini Brainstorm:** Ask candidate to generate ideas about what to do with the free cabins. Ask “what else” till they are out of ideas. Direct candidate towards the idea of free upgrade for loyal economy passengers at price of economy cabin i.e., €75 per person.
- After candidate has explored 20% price hike and realized EuroRail is still not able to meet the target, communicate that you like the idea of 20% price hike but what else can we do. Candidate should go back to the framework and suggest exploring additional revenue streams.

**Goal = €44M over next year = €44M / (20 trains x 360 days) = €6,111 per train per day additional revenue (Okay to go with €6,000)**

| Scenario                                                                                                                                                                                     | Luxury Ticket Sales                                  | Economy Ticket Sales                                | Total Sales                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Current</b>                                                                                                                                                                               | = 10 coaches x 10 cabins x 2 PAX x €125<br>= €25,000 | = 10 coaches x 10 cabins x 2 PAX x €75<br>= €15,000 | = €25,000 + €15,000<br>= €40,000                                                                           |
| <b>10% price increase for luxury segment</b> <ul style="list-style-type: none"> <li>• 5% demand drop</li> <li>• Free upgrade for economy travelers to freed up 5% luxury cabins</li> </ul>   | = €25,000 x 1.1 x 0.95<br>= €26,125                  | = €15,000 x 1.05<br>= €15,750                       | = €41,875<br>= €1,875 additional revenue<br>(not enough, so explore 20% price increase)                    |
| <b>20% price increase for luxury segment</b> <ul style="list-style-type: none"> <li>• 10% demand drop</li> <li>• Free upgrade for economy travelers to freed up 10% luxury cabins</li> </ul> | = €25,000 x 1.2 x 0.90<br>= €27,000                  | = €15,000 x 1.1<br>= €16,500                        | = €43,500<br>= €3,500 additional revenue<br>(still not enough, so explore additional revenue growth ideas) |

# Exhibit 2 (for interviewer)

- After asking candidate about additional revenue streams, inform that EuroRail would like to add a casino coach onboard and give below information about the Casino Coach:
  - Additional Revenue: €25 per PAX
  - Casino coach will be created by converting 1 economy coach
  - Casino will require 6 months to launch



- After candidate calculates net new revenue form Casino Coach, ask them to brainstorm some pros and cons of casino and then move on to the recommendations

## Solution to Casino Coach

| Revenue Lost Due to Coach Conversion                                                                                       | New Revenue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Net New Revenue |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <ul style="list-style-type: none"><li>• 20 Economy PAX x (€75 ticket + €25 onboard purchases)</li><li>• = €2,000</li></ul> | <p>= 380 PAX x €25 x 50% year<br/>= €4,750</p> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• Candidate should realize that there will only be 380 PAX and not 400 PAX since EuroRail will convert one of its economy coach to casino and will lose 20 economy PAX</li><li>• If candidate misses the time factor, direct them to think how time required for launch will impact the new revenue (Casino Coach will take 6 months, therefore it will generate revenue only for 50% of the year)</li></ul> | €2,750          |

# Brainstorming

## What are some pros and cons of each program having a Casino Coach?

Below is an example of answer. This is not an exhaustive list, and any additional reasonable points are acceptable

### Pros:

- High revenue potential in the long term
- Will make train more attractive for all passengers
- Adding a casino could be a good way to justify 20% price increase

### Cons:

- Might face regulatory issues
- Casinos are risky investments and EuroRail might lose money on bad days
- Will require high investment (security, surveillance, dealers, etc.)
- High time to launch

# Conclusion

## Recommendation

- To achieve its revenue growth target of €44M in 2023, EuroRail should increase price of its luxury segment by 20%.
- This will lead to 10% drop in demand. EuroRail can give free upgrades to its loyal economy travelers to backfill freed up luxury cabins. (combined €3,500 additional revenue per train per day)
- EuroRail should also convert one of its economy coaches on each train to a casino (Additional €2,750 revenue per train per day)

## Risks

- Increase in price might not be well received by the travelers
- Price hikes might make this industry more attractive and invite competition
- Casinos are risky and capital- intensive investments

## Next Steps

- Announce launch of onboard casino along with price
- hike – create marketing and communication material
- Perform customer research to decide games to be place in casino
- Consult lawyers to discuss regulations around casino operations
- Hire designers to redesign coach as a casino

# Attack Helicopter

Defense | Market Entry

*Interviewee Led*

Ratings: Quant - 3

Qual - 2

Overall - 3

# Prompt

Our client is a US defense contractor and one of its divisions manufactures attack helicopters for military operations. The company is considering setting up a new plant to meet increasing demand in the attack helicopter space. These helicopters are fully equipped with guns and ammo when delivered to the end customer. Our client has considered three sites to setup operations: Brazil, France, and the US.

How would you make this decision, and where should our client set up the plant based on that analysis?

## Clarifying Information

- Our client has 3 plants in the US; 2 in Kansas and 1 in Michigan
- The plants operate at full capacity today
- One of the US plants can accommodate an additional assembly line at a cost of \$500M; the other 2 are landlocked in residential areas and cannot be expanded

### Notes / Advice to Interview / How to Move Forward:

- Candidate should structure the framework to consider the elements for evaluating 3 countries
- Candidate should start with a financial analysis. If not, nudge them in this direction
- If candidate does not mention examples of variable and fixed costs, prompt them to do so prior to revealing financials

# Potential Framework

## Profitability Analysis of Each Site

### Cost

- VC
  - Materials
  - Hourly Labor
  - Ship/Distribution
- FC
  - PP&E
  - Utilities
  - SG&A
  - Initial Cost

### Revenues

- Price per helicopter
- # of helicopters / year
- How frequently do helicopters get replaced?

## Customers

- Where are they based?
- Need to be close to the customer for design inputs?
- Additional customer points?
- Do governments prefer to buy from domestic companies?
- Are there custom designs? Is each helicopter the same?
- Proprietary info: can you build two helicopters for two countries at once?
- Is there a market for private security and not just military?

## Suppliers

### Supply chain considerations

- Do we have an adequate supply base in each location?
- Assembly inputs
  - Spare parts
  - Raw materials
- Local regulations: can you import what is needed?

## Logistics

- What's it going to take to get the product to the customer?
- Labor
  - Availability of skilled managers & technicians
- Export control restrictions?

# Financial Information

## Cost Information

Initial Plant Set up Cost: \$500M (US) | \$2B (BR) | \$3B

(FR) Fixed Cost: \$100M annually in all three countries

Variable Cost: \$15M (US) | \$20M (BR) | \$25M (FR)

## Market Size and Revenue Information

Defense budgets for next 5 years: \$100B (US) | \$15B (BR) | \$10B (FR)

% of Defense budgets to be spent on our helicopters: 20% (US) | 50 % (BR) | 15% (FR)

## Sales Information

Each helicopter sells for \$100M/unit, but if they are imported into the US, the US government requires them to be certified; the certification process costs \$15M/unit.

In the case of attack helicopters, the US and France each have export control restrictions in place prohibiting the sale of attack helicopters to other countries.

*Implication: If the client manufactures within the US, they will only be able to sell to the US government – no exports are allowed.*

## If plant in the US

### Revenues:

US revenues over 5 years = 20% of \$100B = **\$20B**

### Costs:

# of copters = \$20B / \$100M = 200

helicopters **Variable Cost:** 200 helicopters

\* \$15M = **\$3B Fixed Cost: \$100M**

**annually**

**Initial Cost: \$500M**

$$\text{Profit} = \text{Rev} - \text{VC} - \text{FC} - \text{IC}$$

$$\text{Profit} = \$20\text{B} - \$3\text{B} - (\$100\text{M} * 5) - \$500\text{M} = \mathbf{\$16\text{B}}$$

## If plant in Brazil

### Revenues:

US **revenues** over 5 years = 20% of \$100B = **\$20B**

BR **revenues** over 5 years = 50% of \$15B = **\$7.5B**

### Costs:

# of US-bound copters: \$20B / \$100M = 200 helicopters # of BR-bound

copters: \$7.5B / \$100M = 75 helicopters

**Variable Cost:** (275 helicopters \* \$20M)\*\* + (200 helicopters \* \$15M)\* = **\$8.5B**

**Fixed Cost: \$100M annually Start-up Costs: \$2B**

Profit: \$27.5B - \$11B = **\$16.5B**

**Key Takeaway: Plant in Brazil will give us profits higher than US by \$500M**

## Notes

- Interviewee should notice that France is not an attractive candidate due to having the same exact costs as Brazil but significantly smaller revenues in addition to export issues.
- Once interviewee mentions a profitability calculation, nudge them to calculate profit over the next 5 years
- \*\* = VC to manufacture in Brazil \* = Certification costs to sell to United States

# Brainstorming

Evaluate the potential risks of acquiring helicopters in Brazil.

## Brainstorming Guidance

- If the interviewee does not provide potential risks for moving forward with Brazil, ask this question.

## Best candidates will:

- Drive the case toward potential risks of launching in Brazil without prompting.

## Potential Solutions

- Political stability
- Product quality
- Robustness of supply chain
- Cultural customs are different and may affect business relationships
- Know-how in other countries
- Brand image and trust in helicopters
- National security may be an issue as helicopters/defense equipment will be manufactured outside of US

# Conclusion

## Recommendation

- Based on the financials, Brazil appears to be a more attractive candidate for setting up the new plant because:
  - Our profits over 5 years will be higher by \$500M
  - We won't be entirely dependent on one single country (US) for sales

## Risks

- Those gained from brainstorming

## Next Steps

- What is the potential for selling choppers outside of these 3 countries to the global market?
- What will labor's reaction at our
- existing plants be if we
- Are US relations with Brazil likely to be cordial over the next 5 years for us to benefit from favorable export control policies?

# One Tree Hill

Non Profits | Micro-Economics, Financing  
*Interviewee Led*

Ratings: Quant - 3

Qual - 3

Overall - 3

## General Information:

- This case is similar to a Bain case. It follows their typical format of a company looking to grow and needing multiple different avenues to get there: organic growth and inorganic growth.
- Strong candidates will consistently identify the revenue deficit and use it as a waypoint to know if they have achieved the client's goal or not. Even stronger candidates will get into the nuances of the client and why certain acquisitions are or are not a good fit.

## Exhibit & Quant Summary

- Quant
- Exhibit 1: Two potential acquisitions

# Prompt

The nonprofit organization, One Tree Hill (OTH), has a mission of restoring forest cover globally. They typically give grants (donations) to local NGOs, communities, or startups that reforest (i.e., plant trees) in a given area.

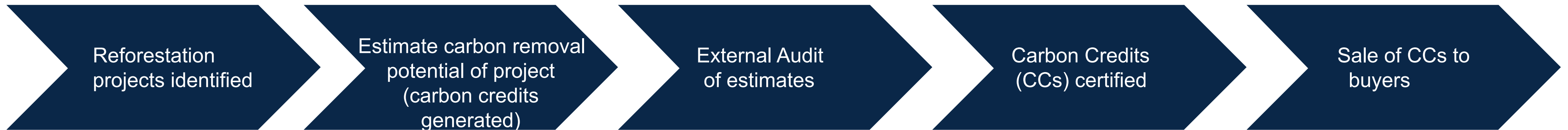
Due to the limited supply of grant funds in recent times, and in order to enter a very lucrative market, OTH is looking for ways to fund some reforestation projects through the sale of carbon credits (CC), also called “carbon financing”. They have a portfolio of projects in need of funding, and they would like to select a few for carbon financing. Should OTH pursue carbon financing, and if yes, for which projects?

## Clarifying information:

- A carbon credit (CC) is a tradable certificate that provides the holder of the credit the right to emit greenhouse gases, typically 1 ton of carbon dioxide equivalents. Corporations and countries purchase carbon credits to “offset” their emissions above allowable limits.
- If the candidate asks about the process for selling carbon credits, show them the flowchart in Exhibit 1 (this is just meant to help them understand the process, candidate is not expected to draw insights from this)
- There is no target amount to be raised for funding these projects (there are many). The case is about whether financing a project via generating and selling carbon credits is a feasible option. If feasible, OTH will proceed with that project.
- The customers are typically the corporations or countries wanting to offset emissions from their operations.
- This is a forward-looking analysis, i.e., only for new projects. We can ignore any past planted trees or projects.

# Exhibit 1: Process for Selling Carbon Credits

Case 15 – One Tree Hill



- 1 ton of CO<sub>2</sub> removed by trees = 1 carbon credit generated
- Done by external agencies/certification bodies

# Potential Framework

Case 15 – One Tree Hill

| Financial Factors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Logistical Factors                                                                                                                                                                                                         | Legal & Regulation Factors                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Cost of carbon financing</p> <ul style="list-style-type: none"><li>• Costs of certification</li><li>• External audit</li><li>• Commissions to CC issuing body tree planting &amp; maintenance costs</li><li>• Labor</li></ul> <p>Revenue generating potential</p> <ul style="list-style-type: none"><li>• Price at which carbon credits are sold</li><li>• Demand for purchasing CCs</li><li>• Market trends – nascent vs growing</li></ul> <p>Opportunity cost</p> <ul style="list-style-type: none"><li>• Does this preclude projects from other forms of funding or other revenue sources (e.g.: sustainable timber sales)</li></ul> | <p>Resources &amp; Capacity to execute</p> <ul style="list-style-type: none"><li>• Manpower, technical assistance, on-field resources, etc. for data collection and ongoing monitoring</li></ul> <p>Execution timeline</p> | <ul style="list-style-type: none"><li>• Geography of project – national policy on CCs? Who can you sell to?</li><li>• Any unique taxation policies to consider?</li><li>• CC Regulatory bodies like the UNFCCC – what is their role?</li></ul> |

## Notes / Advice to Interview / How to Move Forward:

### Expectations of a good candidate:

- Identify that the process in Exhibit 1 involves many costs, and the final goal is to find the projects that are "profitable".
- The framework will reflect that this case boils down to doing a cost-benefit analysis.

### Expectations of an outstanding candidate:

- On top of what a good candidate can identify, also identify the variety of other factors that can go into this decision, such as the timeline, national climate policy, etc.
- Bring in knowledge and examples from this subject in the real world.

# Cost Breakdown

After the framework, the candidate should ask for data on the costs of the various steps – tell them the information in **Table 1**. If they do not ask, guide the interviewee towards thinking about how an organization can generate carbon credits, who would issue these, certify them, etc.

Do not show this information to the candidate, only read the data to them.

**Table 1**

**Fixed costs (do not change with size of project):**

1. Audit cost = \$325,000
2. Project certification cost = \$175,000

**Variable costs:**

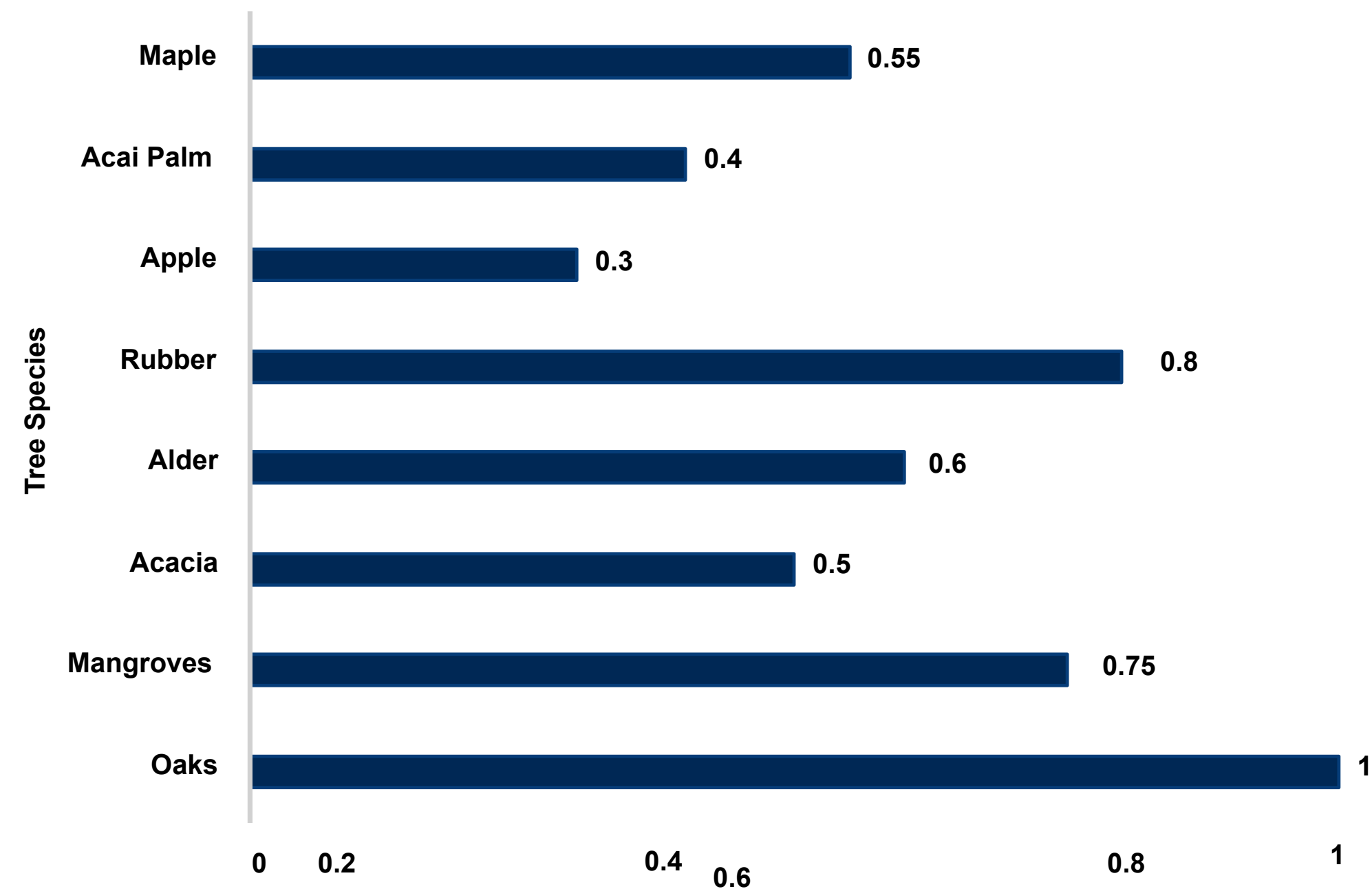
1. Carbon Credit issuance fee = \$2 per CC issued
2. Tree planting & maintenance cost = \$5 per tree (throughout the tree's lifetime)

**Price of CC sold in the market:** \$10 per CC

After receiving Table 1 information, the candidate should realize that the costs are variable and should seek information on the number of trees in various projects and the number of carbon credits that can be sold. **Show them Exhibit 2**

# Exhibit 2: Data by Tree Type

Carbon Credit Potential Per Tree



| Project | # of trees | Tree Species | Location  |
|---------|------------|--------------|-----------|
| A       | 3,000,000  | Oaks         | France    |
| B       | 7,000,000  | Mangroves    | India     |
| C       | 4,000,000  | Acacia       | Ghana     |
| D       | 3,000,000  | Alder        | Peru      |
| E       | 350,000    | Rubber       | Venezuela |
| F       | 2,000,000  | Apple        | Turkey    |
| G       | 3,000,000  | Acai Palm    | Brazil    |
| H       | 4,000,000  | Maple        | Canada    |

Number of Carbon Credits generated\*

\*Carbon credits generated are over the tree’s lifetime of 30 years, not annually

# Cost Breakdown Solution

*(Time value of money can be ignored)*

**Method 1:** Candidates can solve this in the traditional method of calculating variable costs for each of the projects, add fixed costs, then subtract total cost from revenue to find if it's profitable. However, this is extremely lengthy and not advised. [Example for Project

A: Total CCs generated = Number of trees\*CC potential per tree = 3M \* 1 = 3M

Total revenue = Total CCs generated \* \$10 per CC = 30M; Variable costs = \$2 \* Number of trees + \$5 \* CCs generated =

\$21M; Total costs = \$21M + \$0.5M (Fixed cost) = \$21.5M; Profit = Revenue - Cost = \$8.5M. Hence, this project breaks even]

| Project | Number of trees | Tree species | Location  | Total CCs | Total revenue | Variable costs | Fixed costs | Total costs | Profit   |
|---------|-----------------|--------------|-----------|-----------|---------------|----------------|-------------|-------------|----------|
| A       | 3000000         | Oaks         | France    | 3000000   | 30000000      | 21000000       | 500000      | 21500000    | 8500000  |
| B       | 7000000         | Mangroves    | India     | 5250000   | 52500000      | 45500000       | 500000      | 46000000    | 6500000  |
| C       | 4000000         | Acacia       | Ghana     | 2000000   | 20000000      | 24000000       | 500000      | 24500000    | -4500000 |
| D       | 3000000         | Alder        | Peru      | 1800000   | 18000000      | 18600000       | 500000      | 19100000    | -1100000 |
| E       | 350000          | Rubber       | Venezuela | 280000    | 2800000       | 2310000        | 500000      | 2810000     | -10000   |
| F       | 2000000         | Apple        | Turkey    | 600000    | 6000000       | 11200000       | 500000      | 11700000    | -5700000 |
| G       | 3000000         | Acai Palm    | Brazil    | 1200000   | 12000000      | 17400000       | 500000      | 17900000    | -5900000 |
| H       | 4000000         | Maple        | Canada    | 2200000   | 22000000      | 24400000       | 500000      | 24900000    | -2900000 |

**Method 2:** Instead, candidate should take a unit economics approach – what is the minimum carbon credit potential per tree beyond which it becomes profitable? (if candidate is struggling, ask them this question as a hint).

Let “x” be the Carbon Credit potential/ number of CCs per tree (0.5, 0.6, 1, etc).

Our break-even equation **per tree** then becomes:  $2x + 5 = 10x$  [We converted all costs and revenues to a per tree basis;

$2x$  = Cost per CC \* number of CC per tree = Cost per tree; 5 is the planting cost per tree;  $10x$  = Price of CC \* CC per tree = Price per tree]

Solving this, we get  $x = 0.625$ . Projects with CC potential per tree  $> 0.625$  could breakeven, depending on fixed costs (Projects A, B, E). Now candidate can perform the calculations shown above only for these 3 projects and find that Projects A & B are profitable!

# Brainstorming

What other factors would you consider when evaluating if OTH should proceed with carbon financing of Projects A & B?

## Expected Answer:

Candidate should provide a structured list of factors other than financial feasibility to be considered when evaluating this decision, including but not limited to:

- a) Legal ramifications of selling carbon credits generated in reforested areas
- b) Double counting of carbon credits with national commitments of reforestation
- c) Availability of manpower to perform the tasks required
- d) Timeline of cash flows – when does the money from sale of CCs come in? Does this overlap with the requirement of funds?
- e) Availability of grant capital in the country – we may want to explore CCs in regions where grants are limited

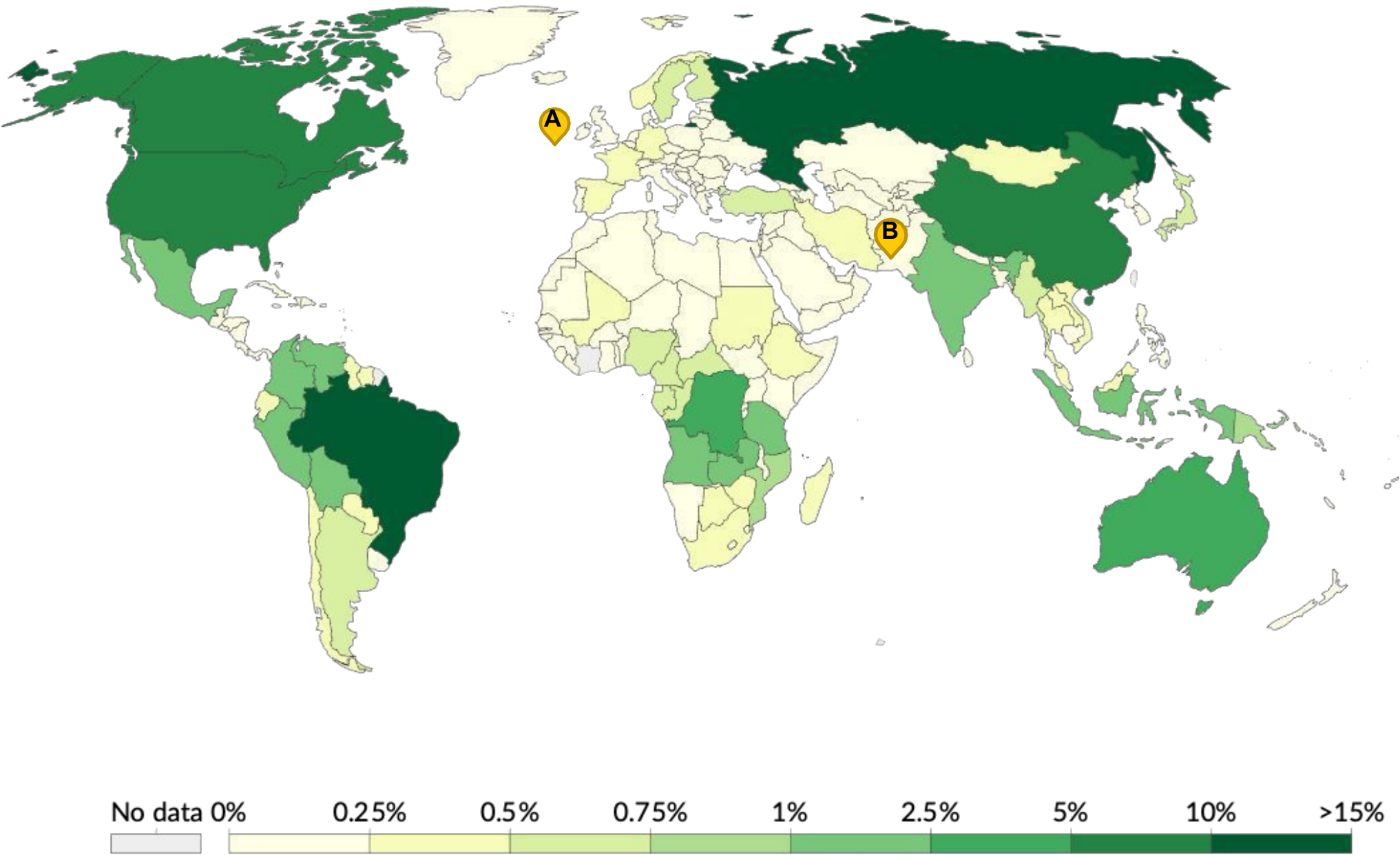
## Interviewer Guidance for Exhibit 3

Once candidate lists a few factors, show them Exhibit 3 and ask for their thoughts. The chart and map in Exhibit 3 is intentionally ambiguous. While India (Project B) has lower availability of grant funding capital and hence may need to do carbon financing more, it has fewer forest protection services available – which makes implementation very difficult. Similarly, existing forest cover and density is very different for both projects. There is no “right” conclusion from Exhibit 3, it is meant to test the candidate’s ability to interpret ambiguous data. Any reasonable conclusion is accepted.

# Exhibit 3: Global Forest Data

Share of global forest area, 2020

Source: 



Source: UN Food and Agriculture Organization (FAO). Forest Resources Assessment. OurWorldInData.org/forests • CC BY

## Resource Availability

Project A



Project B



Availability of capital for grant funding

Strength of forest protection services

# Conclusion

## Recommendation

- There can be multiple recommendations for this case. Project A and B are both profitable, however they differ significantly on other factors. Candidate can recommend to proceed with both projects or either one, but they should give a sound justification for their choice. The justification should summarize all the areas that were covered in this case (financial feasibility, geography, resource availability, and any others from framework /brainstorming).

## Risks

- We have ignored the time value of money in our calculations.
- \$8.5M over a significantly long project period could mean very little returns on a yearly basis to the project.
- The source of revenue is sale of carbon credits in the future. Upfront capital is still required to plant trees and kickstart the project, so OTH will have to find other sources of cash for that purpose.
- Any regional limitations/ policy risk/ other area covered in brainstorming

## Next Steps

- Source start up capital for projects to begin planting of trees
- Work with French and/or Indian governments to understand carbon sale rights, policy, and other legal ramifications of this project
- Gauge interest from potential corporate customers for carbon credits

# Mahogany Wonders

Retail & CPG | Sustainability

*Interviewee Led*

Ratings: Quant - 3

Qual - 3

Overall - 3

Maderas is a Costa Rican premium furniture company, specializing in mahogany wood-based products exported to high-end retailers in Europe. Recently, Maderas' largest European clients commissioned an audit of the company's environmental footprint and have raised concerns about Maderas' environmental practices. The company's leadership is exploring how to respond to these concerns and has asked your team for a strategic recommendation. **How should Maderas respond to its customers' sustainability demands?**

## Clarifying Information

### European customers profile and expectations

- These customers represent approximately 80% of Maderas' profits.
- Clients expect at least a 20% reduction in emissions and broader ecological improvements within 5 years. With no change, Maderas is expected to generate 5,000 metric tons of CO2 over the next five years, but would face consequences if they emit more than 4,000.

### Maderas priorities/constraints

- Maderas has \$17M cash available for this project; multiple initiatives may be pursued if they stay within this budget.

### Maderas operations and product

- Sourcing and logistics are mostly outsourced; manufacturing and warehousing are internal.
- Mahogany represents the bulk of volume from Maderas and is a key competitive differentiator due to quality and cost.
- Maderas sources pre-cleaned mahogany wood from multiple local suppliers.
- Other core functions include product design, R&D, and marketing.

# Potential Framework

## Value Chain Diagnosis

- Sourcing
  - Emissions from transport
  - Supplier emissions (deforestation)
- Manufacturing
  - Emissions due to production processes, waste/byproducts
  - Equipment efficiency and energy sources
- Warehousing
  - Energy efficiency and sources
  - HVAC use and efficiency
- Packaging
  - Type of packaging materials
  - Recyclable/renewable
  - Overpacking
- Distribution
  - Shipping modes used
  - Logistics routes optimized?

## Sustainability Initiatives

- Sourcing
  - Contract reform
  - Evaluate sourcing mix
  - Transport optimization
  - Verticalize own plantation
- Manufacturing
  - Process optimization
  - Waste circularity
  - Change energy sources
- Warehousing
  - Solar installations, battery storage systems
  - Smart energy management
  - Relocation
- Packaging
  - Biodegradable materials
  - Change suppliers
- Distribution
  - Emissions-efficient routing

## Impact Assessment

- Upfront investment
- Implementation time
- Financial performance of initiatives; ROI or breakeven period
- Potential emissions reduction
- Initiative/s fit within the company's \$25M investment budget?

# Exhibit 1

## Results of CO2 Emissions Audit

| Process                              | Emission Intensity<br>(kgs CO <sub>2</sub> /ton wood) | Wood Input/Output<br>(tons/year) | Reduction % Possible |
|--------------------------------------|-------------------------------------------------------|----------------------------------|----------------------|
| Sourcing                             | 400                                                   | 1500                             | 40%                  |
| Manufacturing (Cutting,<br>Assembly) | 250                                                   | 800                              | 80%                  |
| Packaging                            | 80                                                    | 800                              | 60%                  |
| Warehousing                          | 20                                                    | 800                              | 70%                  |
| Distribution                         | 150                                                   | 800                              | 50%                  |

Note: Degree of control reflects the portion of emissions Maderas can realistically reduce through decisions

# Exhibit 1 Guidance

## What do you see in Exhibit 1?

| Kg Co2 / Ton | Tons Of Production     | Kg Co2 Produced (Calculated) | Reduction % Possible         | Total Possible Reductions (Calculated) |
|--------------|------------------------|------------------------------|------------------------------|----------------------------------------|
| 400          | 1500                   | 600,000                      | 40%                          | 240,000 (240t)                         |
| 250          | 800                    | 200,000                      | 80%                          | 160,000 (160t)                         |
| 80           | 800                    | 64,000                       | 60%                          | 38,400 (38.4t)                         |
| 20           | 800                    | 16,000                       | 70%                          | 11,200 (11.2t)                         |
| 150          | 800                    | 120,000                      | 50%                          | 60,000 (60)                            |
|              | Total Carbon Emissions | 1,000,000                    | Total Controllable Emissions | 509,000 (509t)                         |

Candidates should convert the objective unit or the calculated units to match for quick comparison. Candidates should drive to calculate the total emission reduction that is feasible.

Good candidates will note that controllable emissions are significantly greater than the emissions reduction demanded by important clients.

Candidates should drive the case forward by identifying which area they would study further. Good candidates should suggest sourcing, since this single area can account for all required emission reductions. Excellent candidates may or may not recommend this, but will note that sourcing could create additional problems due to it being outside the direct control of Maderas.

# Exhibit 2

## Impact Evaluation of suppliers/sourcing initiatives

| Initiative                                 | Cost (\$M) | Time to implement (yrs) | CO <sub>2</sub> Reduction (tons/yr) | Annual Savings (\$M) |
|--------------------------------------------|------------|-------------------------|-------------------------------------|----------------------|
| A) Supplier Incentive Program              | 5          | 2                       | 80                                  | 1.2                  |
| B) Supplier Consolidation & Re-negotiation | 4          | 1.5                     | 60                                  | 1                    |
| C) Traceability Tech Implementation        | 3          | 1                       | 40                                  | 0.6                  |
| D) Invest in Local Plantation Partnerships | 8          | 3                       | 70                                  | 0.9                  |
| E) Switch to International Suppliers       | 10         | 3                       | 120                                 | 1.5                  |

# Exhibit 2 Guidance

Which sustainability initiative should Maderas pursue?

| Initiative                                 | Cost (\$M) | Savings (\$M) | Net Cost (\$) | Time To Implement (Yrs) | Co <sub>2</sub> Reduction (Tons/Yr) | Reduction Within Timeframe | \$ / Ton Reduction<br>(Suggest Reasonable Rounding) |
|--------------------------------------------|------------|---------------|---------------|-------------------------|-------------------------------------|----------------------------|-----------------------------------------------------|
| A) Supplier Incentive Program              | 5          | 1.2           | 3.8           | 2                       | 120                                 | 360 t                      | \$10,555                                            |
| B) Supplier Consolidation & Re-negotiation | 4          | 1             | 3             | 1.5                     | 90                                  | 315 t                      | \$9,523                                             |
| C) Traceability Tech Implementation        | 3          | 0.6           | 2.4           | 1                       | 95                                  | 380 t                      | \$6,315                                             |
| D) Invest in Local Plantation Partnerships | 8          | 0.9           | 7.1           | 5                       | 80                                  | 0 ton                      | Infinite! Very Bad!                                 |
| E) Switch to International Suppliers       | 10         | 1.5           | 8.5           | 3                       | 190                                 | 380t                       | \$22,368                                            |
| Totals                                     |            |               | 24.8          |                         | 370                                 | 1,435 tons                 |                                                     |

Candidates should calculate the net cost of each proposal first. Then, candidate should calculate the total emissions reduction over the 5 year period. Candidates should note that Maderas will need to pursue 3 of the 4 viable initiatives.

Good candidates will immediately rule out (D) for taking too long and perform calculations without that option. Good candidates will use an efficiency metric to identify which option to drop. Good candidates will select options A-C.

Excellent candidates will mock Option E as absurdly expensive.

# Brainstorming

## What are risks associated with the sustainability initiatives we've identified with Maderas?

### Operational Risks

- Supplier non-compliance
  - Small or remote suppliers difficult to monitor
  - Cost incentives reduce likelihood of compliance
- Implementation Complexity
  - More difficult to implement custom reduction plans across suppliers than in a centralized company
  - Equipment efficiency and energy sources
- Supply chain disruption
  - Changes in sourcing can complicate flow of materials
- Timeline risks
  - Projected 5-year reductions very sensitive to time operating
  - Supplier reluctance can lead to significant delays

### Financial Risks

- Upfront capital expenditures
- Cost Overruns
- Inefficient Spending

### Reputational Risks

- Greenwashing Exposure
  - Shifting initiatives to Scope III removes “responsibility” from the Maderas
- Stakeholder Backlash
  - Shifting obligations upstream could anger local partners
- Unmet commitments
- Competitive parity

# Conclusion

## Recommendation

- Maderas should focus initially on reductions in the **sourcing** scope of emissions.
- Maderas should pursue **Options A, B, and C** (candidates should *name* the options)
- These options (1) achieve the needed reduction in the client-demanded time period while (2) prioritizing cost effectiveness.

## Risks

- Supplier Pushback: Local suppliers may resist traceability or emissions performance requirements.
- Cost Pressures: Upfront investment with slow payback may strain cash flows if client loyalty weakens.
- Data Limitations: Lack of emissions measurement or traceability may impair long-term management.
- Execution Risk: Sequencing and cross-supplier coordination may delay visible progress.

## Next Steps

- Engage clients early to co-design reforms and potentially co-fund initiatives (e.g. plantation partnerships).
- Negotiate performance-based contracts with current suppliers, incentivizing compliance.
- Invest in LCA tools and supplier audit frameworks to baseline and track carbon over time.
- Monitor pilot performance (cost, emissions, client perception) and adapt rollout accordingly.

# New England Trucks

Power & Utilities | Operations

*Interviewer Led*

Ratings: Quant - 3

Qual - 3

Overall - 3

# Prompt

Our client (Power Grid) is a power and utilities company with a national presence. The client has 3 hubs in the New England region where their maintenance activities are centered in. Each hub is identical in size and operations.

## Overview of operations:

- Maintenance operations are done between Monday and Saturday.
- Sunday is a day off with only one driver and one engineer at the hub for emergency calls.
- Each hub operates a fleet of trucks which go on routine maintenance runs on a daily, bi-weekly (2 times per week) and weekly basis.
- Each truck carries around diagnostic equipment, replacement parts and two other maintenance personnel
- Each truck starts at their base location at 8AM and has to be back at their base by 5PM

Power Grid is under pressure from the investors to reduce costs but has been facing a shortage of qualified truck drivers. To keep their workforce Power Grid has temporarily increased the hourly rate of each driver from \$18/hour to \$25/hour, but this rate is not sustainable for the entire workforce. The VP of Operations thinks costs on the power generation and transmission side have been completely optimized and can see no further reduction and needs your help to bring costs down further.

## Clarifying Information

This is a McKinsey style, interviewer-led case and little clarifications are needed except for parts the interviewee missed while the prompt was read out.

### Notes to Interviewer

This case has a lot of details and brainstorming. Try to adjust the pace of the case as needed in question 2. If the candidate takes a long time in Q2 ask them to only find the optimal route for one of the cities and push them to find the optimal case. Once the candidate finds the optimal route give them the optimal number of drivers required.

# Question 1

How would you advise the VP of Operations to bring costs down further?

## Expected framework for Average Performance

| Needs                                                                                                                                             | Usage                                                                                                                                                                                                                                       | New Initiatives                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hubs</b> <ul style="list-style-type: none"> <li>Number of hubs</li> <li>Stations served by each hub</li> <li>Operating hours</li> </ul>        | <b>Personnel</b> <ul style="list-style-type: none"> <li>Driver work hours</li> <li>On-site engineer work hours</li> <li>Hub employees</li> <li>Cost of engineer</li> <li>Cost of driver</li> <li>Cost of hub employees</li> </ul>           | <b>Change Routes</b> <ul style="list-style-type: none"> <li>Increase driving time by adding more stops per run</li> </ul>                           |
| <b>Trucks</b> <ul style="list-style-type: none"> <li>Number of Trucks</li> <li>Distance travelled by each Truck</li> <li>Driving hours</li> </ul> | <b>Assets</b> <ul style="list-style-type: none"> <li>Asset used per maintenance activity</li> <li>Cost of assets per maintenance activity</li> <li>Age of vehicles</li> <li>Fuel usage</li> <li>Vehicle maintenance requirements</li> </ul> | <b>Fuel Efficiency</b> <ul style="list-style-type: none"> <li>Use fuel efficient vehicles or EVs</li> <li>Use more fuel-efficient routes</li> </ul> |
| <b>Maintenance</b> <ul style="list-style-type: none"> <li>Maintenance activities</li> <li>Time required for each activity</li> </ul>              |                                                                                                                                                                                                                                             | <b>Hub Reduction</b> <ul style="list-style-type: none"> <li>Increase or decrease the number of hubs</li> </ul>                                      |
|                                                                                                                                                   |                                                                                                                                                                                                                                             | <b>Asset Allocation</b> <ul style="list-style-type: none"> <li>Allocate the right size of the truck per trip</li> </ul>                             |

## Additional Buckets for Exceptional Performance

| New Technologies                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Autonomous vehicles</li> <li>Drones for diagnostics</li> <li>Sensors to monitor station health and conduct maintenance only when required</li> </ul>                                           |
| Risks                                                                                                                                                                                                                                 |
| <ul style="list-style-type: none"> <li>Layoffs may not be good for company's reputation</li> <li>Longer routes may not be great for driver safety</li> <li>Reduced maintenance runs may increase risk of outages and fires</li> </ul> |

# Question 2

[Give Exhibit 1 and 2] Here is the current network of Power Grid and an overview of the current staffing at each hub. Is there an opportunity to reduce costs here by changing the routes? Give this information when asked:

- Each driver is allowed to drive a maximum of 8 hours by government regulations (this time includes the time to drive back from the last station)
- Daily stations require 30 minutes at each station
- Bi-weekly stations require 45 minutes at each station
- Weekly stations require 1 hour at each station
- When a driver shows up to work he is paid for the entire duration of 8 AM – 4 PM

## Solution

If candidate takes too long to determine the best routes let them do just Boston and Hartford



|                 |                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                   |
|-----------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Daily Trips     | H -> B1 -> B2 -> B3 -> B4 -> B5 -> H<br>30 + 30 + 30 + 30 + 30 + 30 + 30 + 30 + 30 + 30 + 30 = 5.5 Hrs | H -> H1 -> H2 -> H3 -> H4 -> H<br>30 + 30 + 30 + 30 + 30 + 30 + 30 + 30 + 30 + 30 = 4.5 Hrs                                                                                                                                                                                                                                                                                            | (Wed, Sat): H -> N1 -> N2 -> N3 -> N4 -> N5 -> N6 -> N7 -> H<br>30 x 15= 7.5 Hrs<br>(All 6 Days): H -> N8 -> N9 -> H<br>30 x 5= 2.5 Hrs<br>(Mon, Thu) H -> N1 -> N2 -> N3 -> N4 -> N10 -> H<br>(Tue, Fri) H -> N1 -> N2 -> N3 -> N4 -> N11 -> H<br>30 x 8 + 45 x 2 + 60 = 6.5 Hrs |
| Bi-weekly Trips | (Tue, Thu) H -> B8 -> B7 -> B6 -> H<br>60 + 45 + 60 + 45 + 60 + 45 + 60 = 6.25 Hrs                     | (Mon) H -> H1 -> H6 -> H2 -> H3 -> H4 -> H<br>30 + 30 + 45 + 45 + 45 + 30 + 30 + 30 + 30 + 30 + 30 = 6.25 Hrs<br>Wed) H -> H1 -> H2 -> H5 -> H3 -> H4 -> H<br>30 + 30 + 45 + 45 + 45 + 30 + 30 + 30 + 30 + 30 + 30 = 6.25 Hrs<br>This is a little tricky as the weekly stations need to be visited only once per week and each weekly station can be visited as a part of a daily trip |                                                                                                                                                                                                                                                                                   |
| Weekly Trips    | (Mon) H -> B12 -> B11 -> H   (Wed) H -> B10 -> B9 -> H 90 + 60 + 90 + 60 + 90 = 6.25 Hrs               |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                   |
| Asset Req.      | 2 Trucks, 2 Drivers                                                                                    | 1 Truck, 1 Driver                                                                                                                                                                                                                                                                                                                                                                      | 2 Trucks, 2 Drivers                                                                                                                                                                                                                                                               |

# Question 2 Solution Guidance

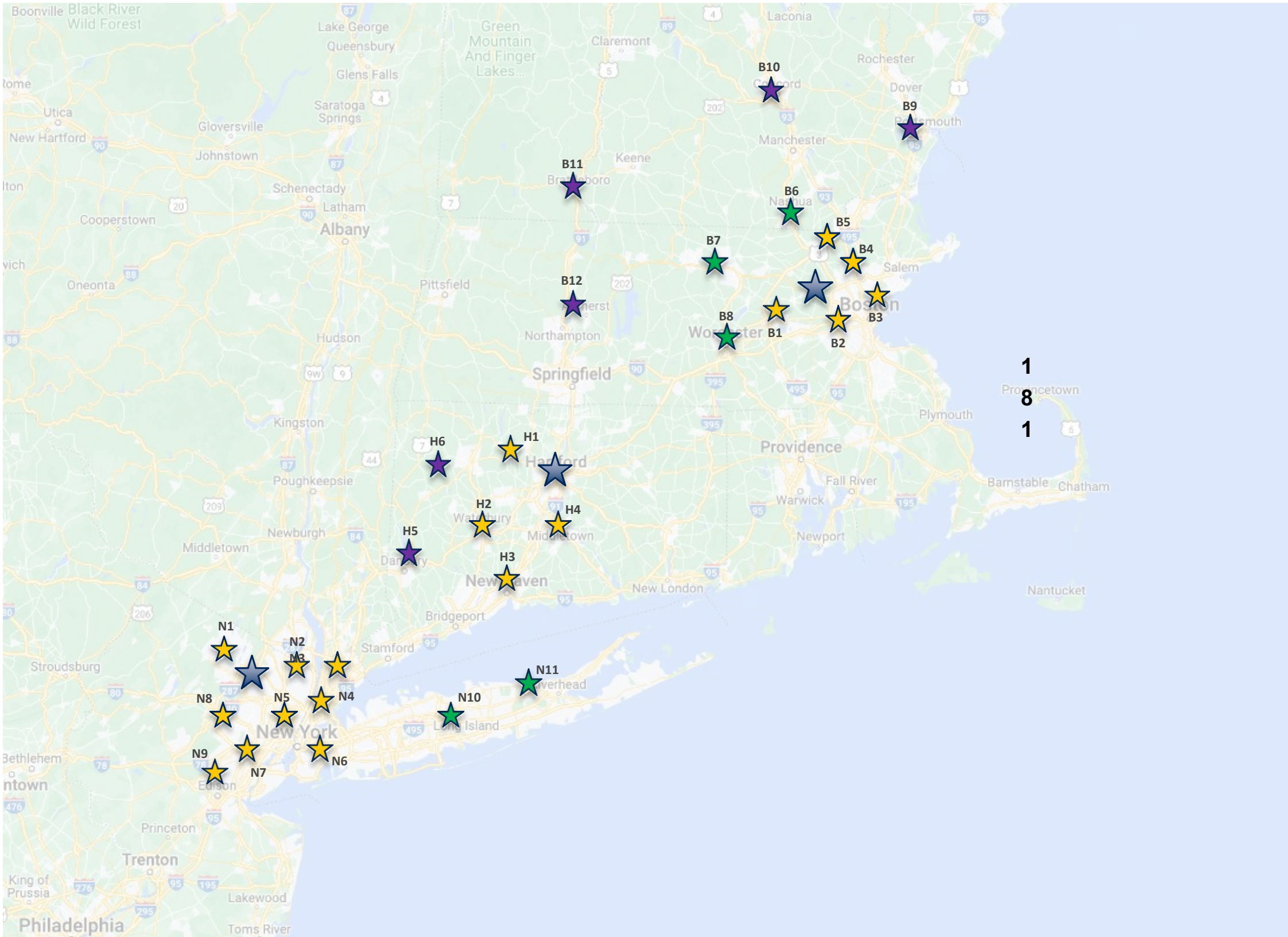
| Days of Week | Mon | Tues | Wed | Thu | Fri | Sat | Sun | Old Costs                                                                  |
|--------------|-----|------|-----|-----|-----|-----|-----|----------------------------------------------------------------------------|
| Boston       | 3   | 3    | 3   | 3   | 3   | 3   | 1   | 19 Drivers Days x 8 hours per day x \$25 per hour = <b>\$3800 per week</b> |
| Hartford     | 2   | 2    | 2   | 2   | 2   | 2   | 1   | 13 Drivers Days x 8 hours per day x \$25 per hour = <b>\$2600 per week</b> |
| NYC          | 4   | 4    | 4   | 4   | 4   | 4   | 1   | 25 Drivers Days x 8 hours per day x \$25 per hour = <b>\$5000 per week</b> |

**Total: \$11,400 per week**

| Days of Week | Mon | Tues | Wed | Thu | Fri | Sat | Sun | Old Costs                                                                  |
|--------------|-----|------|-----|-----|-----|-----|-----|----------------------------------------------------------------------------|
| Boston       | 2   | 2    | 2   | 2   | 1   | 1   | 1   | 10 Drivers Days x 8 hours per day x \$25 per hour = <b>\$2000 per week</b> |
| Hartford     | 1   | 1    | 1   | 1   | 1   | 1   | 1   | 7 Drivers Days x 8 hours per day x \$25 per hour = <b>\$1400 per week</b>  |
| NYC          | 2   | 2    | 2   | 2   | 2   | 2   | 1   | 13 Drivers Days x 8 hours per day x \$25 per hour = <b>\$2600 per week</b> |

**Total: \$6,000 per week Savings:  
\$5,400 per week**

# Exhibit 1



- ★ Hubs
- ★ Daily Stations
- ★ Bi-weekly Stations
- ★ Weekly Stations

**Assume:**

- Each daily station is 30 minutes away from the hub and 30 minutes away from the nearest daily station
- Each bi-weekly station is 1 hour away from the nearest hub and 45 minutes away from the nearest bi-weekly station
- Each weekly station is 1.5 hour away from the nearest hub and 60 minutes away from the nearest bi-weekly station
- Daily station to weekly or Bi-weekly station takes 45 Minutes
- Bi-weekly station to weekly stations stations take 45 Minutes

# Exhibit 2

## Current driver staffing schedule

| Days of Week | Mon | Tues | Wed | Thu | Fri | Sat | Sun |
|--------------|-----|------|-----|-----|-----|-----|-----|
| Boston       | 3   | 3    | 3   | 3   | 3   | 3   | 1   |
| Hartford     | 2   | 2    | 2   | 2   | 2   | 2   | 1   |
| NYC          | 4   | 4    | 4   | 4   | 4   | 4   | 1   |

## Drivers on payroll:

|          | Regular Weekday | Sunday On-Call |
|----------|-----------------|----------------|
| Boston   | 3               | 1              |
| Hartford | 2               | 1              |
| NYC      | 4               | 1              |

# Question 3

For every driver laid off, the client would have to pay severance pay of \$30K. How long would it take to break even on the costs of the layoffs?

## Baseline

|          | Regular Weekday | Sunday On-Call |
|----------|-----------------|----------------|
| Boston   | 3               | 1              |
| Hartford | 2               | 1              |
| NYC      | 4               | 1              |

## New Scenario

| Regular Weekday | Sunday On-Call |
|-----------------|----------------|
| 2               | 1              |
| 1               | 1              |
| 2               | 1              |

## Calculation

| Layoffs | Cost of Layoffs |
|---------|-----------------|
| 1       | \$30K           |
| 1       | \$30K           |
| 2       | \$60K           |

- Total severance payout : \$120K
- Savings per week: \$5,400
- Time to breakeven: ~22 Weeks

Exceptional candidates would recognize that the client no longer requires the extra trucks without the drivers. At this point tell them that each truck would currently sell for \$30K and ask them to re-consider their calculations.

# Solution

4 trucks could be sold for \$30K each, resulting in \$120K cash, which would ensure that there is no breakeven period.

# Question 4

The client is considering selling the current fleet and upgrading to a new fleet of trucks. How would this affect operations and costs?

## Cost Increases

- Training Cost/Hire new personnel
- Learning Curve and initial failures
- Maintain new set of spare parts

## Cost Reductions

- Fuel efficiency
- Less truck maintenance
- Reduced downtime

# Question 5

[Brainstorming question] Is there anything else they could do to reduce costs further?

- Reduce frequency of maintenance
- Remote monitoring of station health
- Train engineer to drive the trucks i.e instead of employing two people send only person can who can do the job for the engineer and the driver

# Mission Artemis

Aerospace & Defense | Supply Chain & Operations

*Interviewee Led*

**Ratings: Quant - 3**

**Qual - 3**

**Overall - 3**

# Prompt

In May 2032, the Artemis Orbital Laboratory (AOL)—a joint mission between NASA, ESA, and JAXA—suffered an aborted resupply mission (RS-19) due to a propulsion malfunction. The station currently houses **four** astronauts and supports **seven ongoing scientific experiments**. The next available launch windows span from **Day 3 to Day 10**, but each comes with differing cargo capacities, failure risks, and agency ownership. Internal reports estimate that the station's critical life-support systems and scientific experiments are under mounting strain. Your team at **NovaLogix**, a space logistics advisory firm, has been contracted to develop an emergency logistics plan. You are expected to recommend a resupply strategy that balances astronaut safety, operational continuity, and stakeholder priorities across multiple international launch options. **(Extra Hint)** You will need to weigh trade-offs across cargo prioritization, timing, risk, and multi-agency coordination to avoid escalation into a full-scale mission failure.

## Clarifying Information

### 1. What are the available launch vehicles, and which agencies control their deployment?

There are currently three launch vehicles available within the next 10 days. We can dive into the specifics later, but each vehicle has unique capacity, risk, and approval requirements. Inter-agency coordination is needed for cross-cargo loading. **(More info in the Exhibit 3)**

### 2. What are the astronauts' daily resource requirements, and how much inventory remains on the station?

Each astronaut consumes 1.4 kg of oxygen, 3.0 kg of water, 2.0 kg of food per day. **We'll discuss more about the inventory remaining in Exhibit 1.**

### 3. How many experiments running on the station and what is the threat? (Answer when asked about scientific experiments)

Answer: There are seven active experiments on board. Two of them are time-sensitive and could suffer irreversible data loss or hardware failure if not addressed within the next 12 to 14 days. The others can tolerate longer delays but may face performance degradation.

### 4. What are the communication protocols and public disclosure policies in the event of mission disruption?

Answer: NASA has released a limited public statement acknowledging the launch delay but has not disclosed specific risk levels. A broader media response will be required if resupply does not occur by Day 8. Internal comms must be carefully aligned with ESA and JAXA press offices.

### 5. What constraints exist in terms of payload volume, weight, and docking compatibility across the launch options?

Answer: All three vehicles are compatible with the station's docking system but have different capacity constraints in terms of weight and volume. Payloads must be pre-approved, and mixed-agency cargo must be prioritized and coordinated in advance.

# Potential Framework

## Internal Assessment

- Resource inventory and consumption rates
- Operational constraints (e.g., station capacity, system reliability)
- Key dependencies and bottlenecks
- Resource inventory and consumption rates
- Operational constraints (e.g., station capacity, system reliability)
- Key dependencies and bottlenecks

## Logistics & Launch Planning

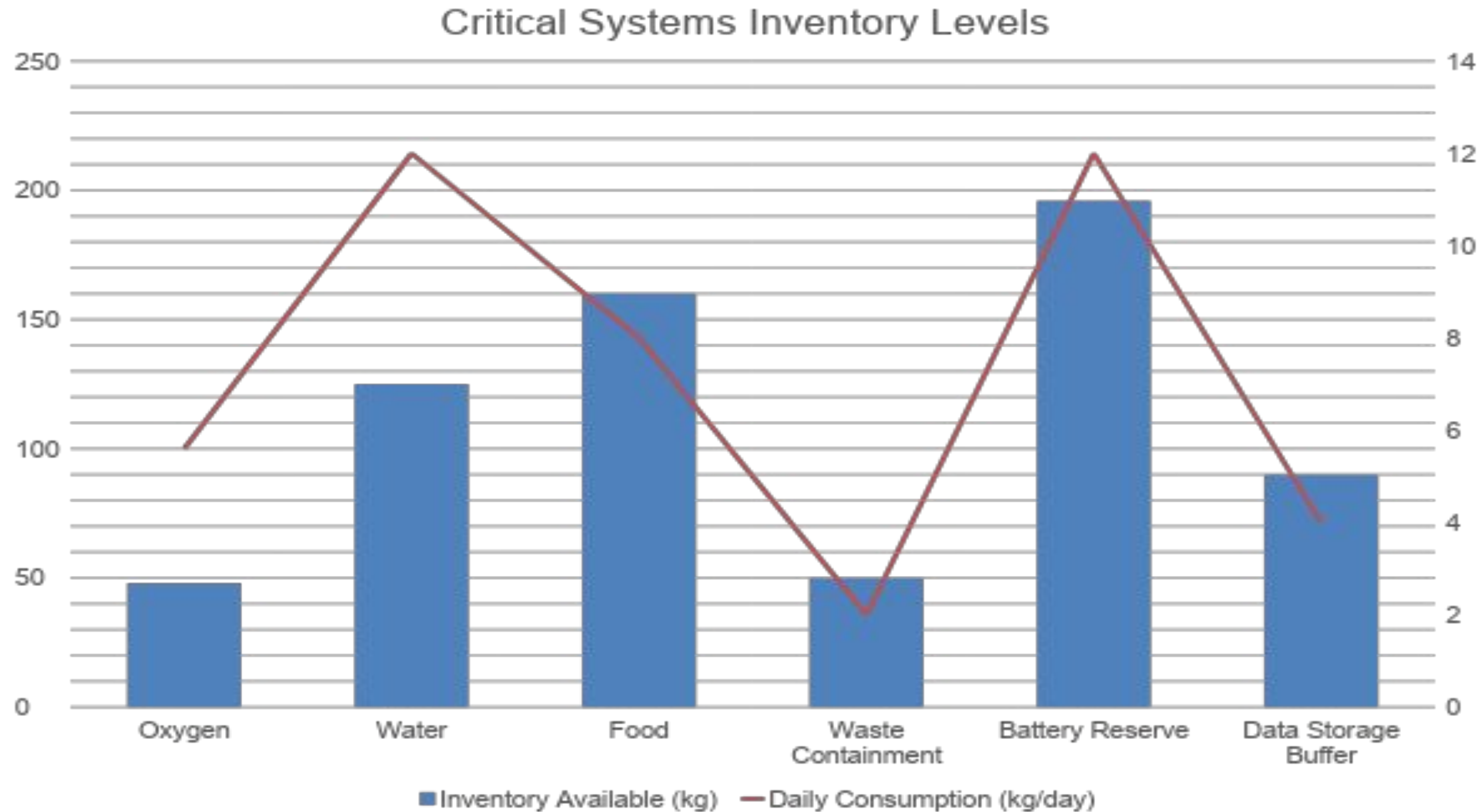
- Supply/delivery capabilities and timing
- Cost, risk, and capacity trade-offs
- Feasibility of alternatives (backups, reallocations)
- Launch windows, vehicle capacities, failure risks
- Cargo weight/volume constraints and docking compatibility
- Allocation strategy under time, risk, and agency constraints

## Stakeholder & Priority Alignment

- Primary and secondary objectives
- Political, financial, and reputational risks
- Communication and coordination requirements
- Inter-agency goals (NASA, ESA, JAXA, private sponsors)
- Contractual and diplomatic obligations
- Scenario planning for trade-offs (life support vs. research)

# Exhibit 1

## Artemis Station – Life Support and System Inventory



# Exhibit 1 Guidance

**Based on the exhibit above, how long will each supply last the astronauts? Which are most urgent to consider?**

| System Metric       | Inventory Available (kg) | Daily Consumption (kg/day) | Days Remaining |
|---------------------|--------------------------|----------------------------|----------------|
| Oxygen Supply       | 48                       | 5.6                        | 8.6 days       |
| Water Supply        | 125                      | 12.0                       | 10.4 days      |
| Food Supply         | 160                      | 8.0                        | 20.0 days      |
| Waste Containment   | 50                       | 2.0                        | 25.0 days      |
| Battery Reserve     | 196                      | 12.0                       | 16.3 days      |
| Data Storage Buffer | 90                       | 4.0                        | 22.5 days      |

Notes for interviewer:

Provide days remaining formula to interviewee if stuck:

Days Remaining = Inventory Available/(Daily Consumption\* Number of Astronauts)

1. Oxygen (8.6 days) and Water (10.4 days) are the **most urgent life-support constraints**.
2. Battery Reserve now has 16.3 days of supply — no **longer immediate risk**.
3. Food (20.0 days) is stable short-term; **resupply can be deferred**.
4. Waste (25.0 days) and Data Storage (22.5 days) are **low priority**.
5. Launch sequencing must address oxygen and water within 10 days to avoid mission failure.

# Exhibit 2

## Payload overview

| Payload Item                           | Weight (kg) | Volume (m³) | Minimum Delivery By (Days) | Priority                                                                              | Notes                                     |
|----------------------------------------|-------------|-------------|----------------------------|---------------------------------------------------------------------------------------|-------------------------------------------|
| Oxygen + Water Resupply                | 240         | 3.5         | 8                          |    | Required for crew survival                |
| Battery Replacement Packs              | 120         | 2.0         | 16                         |    | Not urgent post-inventory update          |
| Biomedical Experiment Kit              | 80          | 1.0         | 10                         |   | NASA-funded; data loss if delayed         |
| Food Buffer (15 Days)                  | 150         | 2.5         | 18                         |  | Optional; improves comfort, not survival  |
| Private Research Payload               | 100         | 1.5         | 20                         |  | Delay results in refund/PR penalty        |
| CO <sub>2</sub> Scrubber Repair Module | 50          | 0.5         | 14                         |  | Needed to restore recycling functionality |

*\*Payloads must be pre-approved for agency-specific vehicles. Mixed agency cargo requires diplomatic coordination 48 hours in advance.*

# Exhibit 3

## Launch Vehicles – Capacity, Cost & Risk Profile

| Launch Vehicle | Agency | Available In (Days) | Max Payload (kg) | Volume Capacity (m³) | Failure Risk | Cost (USD) |
|----------------|--------|---------------------|------------------|----------------------|--------------|------------|
| Falcon-X       | NASA   | 3                   | 500              | 8.0                  | 2%           | \$28M      |
| Vega+          | ESA    | 7                   | 300              | 5.0                  | 5%           | \$18M      |
| Kibo-L         | JAXA   | 14                  | 600              | 6.0                  | 6%           | \$22M      |

# Exhibit 2 & 3 Solution

## 1. Match Launch Feasibility Based on Timing

| Payload                        | Can Launch On Falcon-X? | Vega+?   | Kibo-L?                |
|--------------------------------|-------------------------|----------|------------------------|
| Oxygen + Water (D8)            | Yes (D3)                | Yes (D7) | No (D14 – too late)    |
| Biomedical Kit (D12)           | Yes                     | Yes      | No (D14 – too late)    |
| CO <sub>2</sub> Scrubber (D14) | Yes                     | Yes      | Risky (D14 – too late) |
| Battery Packs (D16)            | Yes                     | Yes      | Yes                    |
| Food Supply (D18)              | Yes                     | Yes      | Yes                    |
| Commercial Payload (D20)       | Yes                     | Yes      | Yes                    |

# Exhibit 2 & 3 Solution

## 2. Optimize Cargo Allocation by Weight/Volume

### Falcon-X Priority Payload Loading

Falcon-X Capacity: 500 kg / 8.0 m<sup>3</sup>

#### Suggested Loadout:

Oxygen + Water → 240 kg / 3.5 m<sup>3</sup>

Biomedical Kit → 80 kg / 1.0 m<sup>3</sup>

CO<sub>2</sub> Scrubber → 50 kg / 0.5 m<sup>3</sup>

Subtotal: 370 kg / 5.0 m<sup>3</sup>

**Remaining Capacity:** 130 kg / 3.0 m<sup>3</sup>

Could optionally include:

**Battery Reserve** → 120 kg / 2.0 m<sup>3</sup> → Total = 490 kg / 7.0 m<sup>3</sup>

or

**Commercial Payload** → 100 kg / 1.5 m<sup>3</sup>

**Decision point:** Can send 1 more item — either Battery (medium priority) or Commercial (low priority, long deadline)

# Exhibit 2 & 3 Solution

## 2. Optimize Cargo Allocation by Weight/Volume

### Falcon-X Priority Payload Loading

Use Vega+ or Kibo-L for Deferred Payloads

#### Remaining Items:

Food (150 kg / 2.5 m<sup>3</sup>, D18)

Commercial Payload (100 kg / 1.5 m<sup>3</sup>, D20)

Whichever wasn't included on Falcon-X

#### Option A – Vega+ (D7):

Capacity: 300 kg / 5.0 m<sup>3</sup>

Food + Commercial = 250 kg / 4.0 m<sup>3</sup>

#### Option B – Kibo-L (D14):

If Vega+ fails or is unavailable, both items can still be delivered here safely

Extra buffer for backup planning

# Exhibit 2 & 3 Solution

## 3. Cost Scenarios

| Scenario                                     | Vehicles Used     | Payloads Sent                                         | Total Cost |
|----------------------------------------------|-------------------|-------------------------------------------------------|------------|
| Falcon-X only (limited)                      | Falcon-X          | Oxygen + Water, Biomedical, CO <sub>2</sub> , Battery | \$28M      |
| Falcon-X + Vega+                             | Falcon-X + Vega+  | Critical payloads + Food + Commercial                 | \$46M      |
| Falcon-X + Kibo-L (buffered launch)          | Falcon-X + Kibo-L | Critical payloads + deferred payloads                 | \$50M      |
| Falcon-X + Vega+ + Kibo-L (risk-spread plan) | All 3             | Spread payloads across launches for safety            | \$68M      |

## 3. Risk Evaluation

- Falcon-X: 2% failure → place Oxygen + Water here
- Vega+ / Kibo-L: 5–6% failure → use for lower-priority or backup items
- Consider splitting non-critical payloads (e.g., food/commercial) to de-risk

# Exhibit 3

## Stakeholder Objectives & Constraints

| Stakeholder    | Sponsor Type   | Primary Objective                          | Critical Payload(s)                            | Flexibility Window       | Penalty or Risk if Delayed                 |
|----------------|----------------|--------------------------------------------|------------------------------------------------|--------------------------|--------------------------------------------|
| NASA           | Government     | Astronaut safety + U.S.-funded research    | Oxygen + Water, Biomedical Kit                 | Low ( $\leq 12$ days)    | Political fallout; biomedical data loss    |
| ESA            | Government     | European research continuity               | CO <sub>2</sub> Scrubber, Shared Food Cargo    | Medium ( $\leq 14$ days) | Contractual obligation + reputational risk |
| JAXA           | Government     | Long-term station operations & payload use | Battery Reserve, Joint Maintenance             | Medium ( $\leq 16$ days) | Tech system inefficiency; funding pushback |
| Commercial Lab | Private Sector | Deliver proprietary experiment module      | Commercial Payload                             | High ( $\leq 20$ days)   | Refunds, reputational loss, PR escalation  |
| Astronaut Crew | Internal User  | Survival + mission stability               | Oxygen + Water, Food, CO <sub>2</sub> Scrubber | None (hard deadlines)    | Health failure, mission failure            |

# Exhibit 4 Guidance

## Key Observations:

- NASA and the Astronaut Crew are aligned on life-support-critical payloads (Oxygen, Biomedical Kit) with non-negotiable deadlines. These must be prioritized on the earliest, most reliable launch (Falcon-X).
- ESA's request (CO<sub>2</sub> Scrubber) is important but has a medium window, making it a strong candidate for a second launch (Vega+ or Kibo-L), assuming early launches are maxed out.
- JAXA's priorities (Battery Packs) are not time-sensitive and can be postponed or bundled with other medium-priority cargo.
- The Commercial Lab has the most flexible deadline, lowest mission impact, but the highest external visibility if delayed — a PR risk, not a survival or technical one.
- When allocating resources, operational risk > reputational risk — especially under tight weight/volume constraints.

# Conclusion

## Recommendation

- Recommend launching the **Falcon-X in 3 days** with a priority load consisting of the Oxygen + Water Module, the Biomedical Kit, and the CO<sub>2</sub> Scrubber Module. This fully utilizes Falcon-X's capacity and ensures on-time delivery of mission-critical systems with minimal failure risk (2%).
- Remaining payloads—including the Battery Reserve Packs, Food Supply Extension, and the Commercial Payload—should be sequenced across Vega+ (Day 7) and, if needed, Kibo-L (Day 14) based on final volume and cost efficiency. This staggered approach balances capacity, timeline flexibility, and inter-agency obligations.

## Risks

- Single-point-of-failure on Falcon-X: If the initial launch fails, critical systems are jeopardized. A backup plan using Vega+ for partial resupply of life-support must be pre-approved.
- Diplomatic friction: ESA or JAXA may push back if their payloads are deprioritized. Proactive communication is needed to align on mission-first principles.
- Commercial sponsor backlash: Private lab's visibility means that delays must be clearly justified with a future recovery plan or partial data retention.

## Next Steps

- Confirm cargo manifest and safety checks for Falcon-X within 24 hours.
- Engage ESA and JAXA to align on staggered resupply strategy and provide visibility on Vega+ and Kibo-L plans.
- Issue formal communication to the commercial sponsor, including risk disclosure and a backup commitment via Kibo-L.
- Prepare contingency protocol to reallocate payloads in case of launch failure or technical docking complications.



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